Formalizing the IUT III 3.11–3.12 Corridor in Lean

IUT Lean formalization project

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Abstract

This paper reports the current state of a Lean formalization project for a narrow part of Mochizuki's inter-universal Teichmuller theory: the corridor from IUT III, Theorem 3.11 to Corollary 3.12. The project is deliberately neutral about the correctness of IUT as a proof of the *abc* conjecture and about the Mochizuki–Scholze–Stix dispute. Its purpose is to make the comparison levels, indeterminacy actions, real-line transports, and source obligations explicit enough that the proof assistant can distinguish proved consequences from mathematical hypotheses still to be constructed.

The formalization now verifies the ordered-real endpoint of Corollary 3.12, separates the representative j^2-level from full-label, averaged, and hull/log-volume comparison levels, and builds a typed route through concrete Hodge-theater/log-theta-style source data, $\mathrm{Ind}_1/\mathrm{Ind}_2/\mathrm{Ind}_3$indeterminacy records, quotient possible images, a selected q-region, Remark 3.9.5-style hull/determinant data, and the existing C_Θ endpoint. The possible-image layer is now witness-bearing: class-level possible images, quotient pullbacks, and the selected q-region carry explicit nonemptiness data through the Step (xi) valuation-ball/log-shell source chain. The class/log-theta-lattice formula and typed fiber-transport sources now carry these witnesses as part of the same Theorem 3.11 spine; the top-level additive-Haar arithmetic-divisor audit consumes the corresponding nonemptiness proof to build the witness-bearing Theorem 3.11 possible-image record, rather than using a vacuous possible-image predicate. The arithmetic-divisor and record Ob3/Ob5 additive-Haar constructors can now obtain all-choice possible-image nonemptiness directly from the class-indexed Theorem 3.11 image law and the record/source image equality; the same witness bridge constructs the central Theorem 3.11 possible-image source-data record and its nonemptiness audit. The current Step (xi) milestone now also has a constructed theta-region completion audit: the preferred public route and the norm-square/Ob7 evidence are built from the same theta-region principal valuation-ball source, localized Step (xi) C_Θ source, and restriction-calibrated p-adic source. The Theorem 3.11 record boundary has also been lowered: a primitive constructor now derives the multiradial source record, the transport-base constructed qualitative/SHE bundle, and the constructed transport guards from explicit input-prime-strip, structured-SHE, algorithmic-parallel-transport, certificate-equality, Hodge-theater/log-theta/ log-Kummer, and target-region lattice source fields. A further packet layer now projects the unified source spine, target-region lattice formula, bridge-certificate constructor, and primitive constructor from those concrete fields. The packet also projects a vertical log-Kummer packet-aligned one-sided multiradial source, so its own theta-pilot images, gluing torsor, selected q-choice, and vertical log-Kummer source feed the finite F_ℓ procession/equality-quotient audit. The packet-route audit checks that the bridge certificate generates the primitive qualitative source and primitive constructor audit used by the public Step (xi) route. The qualitative side has also been linked to the Theorem 3.11 IPL/SHE/APT bridge package by an explicit source-copy and output-family alignment layer. The concrete bridge component can now be constructed from the lower constructed qualitative certificate source, and the packet audit exposes the corresponding no-collapse guards: forbidden direct SHE domain-to-codomain identification, permitted APT quotient transport, and selected IPL source/target matching. A further alignment source now packages the constructed qualitative certificate together with

the packet-specific bridge alignments and builds the primitive packet constructor from that source. The SHE/common-container alignment has been lowered one step further: it is now derived from constructed qualitative common-container data rather than supplied as a standalone packet equality. The input and selected output prime strips can now also be chosen canonically from the constructed qualitative IPL source. The remaining lower task is to construct this packet from the analytic multiradial, log-theta, local arithmetic, and IUT IV estimates of the papers. The current result is a checked conditional Stage 1 corridor, not a closed proof of Corollary 3.12 from Mochizuki's published hypotheses.

Scope correction. Names such as "Theorem311", "HodgeArakelov", "logKummer", "IPL", "SHE", and "APT" in the current Lean declarations label finite or conditional interfaces. They do not certify that the corresponding IUT I–III definitions, constructions, or theorem clauses have been formalized. The machine-readable primary-source ledger currently marks five transitive foundation clauses source-faithful, but no complete M1–M3 milestone or Theorem 3.11 clause source-faithful.

The source-reconstruction branch now contains actual topological mono-theta environments and a categorical inverse limit for IUT II, Definition 2.7(i). For Definition 2.7(ii), the nested groups are inverse images under the ambient homomorphism and the four indicated subquotients are actual topological quotients by normal subgroups. Subquotient restriction is implemented as $f^{-1}(U)/f^{-1}(N)$: the exterior cyclotome comes from the exact kernel in part (i), while the other three narrow-level objects are pullbacks of ambient subquotients. Cyclotomic rigidity is represented by a continuous group equivalence between the derived endpoints. Continuous nonabelian H^1 and its direct limit over open subgroups are quotient constructions, and pullback along a continuous group homomorphism descends to this direct limit. With abelian coefficients, both quotient constructions carry proved commutative group laws and restriction is a homomorphism. Consequently, given the paper's two homomorphisms and the required μ_{2l}- and μ-equivariance, both Corollary 2.8 restriction stages derive their output orbit families and preserve identity representatives at the zero label. The two independent conjugacy actions on inertia and ambient subgroup choices are represented by a product action; any evaluation equivariant for this action is proved to descend to the associated orbit quotients. The exact quotient by torsion in a constant subgroup and its constant/theta multiplication map are also constructed; if every zero-label theta value is torsion, this map is proved bijective. Given a zero-to-base action-pair comparison, the zero-labeled constant monoid is also proved multiplicatively equivalent to its actual diagonal range, compatibly with the transported labeled actions. Construction of the precise geometric subquotients, homomorphisms, actions, input theta orbits, and independent conjugacy compatibility, together with the paper-specific splitting compatibility, remains open.

For IUT II, Corollary 3.5, multiplication by $(\mathbb{Z}/l\mathbb{Z})^\times$ has been descended through the sign-label quotient and factored through the exact quotient by $\{\pm 1\}$; the resulting action on nonzero absolute labels is proved transitive. A label-indexed family of topological group/monoid action pairs now requires coherent identity and multiplication laws for its outer transports. Such transports derive the symmetrizing maps, an injective diagonal monoid, and the heterogeneous Gaussian monoid. The transported base-group action is proved to preserve diagonal units, and profile stability yields an actual action by automorphisms of the Gaussian monoid. Exponent-$2l$ unit orbits also imply equality of the corresponding $2l$-multiple Gaussian submonoids. These are conditional source interfaces: their construction from the paper's outer action, theta-value restrictions, and single-inner-automorphism synchronization remains open.

At the legacy toy boundary, initial theta data constructs typed zero/one histories, owned prime strips, an effective-divisor degree-category realization of the IUT I Hodge theater, and bad-place Hodge–Arakelov q and theta pilots. The theta pilot is the Gaussian q^{j^2} degree family on $\mathbb{Z}/l\mathbb{Z}$, using coordinates from the actual theta-root quotient torsor. The formalization proves q-order positivity and coprimality, zero/one normalization, sign invariance, and translation/orbit-average identities. The next finite toy layer constructs a typed procession, $\mathbb{Z}/l\mathbb{Z}$ $\mathrm{Ind}_1/\mathrm{Ind}_2$ actions and their generated quotient, a directed Ind_3 upper-semi relation, nonempty possible images, and a finite-coordinate bijection from the bad-local quotient torsor. The vertical graph and image are exact, and the finite discrete image is compact-open. Typed theta-link, gluing, and IPL/SHE/APT compatibilities retain distinct zero/one histories. These calculations test formal variance and quotient bookkeeping only: they do not construct the processions, indeterminacies, vertical log-Kummer correspondence, or IPL/SHE/APT algorithms of Theorem 3.11.

The current audit separates the active M1–M3 source reconstruction from the finite Stage 1 regression route. The active manifest contains fifteen exact IUT I–III construction obligations consumed by the assembled source Theorem 3.11 column boundary. A checked theorem excludes the old `packet`, `sourceWithSymmetry`, `compactOpenRealizedExactSource`, and `hodgeEvaluation` names from that active manifest. The same-index q-normalized route with compact-open additive-Haar p-adic-defect/main residual lowering remains separately recorded as a legacy finite adapter for downstream Corollary 3.12 bookkeeping; it does not discharge a source clause. On that downstream route the p-adic finite localized hull-vector-bundle source is derived from the finite-extension restriction-calibrated local arithmetic handoff. The concrete Ind_2 local-tensor source now has a checked record audit and typed-core audit, recording the direct-summand equality, procession and upper-semi transports, normalized log-volume preservation, equality-quotient compatibility, and theta-pilot class invariance. The pointwise determinant boundary is also audited one layer lower: Lean records the determinant-free measure/summand formula source and the full summand-charted construction of the determinant source, deriving determinant log-volume, normalized determinant, family-hull, and theta-signed equalities from the finite adjusted-summand Hodge calibration. On the active IUT II Kummer route, Lean now proves the arithmetic core of Absolute Anabelian Topics III, Remark 1.5.4(i): for every finite extension of a mixed-characteristic local field, the unit group of the integral closure of the base valuation ring is residually finite. The proof uses Krull intersection to find a separating maximal-ideal power and finiteness of the resulting quotient. Deriving injectivity of the constructed unit Kummer map also uses a proved finite-fixed-field bridge: for a compact IUT acting group, the augmented image of an open germ domain has finite index and is compact closed, hence is open; infinite Galois correspondence makes its fixed field a finite MLF extension. Lean now unpacks equality of a unit Kummer germ with one into a common open subgroup and a coboundary, evaluates the cochain at $1/n$, and proves that the resulting adjusted root has n-th power equal to the original unit and is fixed by that subgroup. The root and its inverse are integral in the finite fixed field by the power criterion. Residual finiteness then forces the original unit to be one, proving injectivity of the constructed unit Kummer map without a stored faithfulness assumption. The canonical C_Θ-scale equality is derived from the q-normalized theta/log-volume identity plus q-pilot positivity. Conversely, the checked p-adic finite, principal-product, and same-index scale-synchronized routes now construct the q-normalized residual source internally from canonical scale synchronization, so the audit records this scalar conversion as derived rather than leaving it implicit in endpoint wrappers. The Record-Ob3/Ob5 valuation-ball constructor-built boundary now applies the same conversion locally: the direct scalar equality between the constructed canonical C_Θ scale and the Record-Ob3/Ob5 adjusted summand is derived from $\theta = \mathrm{adjustedSummandLogVolume} \cdot (-q)$ and q-pilot positivity, while the determinant side remains constructor-built from the same Ob3/Ob4 adjusted-determinant source. Its family-hull refinement now lowers this further: the family-hull q-normalized theta comparison is transported across the source-level $\mathrm{familyHullLogVolume} = \mathrm{adjustedSummandLogVolume}$ theorem before entering the q-normalized valuation-ball constructor, and the matching audit pins the transported q-normalization, derived canonical scale equality, valuation-ball formula source, and endpoint. This now feeds a synchronized route-input constructor: the possible-image witness supplies the named-HDD nonemptiness, the valuation-ball source supplies the constructor-built/record Ob3/Ob4 determinant equality, and Lean reconstructs the localized determinant/scale synchronization, arithmetic-divisor endpoint, and Haar lower bound before any compact-open route consumes the bundle. The residual-frontier manifest now has 320 checked entries: 314 derived entries and 6 constructed countermodel diagnostics. The newest determinant-facing audit checks that the pointwise summand formula is reconstructed from determinant/Hodge calibration and the record-native determinant decomposition into adjusted summands. The source-level global C_Θ audit also now records the principal pointwise log-shell route and its exact-Ξ consequences directly, so the family-hull log-volume equality and no-Ind3-generator conclusion no longer pass through the compact-open log-Kummer correspondence as an exposed consequence boundary. The pointwise log-Kummer projection is also audited structurally: the compact-open map, correspondence, image, possible-region, and reconstructed pointwise sources preserve the same principal valuation-ball source, theta labels, and cover places. The compact-open realized-exact and map sources are now constructed directly from pointwise log-Kummer coordinates via the log-shell exact source and compact-open Haar normalization, with an audit for the realized theta-region equality, selected principal hull, and shared source labels. The Step (xi)-to-case-procession equality is now also factored as a reusable arithmetic-degree bridge: if both sides equal $a_v(D_v + C_v)$, Lean derives the direct local comparison. The selected p-adic handoff is now lowered once more: the public selected endpoint consumes a p-adic-normalized case-defined arithmetic-gap source carrying the Step (xi)-to-case-procession equality and the normalized-place arithmetic-minus-main +1 inequality, while the selected Theorem 1.10 ledger fixes the normalized p-adic place as distinguished. Thus the newest public selected endpoint no longer exposes a prime-error split, arithmetic formula-matching source, standalone normalized-place-kind field, or the stronger normalized processional-estimate package at this layer. When that stronger estimate package is available, Lean derives the selected arithmetic-gap source directly from the selected case-defined equality, the p-adic unit-ball local arithmetic identity $upper_v = StepXI_v + defect_v + M_v$, and the normalized Haar-defect lower bound $1 \leq defect_v$ at the selected place. The same adjusted-sum p-adic arithmetic-gap route now has a constructed Remark 3.9.5 hull/determinant-source form: the public path projects q-positivity, normalization, certification, SHE alignment, and hull/determinant obligations from the constructed holomorphic hull/determinant source instead of accepting a separate hull-det obligation package. The stronger normalized-estimate route is threaded through this lowering after it produces the selected case-defined equality. The p-adic-normalized case source used by the Haar-residual route can now be projected from the lower processional pointwise-residual source once its distinguished residual place is identified with the p-adic normalized place, so the unit margin is read from the residual lower-bound theorem rather than from the stronger processional-estimate ledger. The packet-facing p-adic constructor now composes this projection internally, so that boundary can take the pointwise local-analytic IUT IV source, the expanded p-adic Haar residual identity, the processional pointwise-residual source, and the normalized-place alignment directly. This branch has now been lowered again from the processional arithmetic-gap source: Lean projects its Step (v)/(vi)/(vii) normalized local-log-volume identities to the p-adic-normalized case source after the distinguished place is aligned with the p-adic normalized place. This alignment has now been internalized in a p-adic-normalized arithmetic-gap source: its distinguished place is definitionally the p-adic normalized finite place, while the arithmetic-minus-main +1 gap and the Step (v)/(vi)/(vii) local-volume identities remain source fields. The full local-analytic processional estimate source now also projects to the same p-adic-normalized case source: Lean forgets the distinguished +1 processional upper-bound field and keeps only the case-wise normalized local-volume identities after normalized-place alignment. The case-defined processional source now fixes that local value directly from the Theorem 1.10 Step (v)/(vi)/(vii) case split, so the ordinary and p-adic-normalized processional estimate constructors no longer expose the three local identity families as independent fields on this branch. The normalized Haar-residual case source has now been lowered to the same case-defined value: Lean reconstructs the older distinguished, nondistinguished, and archimedean identity families from a single Step (xi) equality against the case-defined value plus the normalized-place kind. The p-adic unit-ball/local-analytic public surface has now been lowered again to the selected Theorem 1.10 case-defined reconstructed p-adic Haar-defect frontier: it consumes the p-adic unit-ball localized Step (xi) source, a selected distinguished-place valuation-ball local-analytic ledger at the p-adic Haar normalized place, the local-evaluation equality, and a single case-defined Step (xi)-to-procession equality. Lean derives the p-adic normalized-place kind from the selected local-kind construction, then reconstructs the expanded p-adic Haar-residual identity, p-adic Step (xi) synchronization, p-adic-normalized estimate and case records, pointwise residual theorem, generic Haar-defect source, and older local-analytic handoff internally. The selected equality can now be constructed from the p-adic arithmetic formula-matching source $StepXI_v = a_v(D_v + C_v)$, the Theorem 1.10 three Step (v)/(vi)/(vii) local arithmetic-degree clauses: distinguished exact procession, nondistinguished zero, and archimedean metric-container formulas. Lean first recombines these clauses into caseProcession$_v = a_v(D_v + C_v)$, then uses the local-evaluation alignment to reconstruct the selected case-defined p-adic source internally. Separately, the valuation-ball Theorem 1.10 layer now has an arithmetic-degree bound source: its distinguished exact-procession, nondistinguished zero, and archimedean metric-container bounds recombine to the all-place inequality caseProcession$_v \leq a_v(D_v + C_v)$. The stronger equality source canonically maps to this estimate source by le_of_eq, while the older projected-procession/arithmetic-minus-main chain remains the independent upper-bound route. Thus the equality route remains a specialized formula-matching refinement, and the paper-level estimate route is now represented by a derived arithmetic-degree inequality object. The formula-gap matched valuation-ball refinement now also audits the local estimate before the additive-Haar projection: the distinguished Proposition 1.4 valuation-ball log-shell source and the archimedean metric source carry their container bounds through the named formula-gap comparisons. The additive-Haar valuation-ball arithmetic-degree-controlled local source now projects to this canonical bound source: the Step (v) and Step (vii) arithmetic-degree comparisons are reused, while the Step (vi) zero bound is derived from nonnegativity of the local different/conductor degrees and nonnegativity of the coefficient a_v. This bound now lowers through the Record-Ob3/Ob5 valuation-ball source, the local-degree formula source, the controlled component source, and the p-adic-defect/main controlled source, so the pointwise Theorem 1.10 bound and residual package live at the same checked lower source boundary. The same source now also projects directly to the canonical local arithmetic-degree residual package: the Step (xi) calibration supplies $StepXI_v = a_v(D_v + C_v)$, the IUT IV/p-adic defect source supplies $E_v = \delta_v + M_v$, and Lean proves $StepXI_v + \delta_v = a_v(D_v + C_v) + E_v - M_v$ with the finite Haar-defect lower bound carried by the defect source. The p-adic-normalized arithmetic-gap branch has now been lowered to the same case-defined local value: Lean reconstructs the older normalized arithmetic-gap source, normalized case source, lower-weight residual package, Haar-residual handoff, and the pointwise/p-adic unit-ball packet constructors from a single Step (xi) equality against the case-defined procession plus the normalized-place arithmetic-minus-main +1 inequality. The same case-defined arithmetic-gap source now also reaches the signed p-adic unit-ball/local-analytic public endpoint; the public route reconstructs the older normalized arithmetic-gap record internally before invoking the Corollary 3.12 dichotomy. When the stronger case-defined p-adic-normalized processional estimate source is available, Lean now derives that arithmetic-minus-main +1 inequality by composing its projected-procession +1 bound with Theorem 1.10's local arithmetic-divisor gap estimate. The case-defined estimate source itself can now be constructed from the lower local-analytic procession-bound residual source plus the case-defined Step (xi) equality: Lean transports the lower Step (xi) +1 procession-bound inequality across that equality, rather than taking the case-procession +1 field independently. The same case-defined equality now reconstructs the lower processional pointwise-residual source from the pointwise defined residual package: Lean projects the three Step (v)/(vi)/(vii) procession-normalization identities from the case function, while the distinguished unit margin remains the pointwise residual theorem. The Haar-defect layer is lowered in the same way: from the pointwise Haar-defect residual source and the same case-defined equality, Lean reconstructs the processional Haar-defect residual source and its downstream lower-weight/local arithmetic-degree endpoints, without carrying the three identity families independently. The local C_Θ constructor now consumes this lower pointwise-Haar/case-procession package directly: Lean rebuilds the processional Haar-defect source internally before entering the established localized C_Θ handoff. The selected p-adic public branch now consumes the selected p-adic-normalized case-defined arithmetic-gap source and reconstructs the selected case-defined Haar-defect source before invoking the same endpoint. Thus the remaining local-estimate obligation at this frontier is the paper-level construction of the normalized-place arithmetic-minus-main +1 gap and the Step (xi)-to-case-procession equality; the stronger normalized processional-estimate source is now a sufficient constructor for that boundary, not the public boundary itself. The stronger valuation-ball formula-gap source and the case-bounded IUT IV residual source are no longer canonical public inputs: Lean constructs the formula-gap package from the local-analytic Theorem 1.10 ledger when needed, and projects case-bounded/p-adic data into the local arithmetic-degree residual package. The adjusted-sum same-index branch now has the matching local-analytic case-source endpoint: its public valuation-ball input is the local-analytic Theorem 1.10 ledger plus the three local case inequalities, while the formula-gap and residual packages are reconstructed internally. The strongest possible-image/compact-open correspondence branch now has the analogous local-analytic case-residual endpoint, so this lowering is available after the symmetry-label transport, log-Kummer correspondence, pointwise arithmetic handoff, and pointwise determinant-formula data have been constructed. A constructed-hull/determinant variant also removes the independent source-obligation-gap input on that branch by building it from the primitive constructed-SHE bundle and the Remark 3.9.5 hull/determinant obligation projection. A principal-HDD variant lowers the same branch further by reading the source-obligation gap from the principal product-hull CanonicalHDD direct common-container source itself. Its procession-bound refinement lowers this once more: the public residual input is the projected case-defined procession bound and the distinguished one-unit gap, from which Lean reconstructs the local case source internally. Its procession-comparison refinement separates the localized Step (xi) equality with procession-normalized local log-volume from the upper-semi projected procession estimate before reconstructing that procession-bound source internally. Its processional-estimate refinement derives that upper-semi estimate from the local Step (v), Step (vi), and Step (vii) data before reconstructing the comparison source. Its case-defined refinement fixes the processional local value to the local-kind case split itself before reconstructing the general processional-estimate source. Its p-adic-normalized case-defined refinement makes the distinguished nonarchimedean place the normalized p-adic unit-ball Haar-defect place before projecting to that case-defined source. The same-index side-condition endpoint now also obtains the q-normalized theta identity from the family-hull comparison and the principal-product p-adic finite hull decomposition, so this checked surface no longer exposes the direct adjusted-sum theta equality. The audit includes one-place diagnostics for the detached p-adic scale failure, the necessity of the signed $q \leq \theta$ comparison for $C_\Theta \geq -1$, and the remaining residual lower-bound obligation at the local arithmetic-degree boundary. The residual frontier is now classified explicitly below the p-adic-defect/main package: the matched local-degree object valuation-ball Record-Ob3/Ob5 source projects to the canonical p-adic arithmetic formula package and local arithmetic-degree residual package after reconstructing the local-degree formula source and then the p-adic-defect/main valuation-ball source internally. Lean identifies the valuation-ball object reconstructed from the p-adic-defect/main source with the original valuation-ball additive-Haar object by transporting the nondistinguished shell payload across $E_v = \delta_v + M_v$, then reconstructs the arithmetic-degree-controlled local source from this p-adic-defect/main object and transports the controlled component. The derived matched-object source projects directly to the canonical residual package and the reconstructed p-adic-defect/main source still feeds a compact-open product-handoff formula-realized route and the compact-open projected-local-degree formula route without separately exposing the valuation-ball local analytic object, Haar defect source, calibration, split law, comparison bounds, local-degree formula projection, p-adic-defect/main package, or component/formula equalities. Lean also now pins the valuation-ball Theorem 1.10 local analytic source by a source-ingredient audit that expands it into arithmetic divisors, valuation-ball additive-Haar Proposition 1.4 constructions, the additive/local analytic/proposition/formula projections, the theta-pilot local evaluation, Step (vi) zero contribution, and the Step (v)/(vii) formula-to-gap inequalities. The named-HDD Record-Ob3/Ob5 valuation-ball boundary now projects through the same p-adic arithmetic formula-matching bridge into the canonical residual package, so this compact-open boundary no longer has to re-expose the valuation-ball source as the residual consumer object. The local-degree formula valuation-ball Record-Ob3/Ob5 source is now derived from the synchronized controlled component layer by projecting the local Step (v)/(vii) formulas and reconstructing the valuation-ball formula package from the controlled local arithmetic source. The remaining boundary has therefore moved upward to the construction of that matched-object controlled component/local-estimate layer from full

IUT IV data; the legacy strict p-adic-Haar and p-adic-defect/main residual projections remain checked as comparison routes. The constructed diagnostics show the failure modes when the p-adic defect/main split, ordinary component formula matching, strict residual synchronization, or valuation-ball projection preservation laws are removed. Future Lean steps should remove further entries from the manifest, especially the weighted determinant synchronizations, q-normalized theta/log-volume identity, the local-analytic Theorem 1.10 source, and the lower Record-Ob3/Ob5 valuation-ball local-estimate construction from full IUT IV data.

We also maintain a Lean blueprint and intend to make the project consumable by Apodeixis as a collaboration and review layer. The blueprint is presently a compact navigational document plus a generated declaration index; it is useful for orientation, but it is not yet a complete mathematical dependency graph. This paper replaces an earlier ledger-style draft. Future updates should revise the relevant sections of this research paper rather than append chronological implementation notes.

1 Introduction

IUT is large enough that a useful formalization cannot begin by translating all four papers in order. The present project instead focuses on the public pressure point around IUT III, Theorem 3.11 and Corollary 3.12. Mochizuki's 2026 formalization report identifies this as the first stage of a possible formalization program:

$$\text{Theorem 3.11} \xrightarrow{\text{APT+IPL}} \text{“3.11.5”} \xrightarrow{\text{SHE+IPL}} \text{Corollary 3.12.}$$

The label “3.11.5” is not a theorem in the IUT III paper. It names the reorganized checkpoint obtained by moving the holomorphic hull plus determinant operation before the final simultaneous comparison of the Θ-pilot and the q-pilot.

The immediate formal target is not the *abc* conjecture. It is the logical shape of the Stage 1 implication. The question is:

> If the source constructions claimed by IUT III supply the multiradial possible images, indeterminacy behavior, IPL, SHE, APT, hull/determinant operations, and local-to-global C_Θ estimates, does the Corollary 3.12 inequality follow in a type-correct way?

The current code answers a substantial conditional version of this question. It proves the final ordered-real implication, constructs many intermediate typed routes, and exposes the remaining source mathematics as named obligations. It does not yet prove that the full Hodge-theater, Frobenioid, log-Kummer, holomorphic hull, determinant, and IUT IV analytic estimates of the source papers instantiate those obligations.

2 Mathematical Target

2.1 The final inequality

The endpoint formalized in Lean is the signed real comparison used in the Corollary 3.12 corridor. In the notation of the project, the route must supply

$$q_{\text{signed}} \le \Theta_{\text{signed}}, \qquad 0 < -q_{\text{signed}}, \qquad \Theta_{\text{signed}} \le C_\Theta(-q_{\text{signed}}).$$

From these hypotheses the ordered-real calculation gives

$$-1 \le C_\Theta.$$

The Lean theorem behind this step is elementary. This is important: the hard question is not whether the final real inequality follows from the displayed signed comparisons. The hard question is whether the source machinery really constructs the signed comparison without collapsing distinct histories or comparison levels.

2.2 The disputed comparison level

Scholze and Stix criticize the proof around Corollary 3.12 on the grounds that the relevant Θ-side data carries j^2-scaled information, while a naive simplification of the comparison appears to remove the room needed for that data. The formalization responds by enforcing separate types for:

$$\begin{gathered}\text{representative } j^2\text{-coordinate data,}\\ \text{balanced sign-compatible data,}\\ \text{full-label and averaged finite-label data,}\\ \text{transported ordered-real copies,}\\ \text{hull/log-volume endpoint data.}\end{gathered}$$

Lean then rejects the most naive collapse: the balanced sign-compatible layer does not by itself supply the final Corollary 3.12 route. This is a diagnostic result, not a settlement of the dispute. The remaining issue is whether the source construction gives the final hull/log-volume object with the required compatibilities.

2.3 Indeterminacy behavior

The current Stage 1 interface treats the three indeterminacy components separately. In the formal model, Ind_1 and Ind_2 preserve procession-normalized log-volume and generate equality for the possible-image quotient. Ind_3 is not an equality generator; it contributes only an upper-semi log-volume inequality. This distinction is now part of the checked Lean interface:

$$\mathrm{Ind}_1, \mathrm{Ind}_2 : \text{equality/invariance}, \qquad \mathrm{Ind}_3 : \text{upper-semi inequality}.$$

At the finite procession layer this has been sharpened from a pointwise same-label comparison to an explicit action statement: Ind_1 and Ind_2 may act by equivalences of the finite label set, and Lean proves that the normalized average is invariant by finite-sum reindexing. The labels are transported rather than identified, while Ind_3 remains the sole one-sided upper-semi input. This action statement is now connected to the Step (x)-to-Step (xi) numerical handoff. The code factors out the invariant upper-ray payload $q_{\mathrm{pilot}} \leq \Theta_{\mathrm{hull}}$, and the permutation corridor constructs that payload directly from $\mathrm{Ind}_1/\mathrm{Ind}_2$ reindexing and the Ind_3 upper bound. Consequently, the corresponding signed C_Θ endpoint no longer requires same-label $\mathrm{Ind}_1/\mathrm{Ind}_2$ pointwise hypotheses. This separation is one of the central design decisions of the formalization. It prevents a proof from turning an upper-semi source relation into an equality merely because both appear in the informal phrase "indeterminacy".

3 Formalization Architecture

3.1 Source, Lean, paper

The working cycle is:

$$\text{Mochizuki source papers} \longrightarrow \text{Lean definitions and theorems} \longrightarrow \text{research paper and blueprint.}$$

The papers in `docs/` are treated as source of truth. The Lean code is the formal artifact. This paper records the mathematical state of the formal artifact; it is not meant to be a chronological log. This distinction matters because previous versions of this draft had grown into an implementation ledger. The present version is organized as a research paper: target, architecture, results, trust boundary, tooling, and next work.

3.2 Main Lean layers

The Stage 1 development is split into modules whose roles are stable enough to describe at paper level:

Module family	Role
`PilotComparison`, `CorollarySchema`, `SourceObligations`	The signed real endpoint, the final Corollary 3.12 predicate shape, and the ledger turning a common-bound comparison into the endpoint.
`IUTStage1SourceCore`, `IUTStage1FiniteLabels`, `IUTStage1Gaussian`	Real-line copies, regions, finite-label and $ZMod(\ell)$ models, Gaussian degree shadows, and j^2-diagnostic infrastructure.
`IUTStage1StepX`, `IUTStage1HodgeSHE`	Procession-normalized log-volume routes, Hodge/SHE data, square/full-label transport, and upper-semi compatibility records.
`IUTStage1Theorem311`, `IUTStage1StepXI`	Typed Theorem 3.11 indeterminacy cores, possible-image quotient structures, concrete Hodge-theater/log-theta source layers, Remark 3.9.5 hull/determinant interfaces, and paper-trace audits. The public `IUTStage1StepXI` module is now a facade over a core implementation module, preparing the Step (xi) layer for canonical-route source slices without changing public names.
`IUTStage1FrobenioidShift`, `IUTStage1Experiments`	Nonarchimedean realified Frobenioid/log-Kummer and vertical-IQ routes, additive-Haar arithmetic-degree endpoints, and compiled public diagnostics.

The names in this table are intentionally coarse. A paper should not reproduce thousands of declaration names. The detailed declaration surface is maintained in the blueprint's generated declaration index.

3.3 Review-handoff strategy

The current development is organized around one canonical corridor, not around proliferating endpoint variants. The review boundary is the typed remaining-assumptions manifest: every public input is classified as constructed, derived, interface-only, or still a source obligation, and new public wrappers are treated as progress only when they lower or remove a manifest item. This strategy incorporates the main external review feedback on the project: keep the trust boundary explicit, keep CI and declaration checks as the truth signal, split large modules only when it makes the canonical route easier to audit, and pair each serious lowering with a weakened-hypothesis diagnostic when the omitted law is mathematically essential.

For the present review handoff, the strongest residual frontier is a zero-source-obligation submanifest with 320 checked entries: 314 derived entries and 6 constructed countermodel diagnostics.

The newest source slice lowers the compact-open realized-exact and compact-open map sources to pointwise log-Kummer/valuation-ball coordinates, with projection audits showing that the principal valuation-ball source, theta labels, cover places, and selected principal hull are preserved. This is genuine source-facing mathematical lowering inside the Step (x)-to-Step (xi) route, but it is not yet a complete formal proof of IUT III Corollary 3.12 from paper-level initial theta data.

3.4 Records as boundaries

The project uses records in two different ways. Some records are ordinary mathematical data structures whose fields are proved or constructed in Lean. Other records are trust-boundary interfaces: they name a source construction that must eventually be derived from the IUT papers. The current paper distinguishes these roles explicitly.

For example, the typed indeterminacy core is a useful formal object: it stores the $\mathrm{Ind}_1/\mathrm{Ind}_2$ invariance laws, the Ind_3 upper-semi law, and the quotient relation. By contrast, the full Theorem 3.11-and-remarks obligation record still contains source-facing fields such as construction of the input prime-strip link, theta-pilot possible images, the log-theta-lattice procession, and the Hodge/SHE/IPL/APT bridge. These fields are not hidden axioms, but they are also not yet source-derived mathematics.

4 Current Formal Results

4.1 Ordered-real endpoint

The final ordered-real implication is formalized. Once the signed comparison $q_{\text{signed}} \leq \Theta_{\text{signed}}$, positivity of $-q_{\text{signed}}$, and the C_Θ-upper bound are available, Lean derives the expected lower bound $-1 \leq C_\Theta$. The code also proves the corresponding strict/boundary dichotomy: either the boundary case is exact, or $-1 < C_\Theta$.

This part is no longer an open mathematical problem inside the project. It is pure ordered-real algebra. All serious mathematical work sits upstream of the signed comparison and the C_Θ-upper bound.

4.2 Anti-collapse diagnostics

The formalization proves negative diagnostic statements about comparison-level collapse. In particular, a label-independent j^2 collapse is rejected at the representative level, and the balanced sign-compatible layer is not accepted as the final route level. These diagnostics are valuable because they test the exact failure mode emphasized in external criticism.

The diagnostics should be read carefully. They show that the Lean development does not silently make the criticized simplification. They do not show that Mochizuki's construction successfully supplies the stronger hull/log-volume data.

4.3 Concrete Stage 1 route

The current strongest route starts with a unified Hodge-theater/log-theta/log- Kummer source spine. This spine carries the 0- and 1-column histories, prime-strip data, theta-pilot possible images, the selected q-choice, and the vertical log-Kummer packet alignment. Its Theorem 3.11 bridge side is also structured: the input-prime-strip link, one-column simultaneous holomorphic expression, and algorithmic parallel transport are projections of one IPL/SHE/APT bridge package. The Step (x) side is represented by a single finite-divisor/vertical-log-Kummer package: finite divisor tensor packets, realified Frobenioid Kummer compatibility, mono-analytic forgetting, vertical IQ, and source-target log-volume calibration are projections of one object rather than five unrelated source flags. Lean projects from the same spine both the Theorem 3.11 typed $\mathrm{Ind}_1/\mathrm{Ind}_2/\mathrm{Ind}_3$ core and the Step (x) payload. The route then forms the equality-quotient family of possible images, selects the q-region, and connects that region to the existing C_Θ/Corollary 3.12 corridor. The preferred paper-trace route now has a named source-spine audit: the Theorem 3.11/Remarks obligation package and the Step (x) finite-divisor/log-Kummer obligation package are projected from one source object. The first Step (xi) hard-math route has also been added: a paper-derived Remark 3.9.5/Step (xi) source records the substeps (xi-a)–(xi-g), ties the selected q-region to the actual Theorem 3.11 possible image selected by the source spine, and projects the existing Step (xi) hull/determinant obligation record internally. Ob7 is no longer only a raw compatibility flag on the newest route: the legacy prime-strip/log-Kummer record and its input/output strip and selected-column equalities are projected from a coric $\Theta^{\times\mu}$/log-Kummer source link aligned with the same quotient hull source. The preferred localized route now exposes this source as a structured audit: q-region containment, Gaussian/environment place agreement, coric unit-character agreement, and the three hull-bridge alignments are checked before the legacy compatibility record is projected. The constructed-paper-trace public boundary now also packages this information together with the constructed Step (xi-a)–(xi-g) and Ob1–Ob9 ledger, making explicit that Ob7–Ob9 are read from the same log-Kummer/bridge source rather than supplied as separate trace facts. Its strictest boundary now derives the non-Step-(xi) audit from the proof-carrying Theorem 3.11, Step (x), and IUT IV source records; the caller supplies only the explicit additive-Haar arithmetic-degree payload that remains below this milestone. The selected q-region containment is now derived from the equality-quotient possible-image family: it is the canonical inclusion of the selected possible region into the family union, followed by the constructed holomorphic hull source. The possible-image family is no longer allowed to be an empty predicate shell at this boundary: the theta-pilot class image source carries a point witness for every class, Lean pulls these witnesses back to concrete choices, and the equality-quotient construction proves nonemptiness of the quotient pullbacks and of the selected q-region. The Step (xi) valuation-ball/log-shell source chain now transports the same theta-region witnesses down to the local exact-source constructors. More recently, the class-region, class-formula, log-theta-lattice formula, lattice-image-law, and typed fiber-transport source records themselves were made witness-bearing, so this nonemptiness data is no longer introduced independently by the valuation wrapper. The paper-derived Step (xi) source also has a quotient-compatible constructor that fixes the selected choice to the source spine, instead of accepting an independently assembled hull record at that point. The remaining IUT IV and additive-Haar layers are then combined with this constructed Step (xi) source. The concrete-packet goal-facing completion audit

`preferredPublicConcreteStepXI311312ConcretePacketTheorem311ConstructedThetaRegionGoalCompletionA`

packages the current milestone into two checked claims built from the same packet-derived primitive constructor: the constructed theta-region preferred public route and the constructed theta-region current product-hull norm-square/Ob7 evidence. Its scope is:

theta-region principal valuation-ball source,
concrete Theorem 3.11 packet projecting the primitive constructor,
theta-pilot-class target-region formula with point witnesses,
component-level Step (x) finite-divisor/log-Kummer source,
bridge-derived IPL/SHE/APT witnesses with structured output-flag calibration,
principal-product HDD Step (xi) source built by Lean,
localized finite-place C_Θ handoff,
restriction-calibrated p-adic source and Ob7 norm-square projection.

The audit is not a proof of the lower analytic source constructions. Its function is to ensure that, at this milestone boundary, the public route no longer combines an arbitrary constructor-built Step (xi) source with unrelated Ob7 evidence or a standalone hull/determinant obligation package. The new primitive Theorem 3.11 constructor further prevents the source record, constructed-SHE bundle, and transport guard from being independent top-level inputs: the primitive endpoint reconstructs them from the source spine and the lattice-formula realization of target regions, then enters the localized Step (xi) layer through a primitive principal-product HDD source whose audit records the induced source gap and transport guard. The concrete packet constructs that primitive endpoint through a bridge-certificate constructor, so the primitive qualitative source is obtained from the transported IPL/SHE/APT bridge package rather than supplied beside it. The packet now also has an output-flag-calibrated source-spine constructor: the deepest component-representative selected-label q-pilot route constructs the Theorem 3.11 subclaims from the package-aligned bridge and the single structured output-flag equality, instead of accepting an independent subclaim component. The same lowering now reaches the one-sided source-spine-data constructor used by the multiradial packet route, so that lower one-sided path also consumes the structured output-flag calibration rather than a standalone subclaim component. Its packet-construction audit checks that the constructor preserves the source-spine selected q-choice, projected bridge component, Step (x) log-Kummer source, and target-region formula source while deriving the subclaim component from the output-flag calibration.

4.4 Local-to-global C_Θ chain

The code now contains a proof-carrying local-to-global C_Θ route at the interface level. The preferred additive-Haar arithmetic-degree and p-adic formula route threads local analytic estimates, arithmetic-degree calibration, finite-place Haar defects, formula-gap matching, and Step (xi) local terms into the endpoint. For this preferred route, the canonical C_Θ-scale comparison is derived from the finite-place arithmetic gap and the IUT IV handoff identity, rather than supplied as a raw endpoint hypothesis. At the Step (xi) source layer, Lean now also proves the sharper handoff for both the constructed Remark 3.9.5 hull/determinant source and the constructor-built exact-theta source: once the local canonical Step (xi) scale is identified with the finite-place scale in the IUT IV summation record, the global C_Θ-dominance follows from the IUT IV plus-one cancellation theorem. The constructor-built finite-place source lowers this boundary again. It records local canonical scales, local arithmetic and main-log contributions, and local Haar normalization defects; Lean sums the local Step (xi) inequalities and uses the total defect bound to derive the global arithmetic gap consumed by the coupled IUT IV C_Θ ledger. The strongest concrete packet route now exposes the resulting signed inequality $\theta_{\text{signed}} \leq C_\Theta(-q_{\text{signed}})$ and the Corollary 3.12 dichotomy as endpoint fields, not only as an internal localized-handoff audit. A further wrapper removes the independent principal product hull argument from this path by taking it from the packet-backed theta-region source. The next wrapper constructs that theta-region source from the principal valuation-ball product-hull cover and the fiber-transport family-union comparison attached to the concrete Theorem 3.11 packet. The current strongest variant lowers the boundary once more: it accepts a single theta-region-defined principal valuation-ball source and projects both the principal cover and fiber-backed family-union comparison internally. A pointwise metric variant now goes below that source by accepting the principal constructed-log-shell/metric-zero valuation-ball data and projecting the theta-region-defined source from it. The pointwise route also projects the theta-region measure/summand calibration from explicit Hodge/measure formulas: the hull-volume identity, the Hodge-Arakelov theta-degree charting identity, and the finite adjusted-summand formula. Thus the packaged pointwise calibration record is no longer a public endpoint object. The same route now constructs the restriction-calibrated norm-square Ob7 handoff from a pointwise residue-module/unit-ball/additive-Haar local arithmetic source. The local-analytic Theorem 1.10 refinement now keeps this proof-carrying pointwise Ob7 source as the public local-arithmetic boundary, so the structure-sheaf/direct-summand and nonzero norm-square projection facts are internal fields of that source rather than separate endpoint arguments. The newer product-handoff refinement constructs the packet-facing pointwise Ob7 source internally from the principal-product restriction-calibrated handoff, so the local-analytic Hodge-formula route can expose that lower product handoff instead of the pointwise wrapper. This branch is now explicitly quarantined: the product-level and packet-facing handoff contradiction theorems show that the unit-ball norm-square source carries both the selected zero-norm computation and the selected nonzero-norm field. A constructor route from the lower restriction-calibrated residue-module/unit- ball/additive-Haar source remains available; at that lower boundary those two projection facts are still explicit local arithmetic bridge fields. A new parallel pointwise branch now avoids that quarantined unit-ball positive-norm input: it couples the concrete pointwise theta-region principal hull to the residue-module inverse-base-prime/additive-Haar source and derives the localized C_Θ signed comparison from that branch together with the localized Step (xi) handoff. A localized Step (xi) refinement now also packages the localized determinant source, its determinant and canonical-scale identifications, local main-log terms, Haar defects, and finite-place summation identities into a single pointwise source object before the public endpoint projects them. The IUT-IV-backed refinement lowers this finite-place layer further by deriving the arithmetic-upper and main-log summation identities from the IUT IV arithmetic-divisor local evaluation source, leaving only the local Step-(xi)-plus-Haar identification as the visible comparison datum. The latest Theorem 1.10 valuation-ball refinement lowers the IUT IV input one more step: the arithmetic-divisor local evaluation source is now projected from the valuation-ball additive-Haar formula-gap source, whose endpoint contains the distinguished Proposition 1.4 log-shell construction, nondistinguished zero case, archimedean Proposition 1.5 metric construction, and named formula-to-gap comparisons. The corresponding inverse-base-prime pointwise route now carries this same lowered Hodge-formula and Theorem 1.10 valuation-ball/IUT IV interface, while using the residue-module inverse-base-prime additive-Haar source instead of the quarantined unit-ball norm-square local arithmetic handoff. This branch has also been lowered one step further: the pointwise route can now construct the restriction/additive-Haar wrapper from finite Hodge-summand/Gaussian-Hodge projected inverse-base-prime data. The Theorem 1.10 valuation-ball side has likewise been lowered: the strongest pointwise branch now constructs the formula-gap matched valuation-ball source from local analytic arithmetic-divisor data before entering the localized Step (xi) handoff, and the local-analytic and Theorem 1.10 C_Θ paper-trace records are now projected from that valuation-ball additive-Haar source instead of supplied as parallel records. This projected source is now threaded into a public pointwise route, so the assembled local-analytic Step (xi) C_Θ wrapper is not a separate endpoint input there. The heterogeneous projected-weighted determinant route now exposes both the signed dichotomy and the $C_\Theta \geq -1$ companion directly from the q-normalized local-analytic residual source, before passing through the formula-gap wrapper. The same lowering now applies on the non-vacuous inverse-base-prime/additive-Haar pointwise branch before the summand-projected signed comparison is invoked, and the compact-open log-Kummer route propagates this constructed wrapper through the possible-region source. It also now constructs the principal pointwise theta-region valuation-ball source from compact-open log-Kummer map possible-region data at the public endpoint. The newest compact-open branch lowers the inverse-base-prime arithmetic input one step further: the public endpoint takes the residue-module valuation cover and finite Hodge-summand Gaussian/Hodge construction separately, and Lean assembles the summand-projected inverse-base-prime source internally. A companion branch also constructs the local-analytic Step (xi) C_Θ wrapper at this valuation-cover/Hodge-summand boundary from the localized determinant source, Haar defect, and valuation-ball additive-Haar local analytic arithmetic-divisor data. A newer variant lowers the Hodge side again: the endpoint takes the localized-adjusted Gaussian/Hodge source, and Lean reconstructs the finite Hodge-summand source by fixing the summands to the localized weighted-adjusted determinant contributions; its constructed-wrapper companion performs the same local-analytic Step (xi) C_Θ construction at this lower localized-adjusted boundary. A further square-direct-summand variant replaces the direct nonnegativity and strict-positivity fields with square-root presentations of the local direct-summand log-volumes and a nonzero selected root, and its constructed-wrapper companion moves the same local-analytic Step (xi) C_Θ construction below these square-root data. The norm-square variant lowers this again by projecting those square-root presentations from the norm-square localized hull decomposition, and its constructed-wrapper companion moves the local-analytic Step (xi) C_Θ construction below that norm-square projection. The compact-open norm-square variant moves the public Hodge datum down to local compact-open tensor factors whose projected log-volumes generate the norm-square decomposition; its constructed-wrapper companion moves the local-analytic Step (xi) C_Θ construction below those compact-open tensor factors. The additive-Haar variant lowers the same branch to Haar-normalized compact-open local factors and projects the compact-open norm-square source from their normalized log-volume identities; its constructed-wrapper companion now performs the local-analytic Step (xi) C_Θ construction at this lowest compact-open Hodge boundary. Its pointwise-source strengthening now reads the localized determinant/scale synchronization, formula-gap Theorem 1.10 valuation-ball source, Haar defect, local Step (xi)/Haar/main-log identity, and finite Haar lower bound from a single localized Theorem 1.10/IUT IV Step (xi) source. The restriction-calibrated local arithmetic branch now has the same local-analytic Theorem 1.10 valuation-ball/IUT IV lowering as the inverse-base-prime branch. Its compact-open wrapper derives the pointwise valuation-ball source from log-Kummer map data and now also has a pointwise formula-gap variant that reconstructs the local-analytic wrapper from a single localized Theorem 1.10/IUT IV source. At the formula-realized evidence level, the same compact-open restriction-calibrated boundary now constructs the pointwise valuation-ball source and Ob7 local-arithmetic handoff before applying the pointwise Theorem 1.10/IUT IV packet route; its product-handoff variant now takes the typed Ob7 handoff without separately exposing the restriction-calibrated source, structure-sheaf/direct-summand equality, or selected norm nonzero field. A further local-arithmetic-degree variant now constructs the pointwise Theorem 1.10/IUT IV localized source from the localized local-term Step (xi) source, formula-gap valuation-ball source, main-log matching, and local arithmetic-degree identity. The synchronized variant removes that localized local-term source from the compact-open public boundary by constructing it from determinant synchronization, Haar defects, and the finite-place arithmetic-degree identity. The constructed-synchronization variant also removes the packaged determinant/scale synchronization source by constructing it from the localized Step (xi) determinant source and the two constructor-built comparison identities. The explicit-formula variant removes the packaged pointwise measure/summand formula record by constructing it from the volume/hull-log-volume, charted-theta/Hodge-degree, and finite adjusted-summand identities. The realized-exact compact-open variant also constructs the compact-open log-Kummer map source from the statement that each theta-region cell is the realized image of the selected compact-open local subset. Its restriction-calibrated variant constructs the product Ob7 handoff from the normalized local-arithmetic source, structure-sheaf/direct-summand synchronization, and nonzero norm-square summand.

The phrase "proof-carrying" is deliberate. The route no longer consumes a single unstructured numerical inequality at the public endpoint. It consumes a structured chain of local and global source objects. Nevertheless, several of those source objects still stand for arithmetic constructions not yet derived from number-field valuations, conductors, differents, holomorphic hulls, and the local estimates of IUT IV.

5 What Remains Mathematically Open

5.1 Source-faithful construction before the Theorem 3.11 boundary

The machine-checked source-closure ledger now contains 113 clauses from IUT I–III together with transitive dependencies in *Frobenioids*, *The Etale Theta Function*, and *Absolute Anabelian Topics III*. It classifies 84 clauses as partially implemented, 5 as toy models, 19 as unformalized, and 5 as source-faithful. None of the complete M1–M3 milestones or Theorem 3.11 clauses is classified as source-faithful. The new partial Definition 3.1 path constructs etale stacks and strong-transformation morphisms, a coherent C_2-action pseudofunctor and coherent invariant maps satisfying the C_2 cocycle. Both sign quotients carry the bicategorical universal property: invariant maps factor through the quotient, compatible 2-morphisms lift, and equality is detected after precomposition. The path also constructs certified fundamental groups, exact profinite sequences and compatible open immersions. It also ties the elliptic curve and puncture to a scheme over the field, its origin section, and the complementary open subscheme with its open immersion; deriving this realization from a concrete projective Weierstrass model remains open. Its X_F stack is bicategorically equivalent to the scheme's representable Yoneda-style pseudofunctor, so pullback, identity, and composition coherence are intrinsic, and the coherent sign action is identified with an involutive scheme map that acts by elliptic negation on rational points. The path constructs the lower-central and exponent-l theta quotient, its distinct rank-two arithmetic elliptic quotient, and the theta center. It supplies linked arithmetic and cuspidal exact sequences, a surjective Lagrangian quotient on the elliptic quotient, its kernel cover, and the Proposition 2.2 direct-product inversion eigenspace interface with actual arithmetic-conjugation compatibility. It also supplies the theta-root subgroup derived from a Galois splitting and records the normalizing order-two lift, unique admissible coset, induced conjugation, and generation statement of Proposition 2.2(iii). Global and bad-local theta-root/sign-quotient stack realization interfaces are present. The cusp data is a complete atlas: nonzero X-cusps are indexed by nonidentity Q-labels and C-cusps by inversion orbits, with rational inertia and decomposition exact sequences, sheet conjugacy, and sign compatibility. The selected $Q/\{\pm 1\}$ label is derived from an actual sheet representative. The transitive Definition 1.1 construction forms the normal closure of all nonselected inertia, $\Delta^{\dagger}$, and its positive and negative eigenspaces. The doubled label 2ϵ is derived and proved distinct from both selected sheets. Continuous exact quotients $J_X \hookrightarrow J_C$ are recorded; their section is pinned to the doubled-cusp decomposition group, and Lean proves its image normal from exactness and the trivial kernel action. The arrowed stack subgroups are required to be the preimages of $\mathrm{Im}(\sigma)$ and $\mathrm{Im}(\sigma) \times \mathrm{Gal}(X/C)$. The absolute Galois quotients are definitionally mathlib's absolute-separable-closure automorphism groups with their Krull topology; no arbitrary profinite replacement or equality certificate is accepted. Inside the selected $\overline{F}$, the embedded field of moduli is proved equal to the fixed field of the geometric-j automorphism stabilizer, and its maximal solvable extension in that same ambient field is defined as the compositum of all finite solvable Galois layers. The embedded K is required to equal, as an intermediate field, the fixed field of the kernel of the mod-l representation. Restriction from K-places to F_{mod}-places is canonical; only a section is chosen. The finite and infinite sections assemble into a single section whose inverse is restriction, yielding the actual source bijection $V_{\mathrm{mod}} \simeq V$. Lean also proves the disjoint decompositions $V = V^{\mathrm{non}} \sqcup V^{\mathrm{arc}}$ and $V^{\mathrm{non}} = V^{\mathrm{bad}} \sqcup V^{\mathrm{good}}$. The Lagrangian quotient is continuous between profinite groups, so its finite-cover kernel is proved open rather than stored as an assumption. The quotient data is attached only after scalar extension of the curve and X_F/C_F stacks to the K-core through the actual restriction-of-scalars pseudofunctor. This K-core is constructed from the two stack extensions and the extended quotient morphism, so its X-stack, C-stack, and quotient map agree with those extensions definitionally; it accepts neither a separately assembled target pair nor three comparison certificates. The morphism extension carries an invertible modification whose components and naturality cells are pinned to the restricted global map. The transported sign action, quotient universal property, and fundamental-group realization remain explicit obligations. It does not yet derive the explicit type-$(1, l\text{-tors})$ cover/sign-quotient stacks as finite-etale covers, though their arithmetic cover group is fixed to be the profinite group of the open Lagrangian kernel. The global and local theta-root stack interfaces likewise fix their arithmetic groups to the derived open theta-root subgroups and supply the open subgroup chains. The construction and finite-etale representability of both kinds of stack realization remain open. Their cover squares carry the full pseudo-cone cartesian universal property: compatible cone 2-morphisms lift, and the lift is unique.

At every finite and infinite K-place, the base curve and core stacks are scalar-extended over the actual completion. At selected finite places, the type-$(1, l\text{-tors})$ cover stacks are also scalar-extended. Selected infinite places carry the analogous cover, stack, absolute-Galois/decomposition-subgroup, and exact-sequence data over mathlib's archimedean completions. At every selected place, compatible local-to-global cover fundamental groups and rank-two elliptic quotients are recorded. At every finite and infinite place, one sign-quotient scalar-extension package constructs the local X-stack, C-stack, and quotient map definitionally, so separate local target pairs and comparison certificates are not supplied. The local quotient map retains the morphism-level scalar-extension comparison, and the complete cuspidal atlases commute with the cover diagrams. At each selected place the local epsilon cusp is constructed at the global Q-label in that localized atlas, and its X/C decomposition diagrams and sign label are inherited. Thus no independent local epsilon, atlas-equality certificate, or exact synchronization of arbitrary sheet representatives remains. At good finite places the $\Delta^{\dagger}$ quotient, positive profinite quotient, J_X/J_C sequences, doubled-cusp sections, arrowed stacks, and their fundamental groups all commute with scalar extension. At bad places, the standard rank-one quotient is required to factor continuously through the profinite completion $\widehat{\mathbb{Z}}$ and its canonical reduction modulo l. Their absolute Galois groups are identified with closed decomposition subgroups of the global group, with compatible embeddings of fundamental exact sequences. Each closed subgroup is identified with the literal stabilizer of a chosen prolongation of the corresponding place to the absolute separable closure. The local embedding is derived from the equivalence with that closed stabilizer, and the images of the Galois maps in both fundamental-group diagrams are proved equal to it; an arbitrary subgroup and a later equality certificate are not accepted. The localized cusp label is proved to be the canonical sign orbit of 1 mod l. Constructing the algebraic boundary atlas, compactifications, and the supplied finite-etale stack realizations remains open; the records above specify their source contracts but do not prove geometric existence. At every F-place above the bad moduli locus, and at each selected bad K-place, the q-parameter is a nonzero contracting element of the adic completion equipped with a Tate uniformization over its algebraic closure; its group-level uniformization is equivariant for the local absolute Galois action, its positive local height is derived from the discrete valuation exponent, and this height is required to be prime to l. Analytic/topological realization of this uniformization remains open. The cuspidal involution fixes inertia and the Galois quotient, the selected splitting is sign-compatible, and the bad cusp label is the sign orbit of 1 mod l. Each bad place carries source-shaped theta-root and sign-quotient stacks, a coherent quotient universal property, a full two-pullback witness, the derived arithmetic theta-root group, and the two open fundamental-group inclusions. Producing these structures from the source hypotheses and proving finite-etale representability remain open.

The current finite route from `InitialThetaData` is an explicitly classified toy model. Its $\mathbb{Z}/l\mathbb{Z}$ translations, natural-number preorder, finite-coordinate bijection, and transport laws do not instantiate Theorem 3.11. The paper-level procession, Ind_1, Ind_2, Ind_3, vertical log-Kummer, and horizontal IPL/SHE/APT clauses remain source obligations.

At the source boundary, a theta-Hodge theater no longer yields an $F^{\vdash}$-prime-strip from bare equivalences of carrier categories. It requires explicit transport of rational monoids, characteristic splittings, and the selected archimedean splitting, as in IUT II, Corollaries 4.5 and 4.6. For Definition 4.9(ii)–(iv), the finite-place input further ties the selected splitting at an isotropic reference object to an actual MLF-Galois TM-pair, a stable characteristic factor, and the genuine Ism Kummer orbit. Lean constructs the bad/good $O^{\square\times\mu}$ monoid and proves that its complete unit group is $O^{\times\mu}$. A reconstruction output must retain the original base category and realize this exact monoid as its reference rational monoid. At an archimedean place, the exact N-stage replaces the compact factor by S^1/μ_N, retains the noncompact factor, and uses the concrete quotient transition on the first factor. Reconstructed stages are required to be actual split Frobenioids with this exact split topological monoid and the original base. Their transition functors preserve the base projection, FSM morphisms, divisors, and Frobenius degrees; their carrier categories, bases, and reference rational monoids form strict divisibility-indexed diagrams. Every multiplicatively cofinal restriction is proved final, hence preserves and reflects the universal colimit cocone. Construction by Frobenioids I, Theorem 5.2(ii), and derivation of the archimedean stages and transition functors from the original data remain explicit source obligations. Definition 4.9(vi)–(viii) now has a universe-polymorphic $F^{\vdash\times\mu}$-prime-strip record. It selects the exact bad/good finite reconstruction and the complete archimedean quotient system at every place. Its $F^{\times}$- and $F^{\times\mu}$-coarsifications are literal wide subcategories of divisor-zero isometries. The globally realified record retains the global Frobenioid, prime/place bijection, and all currency maps ρ_v. Typed cyclic bad-place generators are transported through those maps to form a local character tuple; the pilot object is the output of a global reconstruction function on that tuple and has negative arithmetic degree. Deriving and populating this record from Corollaries 4.6 and 4.10, remains open. At a finite place, a structure-preserving isomorphism now transports the topological Galois group, arithmetic action, unit-mod-torsion quotient, every open-subgroup invariant Kummer image, both product generators, the reconstructed split Frobenioid, its base, reference object, and rational monoid. Identity and composition are constructed, and preservation of the genuine Ism-orbit is derived. The archimedean system isomorphism transports every torsion quotient, quotient and orientation map, level transition, reconstructed stage, base, divisor, Frobenius degree, FSM morphism, split product, and rational transition. Identity and composition are constructed. Its level maps restrict to the isometry coarsifications and form a natural transformation of the complete divisibility diagrams with equivalence components. The assembled globally realified isomorphism combines these local equivalences with a structure-preserving equivalence of the global Frobenioids, the prime/place and ρ_v squares, bad-place generator and currency compatibility, reconstruction for every character tuple, and arithmetic-degree compatibility. Identity and composition are constructed. Compatibility of the distinguished pilot character, pilot object, and pilot degree is derived. Following IUT I, Section 0, the place-indexed and globally realified full poly-isomorphisms are literal quotients of all complete representatives. Categorical equivalences are identified by natural isomorphism, while group, monoid, quotient, splitting, prime, and local currency maps remain pointwise data. Identity, composition, unit laws, associativity, representative existence, and exact nonemptiness criteria are proved. Constructing and populating these records from the source evaluation and reconstruction algorithms remains open.

The source route now has a separate assembled column boundary for Theorem 3.11. It requires one actual absolute-label Gaussian log-theta lattice, one mono-analytic log-shell algorithm, a vertical family whose common local construction is definitionally the local input of the multiradial presentation, the multiradial functor, full selected-place log-links, site splitting and labeled Kummer data. Horizontal compatibility is assembled in a separate corridor containing two column boundaries on the same lattice, with indices proved to be n and $n+1$. Its triangle and environment prime-strip squares are fixed to the actual (n, m), $(n+1, m)$, $(n, \circ)$, and $(n+1, \circ)$ endpoints for every

m. The active clauses (iii)(a)–(b) interface now uses only the IUT II, Definition 4.9 $F^{\vdash\times\mu}$-prime-strips. Every triangle strip is definitionally computed by one functorial reconstruction algorithm from the mono-analytic transport of its actual site or vertically coric theater. The same algorithm computes each environment strip and comparison from the Proposition 2.1(vi) mono-analytic data and derives the inverse laws. The older ordinary-F square remains only as precursor scaffolding and cannot inhabit the corridor boundary. The times-μ arrows use the IUT I, Section 0 coarsification: componentwise equivalence maps are quotiented by natural isomorphism on their categorical components, rather than treated as strict categorical isomorphisms of the prime-strip records. Identity, composition, associativity, and cancellation of selected inverse pairs are proved on these classes. The square quantifies in both directions over every horizontal class; its chosen upper and lower classes witness nonemptiness but do not by themselves discharge the full-poly claim. Corollary 4.10(iv) is now retained as a representative-level rigidity law for every actual adjacent horizontal pair: lifting the underlying mono-analytic map of any output equivalence recovers that output up to structured natural isomorphism. Lean derives from this law that every times-μ horizontal class has a mono-analytic preimage. The corridor accepts neither a clause (a) square, a chosen theta-link member, an arbitrary quotient-level fullness witness, nor chosen Kummer maps. Every mono-analytic strip carries structure-preserving comparisons to and from the common model with two-sided Section 0 inverse laws. Lean derives the canonical theta-link and both Theorem 1.5 Kummer directions, proves their cancellation, reconstructs the upper and vertical maps, and defines the lower map by Kummer conjugation, and proves commutativity and both all-class transport directions. The environment square is derived class-by-class in both directions by conjugating the triangle square with explicit two-sided natural environment-to-triangle classes and cancelling the inverse pairs; it is not an independent input. This proves the exact typed quotient, functorial reconstruction, fullness boundary, and conjugation mechanism. Constructing the finite and archimedean component outputs, proving representative-level reconstruction fullness, and deriving the coherent model comparisons from Corollaries 4.5, 4.6, 4.10 and Theorem 1.5 remain open. Its mono-theta projective-system squares additionally range over every selected bad place. Clause (iii)(d) no longer represents the displayed $\{\pi_1^{\kappa\text{-sol}} \ltimes M_\infty^\kappa\}$ localization by a single morphism of two profinite groups. For every site, common column, absolute label, and selected place, Lean retains a topological group/pseudo-monoid action pair for the global datum, action pairs for $M_{\infty\kappa v}$ and $M_{\infty\kappa\times v}$, an equivariant continuous localization map, and an equivariant topological embedding for the displayed local inclusion. Here "pseudo-monoid" has its exact IUT I, Section 0 meaning: the carrier is embedded in a topological abelian group, and multiplication is defined only when the ambient product remains in the carrier. The restricted operation is proved continuous and associative wherever both composites are defined; diagram morphisms preserve its domain, and diagram isomorphisms reflect it. For the nonarchimedean mechanism of IUT I, Definition 5.2(v), Lean now uses the arithmetic fundamental group of X_v for the group represented by D_v, as required by Example 3.2(i), rather than identifying it with the C_v group. The sign-quotient geometry constructs the open embedding into the arithmetic fundamental group of C_v, and Lean proves injectivity and openness of its image. This constructs the geometric target; recovering it functorially from the abstract D_v group by Absolute Anabelian Topics II is still open. For each actual finite place, Lean constructs the algebraic closure as the filtered topological-module colimit of its finite Galois local subfields. The nonzero integral submonoid is exhausted by continuous finite-stage maps and is multiplicatively equivalent to $O^{\triangleright}_{\overline{K_v}}$; the equivalence intertwines the literal absolute-Galois action. Since every finite stage is Galois, restriction of each absolute-Galois automorphism acts continuously on every stage. The colimit universal property produces a continuous automorphism of the ind-topological carrier, proves that it is the literal algebraic action, and restricts it continuously in both directions to the integral monoid; identity and composition are proved. Lean now retains the source meaning of the prefix "ind-" as well: the nonzero integral monoids in all finite Galois subfields form a filtered system of topological monoids. Its transition maps are continuous and equivariant, and continuous Krull restriction to each finite Galois group, together with its discrete topology, makes evaluation jointly continuous at every stage. The stage inclusions are injective, exhaust the algebraic limit $O^{\triangleright}_{\overline{K_v}}$, and commute with its Galois action. The global number field is dense in K_v. Primitive elements, same-degree polynomial approximation, continuity of roots, and Krasner's lemma show that every finite Galois stage is generated by a root of a polynomial over the global field. These polynomial-root witnesses form a countable type and surject onto the finite-stage type. Lean's countable filtered-category theorem therefore constructs the cofinal natural-number subsystem required by the strict positive-integer convention of Absolute Anabelian Topics III, Section 0. Reconstruction from the abstract D_v group remains open. Lean also constructs the fixed carrier as an actual sub-pseudo-monoid and proves that every product defined in the original carrier remains defined after passage to invariants. It also constructs the descended action through a surjective profinite quotient from kernel triviality, derives continuity from the compact-to-Hausdorff quotient topology, and proves that pullback recovers the original action. This formalizes the invariant and $\pi_1^{\kappa\text{-sol}}$ factorization steps only. The Absolute Anabelian Topics II/III algorithms that reconstruct the overgroup, rational fundamental group, integral monoid, and coric rational-function carriers from the abstract group remain source obligations. Horizontal and vertical Kummer isomorphisms preserve both arrows; their site/common compatibility is equality of complete diagram transports. Each naturality square is an isomorphism in an arrow category, so equivariance under arbitrary automorphisms follows formally. Constructing these diagram isomorphisms from IUT II, Corollary 4.8 and the cyclotomic-rigidity algorithm of IUT I, Example 5.1(v), and proving evaluation compatibility up to Ind_1–Ind_3, remain open. A full local log-link consists of a continuous rational-linear post-composition between the logarithmic modules, a compatible ring equivalence of their realized logarithmic fields, and continuous injective realizations of the adjacent target units. Thus the finite invariant-unit and archimedean circle-unit relations are derived from equality between a source logarithm and a target unit; they are partial, but their target is unique. At a bad prime, cross-log admissibility is likewise derived rather than supplied as an arbitrary predicate. The actual unit groups of the splitting monoids map multiplicatively into the log-link domain; both representatives must lie in the appropriate domains, the current representative must lie in the adjacent overlap, and their quotient is witnessed by a root of unity in the next domain. From those inputs Lean derives the genuine procession-automorphism action, independent summand action, generated $\mathrm{Ind}_1/\mathrm{Ind}_2$ quotient, vertical local Kummer comparisons, Ind_3 system, vertical log-volume independence, roots-only bad-prime logarithm equality, and commutativity of every indexed horizontal square. Constructing those isomorphisms from the actual horizontal theta links remains an explicit source obligation. This is a coherent source boundary, not an existence proof for its inputs; all upstream constructions remain listed individually in the active manifest. In particular, constructing the full log-links from the cited Ψ_{cns}, Frobenioid, and poly-isomorphism data remains open.

The separate source-shaped analytic interface for Proposition 3.5 and Theorem 3.11(ii) now derives the mono-analytic packet log-shell region from the packet integral construction itself. At finite places this is the constructed closed tensor lattice. At an infinite summand Proposition 3.2(ii)'s Hermitian seminorm is normalized by π, so its unit ball is exactly the raw complex radius-π disk of Definition 1.1(ii) and Remark 1.2.2(ii); it is not a second radius-π ball of an unnormalized metric. At a finite place, each explicit local-unit group carries a continuous profinite Galois action, and its invariant-unit type is definitionally the subgroup fixed by every Galois element, not an arbitrary compact type with an invariant name. The resulting compact fixed subgroups and their p-adic logarithms construct factorwise unit-log direct sums, pure tensors, and the generated packet subgroup. Lean proves that subgroup lies in the closed tensor lattice and that vertical Kummer transport preserves it exactly; the concrete Ind3 constructor therefore derives direct finite-place containment instead of accepting an arbitrary packet subgroup and containment proof. An adjacent finite log-link is now represented by a partial correspondence on the actual fixed local units in every summand. Lean applies the summand logarithms, tensors related unit tuples, takes the additive closure of the resulting graph, and recursively composes adjacent steps. It proves that an (r)-fold iterate from (m-r) ends at (m), and derives containment of the endpoint's Kummer image from endpoint unit-log membership. The concrete Ind3 boundary therefore accepts neither arbitrary multi-step maps nor final iterated-containment proofs. At an infinite place, Lean models the source units as genuine elements of S^1. The principal logarithm $\arg(z)i$ lies in the normalized local unit ball and exponentiates back to z. Selecting one place in each tensor factor produces actual direct-sum and pure-tensor representatives; their packet-unit-ball membership follows from the local, direct-sum, and tensor metric equations. Each unit is also canonically a point of the exact S^1 core of an antitone projective system of open annuli. Lean proves that the annuli have intersection S^1 and are cofinal among all neighborhoods of S^1, restricts ambient complex differentiability to every level, proves that ambient multiplication and inversion preserve the neighborhood filters, and proves that deleting S^1 leaves the two radial connected components. The component determined by $\mathcal{O}^{\triangleright}_{\mathbb{C}} \setminus S^1$ is selected by a connected-component theorem rather than an arbitrary input. For every connected annulus U_n, Lean derives the actual group $\mathrm{Aut}_{\mathrm{hol}}(U_n)$ from biholomorphic self-maps and proves its identity, composition, inversion, and restriction laws. For a groupification-induced holomorphic automorphism of $\mathbb{C}^{\times}$, pullback along the exponential covering yields an integer exponent; bijectivity restricts it to 1 or -1, while preservation of the selected inner component rules out inversion. Lean consequently proves that the induced filter germ is the identity. An adjacent log-link is a partial relation on these semi-germ core points. Lean recursively derives every positive iterate, its domain and range, the participating endpoint-ball subset, and relational surjectivity onto the source domain. Direct Kummer containment of both principal unit tensors and arbitrary selected local radius-π tensors follows from the proved full vertical unit-ball image. Consequently the concrete Ind3 boundary accepts no arbitrary archimedean subgroup, multi-step domain, preimage set, iterate map, surjectivity proof, or direct containment. The general Riemann-surface and RC-holomorphic local-morphism classification of *Absolute Anabelian Topics III*, Definition 2.1 and Corollary 2.3, extension of rigidity from groupification-induced maps to every purely local group-germ automorphism, and co-holomorphic Kummer reconstruction are not yet formalized. Construction of the finite summand unit/log data from the IUT II Ψ_{cns} objects and of the adjacent finite and infinite unit correspondences from the Gaussian log-theta lattice remains a source obligation.

The largest remaining IUT III task is to enrich this finite route with the analytic common-container, holomorphic-hull, determinant, local-field, and Haar data consumed by Step (xi). The older downstream route still has a unified Hodge-theater/log-theta/log-Kummer source spine that projects to its analytic bridge and finite-divisor payload. It is now a legacy adapter target, not the canonical M3 input boundary. The current bridge component is no longer a fully free qualitative packet: Lean can construct it from the lower constructed qualitative certificate and audits the SHE/APT/IPL no-collapse projections at the primitive packet boundary. The packet has also gained a constructed-qualitative bridge-alignment source. The SHE/common-container alignment is now a projection from constructed qualitative common-container data, and the input/selected-output prime strips are canonical projections from the constructed qualitative IPL source. The one-column Hodge theater can now be fixed as the constructed SHE codomain, with the one-column history read from the selected q-choice. A still lower q-pilot-canonical bridge source now constructs the selected concrete q-choice with the constructed SHE q-pilot theater; Lean then derives its codomain placement from the structured SHE q-pilot/codomain law. A q-pilot-class bridge source goes below the concrete-coordinate boundary by rebuilding the selected q-choice from a theta-pilot lattice class and representative data for the forgotten finite label, procession state, local tensor state, and upper-semi state. A component-representative bridge source now lowers that representative section to the generated log-theta/vertical-log-Kummer/Frobenioid divisor-column component kernel. A source-spine q-pilot component bridge now reads the q-pilot history label, label-forgotten log-theta lattice coordinate, and coric datum from the selected q-choice of the unified Theorem 3.11 source spine. A new component-kernel averaging constructor now also builds the log-theta label-procession upper-semi source directly from this Frobenioid base-volume data: the label log-volume is the constant function on $\mathbb{Z}/l\mathbb{Z}$, Lean proves

$$\frac{1}{l}\sum_{j\in\mathbb{Z}/l\mathbb{Z}} V = V,$$

and transports this equality through the constructed representative upper-semi state. Thus the finite label average used by the old procession-normalized source is recovered from the generated divisor-column source rather than supplied as an independent averaged field. The vertical log-Kummer/log-link layer has also been tightened one step below this boundary: the transported $\mathbb{Z}/l\mathbb{Z}$-labelled log-shell family is now proved equal to the constant family on the base log-shell object, and its normalized log-volume is read directly as the base finite log-volume. The generated full-label quotient layer now records the same construction before projection: Lean builds a constant $\mathbb{Z}/l\mathbb{Z}$-indexed averaged log-volume object over each generated theta-pilot class and proves that the typed-core generated full-label log-volume is exactly this finite average. Thus the quotient possible-image audit is tied to the actual finite label average on the generated source space, not only to a class-wise base-volume function. The selected generated F_l-orbit now carries an explicit averaged log-volume object too: its normalized entries are the generated choices with the same theta-pilot class and each finite label, Lean proves the $\mathbb{Z}/l\mathbb{Z}$-average formula, and the orbit audit identifies this average with both the selected generated log-volume and the theta-class average. The generated Ind1/Ind2 audit now preserves this actual orbit average, including the pointwise normalized label family, so the finite average is carried through the quotient construction itself rather than only after reading off the scalar log-volume. Ind3 now gives the corresponding upper-semi inequality on the orbit average, and the generated equality quotient has an exact no-extra-collapse statement at the setoid quotient level: equality there is equivalent to equality of the forgotten theta-pilot class. The canonical selected-q Ob5–Ob6 audit now carries the concrete full-label translation log-volume equality as a field, and the same field is projected through the global C_Θ audit. Thus the selected q-region, its equality-quotient possible image, and the full-label Ind1 log-volume preservation are checked in one source object before the hull estimates are used. The global audit now also projects the selected-region equality with the quotient possible image, its full-label translated possible-region equality, its canonical-hull containment, its equality with the record theta possible image, and its containment in the record theta union, so the final C_Θ estimate retains the source-side region trace and containment data. The same global audit exposes the no-Ind3-generator statement for the equality quotient, so the final comparison layer does not hide an additional Ind3 identification. The exact-$\Xi(P_B)$ version of this audit is now checked at the indexed, hull-fixed principal valuation-ball boundary: Lean derives the exact family-hull log-volume equality from the Remark 3.9.5 principal-product source and carries the same no-Ind3 statement through the global comparison. This has also been pushed down to the compact-open log-Kummer correspondence boundary: the local relation supplies left-totality, realization compatibility, and compact-open soundness, and Lean carries the resulting exact-$\Xi(P_B)$ log-volume equality and no-Ind3 fact into the global C_Θ audit. The possible-image factorization has also been made exact at the typed quotient boundary: equality of equality-quotient indices forces equality of the pulled-back possible-image regions, and the concrete theta-pilot source audits carry the quotient pullback equation through the finite-label, upper-semi, and vertical log-Kummer packet layers. A generated-full-label source-spine bridge lowers the last equality further by matching the source-spine selected q-choice to the concrete projection of the generated full-label representative over the q-pilot theta class; Lean derives the component-kernel representative equality from the full-label kernel theorem. A generated-q-choice source-spine bridge now lowers this once more: it supplies the generated q-choice itself, proves that its theta class is the q-pilot theta class and that its finite label is the component kernel's representative label, and derives the generated-full-label representative projection. A selected-label source-spine bridge removes that generated q-choice as a public object: Lean constructs it internally as the kernel representative full-label choice over the q-pilot theta class read from the source spine, proves that its finite label is the selected source label, and derives the selected q-choice projection from fieldwise source equalities for the q-pilot theater, representative finite label, procession state, local tensor state, and upper-semi state. A component-representative selected-label bridge lowers those fieldwise equalities to the vertical-log-Kummer/Frobenioid component representative source: applying its constructed-representative theorem to the selected q-choice and transporting across the q-pilot theater alignment yields the representative finite label, procession state, local tensor state, and upper-semi state internally. A one-sided component selected-label bridge now removes the independent component-representative source at this boundary: Lean projects it as the `volumeSource` of the concrete one-sided multiradial component package, and retains only synchronization of the one-sided selected q-choice with the source spine and constructed q-pilot theater. Its SHE-codomain successor derives that q-pilot theater placement from the source-spine one-column law, the constructed SHE codomain/one-column alignment, and the structured SHE q-pilot/codomain equality. A source-spine-selected successor now constructs the one-sided component source with the source-spine selected q-choice, making that synchronization definitional. A source-spine-data successor also projects the theta possible-image source and gluing torsor from the unified source spine. Its output-flag-calibrated packet audit now verifies preservation of the selected q-choice, bridge component, Step (x) log-Kummer source, and target-region formula source while constructing the subclaims internally. The primitive-packet successor now projects the multiradial record and target-region-calibrated source spine from the same primitive packet, so the record/source image-family agreement and the SHE codomain/one-column alignment are inherited from that packet rather than supplied as independent bridge inputs. A component-kernel successor now constructs the one-sided component volume source from the Frobenioid divisor-column representative kernel. The generated full-label package layer then avoids a global ambient-coverage assertion: Lean pulls the concrete pre-ledger and source package back along the kernel projection from generated full-label choices, provided only a selected generated q-choice whose projection is the pre-ledger chosen output. A selected-label package-reindex alignment now factors this through the source spine: Lean constructs the selected generated q-choice from the selected-label q-pilot bridge and derives its chosen-output projection from the single equality identifying the source-spine selected q-choice with the pre-ledger chosen output. A primitive-packet selected-label reindex alignment lowers this once more: the selected-label source is projected from the primitive packet's component-kernel one-sided source, and the selected q-choice of that projected source spine is definitionally the primitive packet's selected q-choice. A chosen-output primitive packet reindex alignment removes that equality as an external field by constructing the primitive packet with the pre-ledger chosen output itself. The chosen-output generated full-label quotient corridor then projects the generated kernel, theta-class image family, gluing torsor, and selected generated q-choice from the same chosen-output packet route. A constructed generated-corridor source now rebuilds the generated multiradial record with the full-label image family forced by that packet, so the image-family equality is no longer an external field. The next bridge boundary is therefore the chosen-output one-column/history placement, the bridge/component data indexed by that chosen output, and a generated SHE-structured

Theorem 3.11 bundle: the base generated multiradial record is now constructed from this bundle before the forced image family is installed. This generated bundle is now itself obtained by reindexing the concrete structured bundle along the chosen-output full-label projection. At the direct product-hull/HDD surface, the chosen-output generated full-label wrapper now consumes this chosen-output corridor and projects the older generated-HDD source internally, while its reindexed-structured strengthening lowers the public route through this concrete-to-generated bundle constructor. This replaces the earlier public boundary of prebuilt q-pilot class fields, a prebuilt bridge component, an independent one-column/codomain equality, a supplied representative-data section, a direct constructed-representative equality, a direct generated-full-label projection, a source-provided generated q-choice, fieldwise selected-label compatibilities, an independent component representative source, a selected-q synchronization equality, an independent theta possible-image source, an independent gluing torsor, an independent record/source image-family synchronization, a monolithic one-sided component volume source, a raw q-pilot theater placement field, or an assertion that every ambient concrete choice is a kernel representative. The missing source construction includes:

the richer Frobenioid/Kummer structure beyond the degree model,
the derivation of the structured input link from the typed prime strips,
the multiradial algorithm and its possible images,
the log-theta-lattice procession and F_ℓ-symmetry,
the derivation of the 1-column SHE/IPL/APT bridge,
the derivation of the Step (x) package from vertical log-Kummer data,
the proof that $\mathrm{Ind}_1/\mathrm{Ind}_2$ preserve normalized log-volume,
the proof that Ind_3 has only the stated upper-semi orientation.

The Lean records now say exactly what must be supplied. They do not yet replace Mochizuki's proof of Theorem 3.11.

5.2 Remark 3.9.5 and Step (xi)

The second major gap is the genuine holomorphic hull and determinant construction. The current code now has a paper-traced Step (xi) source route: the old Step (xi) obligation record can be projected from a structured source that records Corollary 3.12 Step (xi-a)–(xi-g), Remark 3.9.5 Ob1–Ob9, the selected Theorem 3.11 possible image, determinant data, and Ob7 link pinning. At the hull-formation boundary, the selected q-region is now constructed from the equality-quotient possible-image family, so its containment in the possible-image union is no longer a separately supplied Step (xi) field; the paper-derived source constructor uses the source spine's selected q-choice directly. This boundary has now been lowered one step further: the Step (xi) hull source may be built from the Remark 3.9.5 minimal-hull system $\phi(P) = \bigcap_{P \subseteq H} H$. Lean derives the holomorphic hull operator, the canonical hull of the possible-image union, closedness of this canonical hull, containment of the selected q-region in the union, and the Ob1/Ob2 absorption statement from this minimal-hull source rather than from a free hull-operator witness. The current lower constructor also accepts the principal product presentation $\lambda \cdot O$ of Remark 3.9.5(i): the base integer/valuation-ball region O, nonzero scalar parameters, scalar-image principal hulls, the intersection-parameter hull, and the principal-hull log-volume are projected into the concrete Step (xi) hull source. At the determinant boundary, the Step (xi) determinant comparison record can now be constructed from the weighted arithmetic-vector-bundle determinant source already present in the Step (xi) layer: the finite adjusted summands, determinant sum, normalization, and positive tensor power are projected from that lower source rather than supplied as independent fields. This has now been lowered to the localized arithmetic-vector-bundle decomposition source: the localized hull cover supplies a finite sum of localized hull-region log-volumes, each localized log-volume is identified with the corresponding structure-sheaf-adjusted vector-bundle contribution, and Lean projects the result to the Step (xi) determinant comparison data. The selected q-region-to-normalized-determinant inequality is then derived from hull containment plus the localized family-hull/normalized-determinant equality; the localized projection audit now also exposes the adjusted Ob3/Ob5 endpoint, the induced Ob3/Ob5 compatibility source, and the bounded-family hull/determinant log-volume endpoint proved in the lower SourceCore layer. This localized decomposition now also constructs the record-pinned Ob3/Ob5 adjusted determinant source itself, after an explicit identification of its possible-region family with the multiradial record family. Consequently the obligations-backed exact-theta possible-image Step (xi) constructor can be entered directly from localized hull/vector-bundle data, with the finite localized sum providing determinant compatibility and the obligations package providing the bridge, q-positivity, and normalization. The strict localized route now packages this with the equality-quotient Ob5 statement: choices in the $(\mathrm{Ind1}, \mathrm{Ind2})$ equality orbit have identical quotient possible regions and quotient-region log-volumes, while the same localized family-hull construction supplies the determinant and bounded-family tensor endpoint. The record Ob3/Ob5 boundary has also been lowered through the theta-region-defined principal valuation-ball source to the typed fiber-transport valuation source. Lean pulls the theta-pilot valuation cover back to concrete choices, proves that its adjusted possible-region family is the multiradial record possible-image family, builds the record Ob3/Ob5 adjusted determinant/log-volume source from this cover, and then feeds the resulting source into the exact-theta constructed hull/determinant API. For a lattice-image-law-backed source the remaining explicit input is the identification of the supplied lattice-image law with the concrete fiber-transport law; for the fiber-transport-backed source this equality is internal to the constructor. This boundary has also been lowered through the local log-shell/log-Kummer layer: a principal pointwise constructed log-shell metric-zero source now constructs the same record Ob3/Ob5 source, and compact-open log-Kummer image, correspondence, and map possible-region sources construct both the record Ob3/Ob5 source and the exact-theta Step (xi) hull/determinant source. Lean passes through the local log-Kummer graph or relation-level image laws, compact-open representatives, and the converse realization law before entering the principal valuation-ball constructor. The compact-open realized-exact source now also exposes the literal image equality $\Theta_v = \mathrm{realization}(U_v)$ and constructs the log-Kummer map source directly from it; the same realized-exact source now also constructs the Record Ob3/Ob5 adjusted determinant/log-volume source and the exact-theta Step (xi) hull/determinant source through that map layer. The Hodge-calibrated compact-open route has been lowered to this same realized-exact boundary: realized exactness constructs the Hodge-calibrated Step (xi) source, its side-conditioned form, and its source-hull-det-data-backed form. The exact possible-region, log-Kummer map, image, and correspondence endpoints now also have a Record Ob3/Ob5 valuation-ball source form: the Ob3/Ob4-to-Step (xi) determinant equality is projected from the record-canonical source rather than passed as a separate endpoint hypothesis. The corresponding choice-indexed arithmetic-formula routes at the exact possible-region, log-Kummer map, image, and correspondence boundaries now also construct the raw determinant compatibility from this Record Ob3/Ob5 source; their remaining synchronization input is the lower equality identifying the adjusted valuation-ball hull operator with the principal product hull. These four choice-indexed compact-open boundaries now also have constructor-sourced bridge variants at both the arithmetic-formula and formula-gap endpoints: the package hull/determinant bridge equality is projected from a Theorem 3.11 constructor-sourced Step (xi) source, while the remaining lower synchronization identifies the constructor bridge datum with the corresponding Record Ob3/Ob5 principal-product hull/tensor-power construction.

The same constructor-sourced bridge form now reaches the choice-indexed arithmetic-degree-controlled, direct p-adic unit-ball Haar, and derived p-adic unit-ball Haar compact-open map, exact possible-region, log-Kummer image, and log-Kummer correspondence endpoints: the lower comparison, local Haar, and derived formula-matching data still supply the formula-gap source and the determinant/C_Θ-sum identifications, but the hull compatibility and package bridge are reconstructed from the Record Ob3/Ob5 source and constructor bridge. The correspondence route now also has an obligations-backed entry point: the package bridge equality, q-pilot positivity, and normalization are projected from 'IUTStage1SourceHullDetObligations'. Its Hodge-calibrated variant removes the raw theta/family-hull scalar equality from this boundary: Lean derives Θ_{signed} = family-hull log-volume from 'IUTStage1HodgeFamilyHullLogVolumeCalibration', via the Hodge–Arakelov theta monoid degree. There is also a direct bridge-based compact-open log-Kummer map entry point for this Hodge-calibrated route: the realization map and local sections construct the correspondence relation internally, and the constructor uses the Hodge-derived theta/family-hull equality without an ambient 'IUTStage1SourceHullDetObligations' object. A side-conditioned form of this map route now projects q-positivity and normalization from 'IUTStage1SourceSideConditions'. A further source-data-backed form projects the package hull/determinant bridge equality from 'IUTStage1SourceHullDetData', leaving only the synchronization of that stored bridge datum with the record-canonical hull/tensor-power construction, together with measure calibration, as explicit lower input. The older obligations-backed map entry point is retained as a compatibility projection. The same calibration now reaches the finite-divisor constructor-backed source: Lean converts the Hodge-calibrated data to the constructor-backed spine, forms the constructor-generated hull/determinant obligations, and builds the possible-image Step (xi) source without reintroducing a raw Θ_{signed}-to-family-hull equality at this boundary. The target-charted exact-theta adjusted-summand source now enters this Hodge-calibrated finite-divisor constructor first and projects the older constructor-backed source only for legacy consumers. The all-in-one target-charted Hodge/IPL possible-image route audit also exposes the record-canonical Theorem 3.11 hull/tensor-power bridge equality beside the ambient-obligations bridge equality, making this synchronization visible at the public route boundary instead of only through the obligations projection. It now also carries the target-charted reduction to the constructor-built Step (xi) source and the resulting Remark 3.9.5 Ob1–Ob5 bridge audit, so the public route records the actual possible-image hull absorption and determinant-compatibility chain below the canonical bridge equality. At the constructor-built boundary itself, Lean now converts the constructor-built possible-image Step (xi) source into the constructed Remark 3.9.5 hull/determinant source while recovering the record-canonical bridge equality through the built constructor's bridge datum, not through a new synchronization argument. The same route now has a parameterized Ob5 quotient audit. On the non-vacuous additive-Haar branch, the residue-module/unit-ball route now has a localized constructor-built C_Θ boundary audit: it records the projected Haar-modulus source, the constructor-built Remark 3.9.5 projection, the localized Step (xi) local-term summation endpoint, the local-to-global C_Θ audit, and the resulting remaining additive-Haar payload audit. For any nonempty comparison possible image, the Ob5 audit projects the constructor-built bounded-family hull inclusions, collapsed upper-semi quotient equality, determinant bound, and local canonical-scale chain. The additive-Haar source now constructs the bundled constructor-built/localized determinant-scale synchronization from arithmetic-degree calibration; the Record-Ob3/Ob5 wrapper derives the same synchronization via the record Ob3/Ob4 determinant handoff and now audits the localized Step (xi) C_Θ endpoint and residual arithmetic handoff at that boundary. Its compact-open synchronized route audit also exposes the derived signed C_Θ inequality and the Corollary 3.12 dichotomy before comparing definitionally with the established synchronized concrete endpoint. The normalized additive-Haar public route can now consume this Record-

Ob3/Ob5 source directly and rebuild the localized constructor-built C_Θ route internally. The compact-open additive-Haar concrete branch now has the parallel synchronized wrapper: the packet first constructs the canonical HDD record, the Record-Ob3/Ob5 source is indexed over that canonical record, and the localized determinant-scale synchronization is projected before the synchronized concrete endpoint is invoked. The compact-open packet-to-HDD construction is now a named Lean boundary: from the compact-open log-Kummer map and pointwise calibration source it constructs the principal HDD source and canonical constructor-built source, and the synchronized concrete route now consumes this named boundary internally instead of rebuilding it inline. The lower valuation-ball Record-Ob3/Ob5 source is now packaged over this named HDD boundary: one source object carries the constructor-built/record Ob3/Ob4 determinant equality and projects the component Record-Ob3/Ob5 source, the localized determinant-scale synchronization, the Theorem 1.10 valuation-ball local-analytic arithmetic-divisor source, the finite Haar defect, the total Haar-defect bound, and both synchronized-source and component-source forms of the pointwise Step (xi)/Haar/main equality. Thus these local inputs are no longer separate mathematical obligations at this boundary; Lean transports the pointwise equality across the component/valuation-ball formula-source identification, so the component Record-Ob3/Ob5 route sees the same decomposition. These projections are also bundled as a synchronized route-input record, giving the later concrete endpoint a single lower source artifact instead of separate component, synchronization, local-analytic, Haar-defect, and pointwise-equality inputs. The bundle also exposes the source-data determinant equality and, when the source data is the constructor-built HDD source, the corresponding component synchronization endpoint. The Theorem 3.11 possible-image nonemptiness needed for the component projection is now derived from the compact-open log-Kummer theta-region source: the principal transported theta-region cover supplies a point in every theta class, and the record/fiber-theta-region synchronization rewrites these witnesses to the record Θ-possible-image family. The component Record-Ob3/Ob5 source now has a direct compact-open constructor; its audit records this nonemptiness field as the compact-open log-Kummer-map theorem rather than as an independent component hypothesis. In the constructor-built source-data case, the same boundary is now constructed without a separate constructor-built/record Ob3/Ob4 determinant equality: that equality is read from the Record-Ob3/Ob5 valuation-ball source whose source-data field is the constructor-built HDD source. The synchronized route-input bundle is also constructed directly from the compact-open log-Kummer source and this valuation-ball source, so the named-HDD boundary no longer has to be exposed as a separate input at that layer. For the canonical-HDD compact-open endpoint, Lean now also transfers theta-possible-image nonemptiness from the primitive Theorem 3.11 record to the canonical HDD record by the direct canonical-HDD audit, constructs the canonical-HDD named boundary from the compact-open log-Kummer source plus the Record-Ob3/Ob5 valuation-ball source, and immediately threads that constructed boundary into the compact-open inverse-base-prime public route. Thus the canonical-HDD possible-image witness and named boundary are no longer public inputs at this compact-open valuation-ball handoff. This boundary has now been lowered one more source layer: Lean can construct the Record-Ob3/Ob5 valuation-ball source from the Record-Ob3/Ob5 arithmetic-divisor component source together with a Theorem 1.10 valuation-ball formula-matching source and the equality identifying its additive projection with the component formula source. The same canonical-HDD valuation-ball source now also projects the synchronized route-input bundle directly: the component Record-Ob3/Ob5 source, localized determinant/scale synchronization, Theorem 1.10 valuation-ball local-analytic source, Haar defect, and pointwise Step (xi)/Haar/main equality are read from the constructed canonical boundary before the inverse-base-prime comparison is invoked. The canonical-HDD route audit now also reconstructs, from this route-input bundle, the inverse-base-prime summand source and the pointwise local-analytic IUT IV Step (xi)

C_Θ source, recording their endpoints together with the localized C_Θ handoff, signed comparison, and dichotomy at the compact-open handoff itself. A further canonical-HDD compact-open route now lowers the p-adic/additive-Haar input side to the product-level restriction-calibrated Ob7 handoff, so the structure-sheaf/direct-summand synchronization and nonzero norm-square projection are fields of the local-arithmetic source rather than separate public arguments. This route still constructs the same local-analytic C_Θ handoff from the Record-Ob3/Ob5 valuation-ball source before invoking the restriction-calibrated Step (xi) endpoint. The strongest canonical-HDD compact-open wrappers now combine both reductions: their public boundaries take the component Record-Ob3/Ob5 source, the valuation-ball formula-matching refinement, their projection equality, and either the inverse-base-prime valuation-cover/additive-Haar inputs or the product-level Ob7 handoff, then construct the Record-Ob3/Ob5 valuation-ball source internally before invoking the same signed Step (xi) endpoint. The product-handoff route has now been lowered once more: it may take the arithmetic-degree-controlled valuation-ball local arithmetic source instead of the assembled valuation-ball formula-matching record, construct that formula-matching record internally, and then apply the same component-to-valuation-ball bridge. The route-input bundle now also projects directly to the pointwise local-analytic Theorem 1.10/IUT IV localized Step (xi) source consumed by the lower C_Θ route: the bundled localized synchronization, valuation-ball local-analytic source, Haar defect, synchronized pointwise equality, and total-defect bound construct that source before the larger compact-open public endpoint is invoked. The route-input audit now records this constructed source endpoint, its localized C_Θ handoff audit, the signed C_Θ comparison, and the Corollary 3.12 dichotomy at this layer. The compact-open route-input audit also records the internally constructed inverse-base-prime summand source, the same local-analytic C_Θ source and signed endpoint, and the definitional equality between the compact-open route and the corresponding pointwise route. These route-input audits, together with the p-adic-defect/main compact-open route-input constructor, the local-degree formula compact-open route-input constructor, the local-degree formula canonical-HDD compact-open inverse-base-prime route, the two p-adic-defect/main compact-open product-handoff routes, the p-adic-normalized processional case local-estimate endpoint, the projection from full p-adic-normalized processional estimates to the weaker Haar-residual case source, the case-defined normalized Haar-residual case projection, the projection from normalized case-defined estimates to that Haar-residual case boundary, the normalization of a general case-defined estimate along the p-adic place alignment, packet handoff, and six public signed endpoint lowerings, and the constructor-level p-adic-defect/main formula-gap residual endpoint, are now registered in the canonical residual-frontier manifest. The same manifest also records the Record-Ob3/Ob5 component and valuation-ball localization-anchor-tensor constructors and the constructor-built valuation-ball determinant-source lowering, together with the possible-image named-HDD and family-hull q-normalized synchronized route-input lowerings, so it is now a 320-entry, zero-source-obligation sub-manifest of the preferred route. The same frontier now records the finite-extension inverse-base-prime valuation-ball construction: Lean proves $p^{-1}O_K$ is the valuation ball of radius p, transports it componentwise to the compact-open inverse-base-prime additive-Haar normalization, lifts the compact-open inverse-base-prime and product-handoff formula-realized routes through possible-image witness and log-Kummer correspondence data, and includes a weakened-boundary countermodel showing that a detached compact open may have unit mass while the true inverse-base-prime ball has mass > 1. That source now also projects the localized C_Θ constructor source, its local-term IUT IV handoff audit, the signed bound $\Theta_{\text{signed}} \leq C_\Theta(-q_{\text{signed}})$, and the resulting Corollary 3.12 dichotomy. These facts are therefore visible at the Theorem 1.10 valuation-ball local-analytic boundary, not only after entering a larger inverse-base-prime wrapper. The current product-hull goal-evidence audit now has the same public-surface lowering: it consumes the Theorem 1.10 valuation-ball local-analytic source and projects the localized C_Θ source, handoff audit, signed bound, and dichotomy internally. The pointwise Hodge-formula companion lowers this one step further at the same current product-hull surface: the public input is the pointwise Hodge formula source, and Lean projects the measure/summand calibration source internally before entering the Theorem 1.10 localized C_Θ handoff. A local-arithmetic companion now also constructs the restriction-calibrated source at this endpoint from the residue-module/unit-ball Haar-character valuation-cover source, the local-global realified Frobenioid restriction package, the structure-sheaf normalized Haar identities, the distinguished non-structure summand, and the theta-degree/localized-adjusted sum comparison. At the synchronized Record-Ob3/Ob5 route-input boundary, the formula-realized product-handoff route now combines this product-level Ob7 handoff with the bundled valuation-ball formula-gap data to construct the pointwise Ob7 source and the pointwise Theorem 1.10 localized Step (xi) source internally. Thus the same formula-realized route is now available at the compact-open named-HDD and component/formula boundaries: the compact-open log-Kummer map supplies the pointwise valuation-ball source, and the component/formula bridge constructs the Record-Ob3/Ob5 valuation-ball source before entering the product-handoff evidence audit. The same compact-open branch now also accepts the arithmetic-degree-controlled local source directly and derives the formula-matching source before invoking the formula-realized handoff. The component-level public route now uses this controlled/product handoff surface as well, so it no longer needs to expose the inverse-base-prime valuation cover, additive-Haar direct-summand source, local-analytic Theorem 1.10 source, Haar defect, or local Step (xi)/main-log equality. Consequently the current product-hull/Hodge-formula public theorem no longer takes the completed restriction-calibrated source itself, although its local arithmetic ingredients remain source-paper obligations. A further current-product-hull companion now constructs the assembled Theorem 1.10/IUT IV localized source from the localized Step (xi) determinant source, determinant and canonical C_Θ-scale synchronizations, valuation-ball additive-Haar local analytic arithmetic-divisor source, local Haar defects, the local Step (xi)/Haar/main-log identity, and the finite defect lower bound. Its synchronized companion lowers this further by taking a constructor-built localized Step (xi) determinant/scale synchronization source and projecting the localized determinant source and the two determinant/C_Θ-scale equalities internally. A formula-gap matched companion lowers the IUT IV input at the same boundary: the valuation-ball local analytic source is projected from named distinguished Proposition 1.4 log-shell and archimedean Proposition 1.5 metric comparison data before the localized C_Θ handoff. The source-level projection theorem now records that this Proposition 1.4/1.5-backed formula source satisfies the older Theorem 1.10 local-case endpoint once the distinguished and archimedean formula-to-gap comparisons are supplied. A route-input companion now packages the determinant/scale synchronization, formula-gap source, Haar defects, local Step (xi)/Haar/main-log identity, and defect lower bound into the typed pointwise Theorem 1.10 localized Step (xi) source, so these data are no longer separate arguments at this current-product boundary. The synchronized route-input bundle now preserves the valuation-ball formula-gap Theorem 1.10 source itself together with its local-analytic projection equality, and the pointwise inverse-base-prime public route now consumes this route-input bundle directly, so the local-analytic localized Step (xi) source is no longer a separate public input or endpoint at that boundary: the route constructs it from the bundle and then builds the formula-gap IUT IV localized source from the preserved formula-gap field consumed by the signed C_Θ comparison. The compact-open log-Kummer/valuation-cover route now makes the same handoff after deriving the principal pointwise valuation-ball source and inverse-base-prime summand source internally. Its named-HDD boundary variant goes one step lower, constructing the synchronized route-input bundle from the valuation-ball boundary before invoking this compact-open handoff. The route-input layer now also has a p-adic-defect/main variant: Lean reconstructs the Record-Ob3/Ob5 valuation-ball source from the stricter p-adic-defect/main controlled component source, then audits the resulting component source, localized determinant/scale synchronization, formula-gap endpoint, and finite Haar-residual lower bound. This lowers the compact-open handoff from a valuation-ball source boundary to the p-adic-defect/main boundary when that lower source is available. The same route-input layer now has a local-degree formula variant. Lean reconstructs the p-adic-defect/main controlled source from the Step (v)/(vii) local-degree formula Record-Ob3/Ob5 source via valuation-ball projection preservation, then audits the compact-open route input together with the distinguished, nondistinguished, and archimedean local formula package. The canonical-HDD inverse-base-prime route can now consume the same local-degree formula boundary by projecting its controlled component source and the reconstructed valuation-ball formula package before entering the signed compact-open comparison. The same public route now exposes the Ob6 exact-$\Xi(P_B)$ boundary. It constructs the Remark 3.9.5 possible-image family from the constructor-built hull source, identifies the canonical $\Phi(P_B)$ approximant with the family hull, and derives the selected and comparison log-volume bounds from a source-calibrated exact-$\Xi(P_B)$ log-volume equality rather than from an arbitrary supplied approximant family. This target-charted route boundary now also has a constructor-built Ob7 product-formula source. The input is finite weighted determinant prime-strip/product-formula data plus the coric unit-character pullback; Lean requires its determinant source to be the same determinant source used by the constructor-built Step (xi) bridge, constructs the coric log-Kummer Ob7 source, and then packages the combined Ob5/Ob6/Ob7 audit. Thus the route no longer needs a standalone determinant/global Ob7 equality at this level, although the lower local product-formula and coric prime-strip data remain explicit source inputs. The determinant-calibrated public audit now also exposes the product-formula endpoint itself: the finite summand formula, determinant/global prime-strip equality, theta-derived tensor-power bound, and retained source/target/log-Kummer-column identities are named fields. At the principal-product localized public route, the Step (xi)/Remark 3.9.5 paper trace is now also constructed internally. Its Step (xi-a)–(xi-g) and Ob1–Ob9 entries are concrete statements about the unified Theorem 3.11 source spine, the $\lambda \cdot O$ hull source, the localized determinant projection, the theta-equals-family-hull equality, and the shared Ob7 log-Kummer source. The strict public route therefore no longer accepts a standalone paper-trace proposition bundle. A localized-cover variant now lowers the determinant alignment boundary: its source fixes the hull system and possible-image family to the principal-product Step (xi) source, so Lean projects the localized decomposition with the old hull-operator and possible-region equalities by reflexivity rather than accepting them as separate public inputs. Its canonical-theta refinement also takes Θ_{signed} to be the projected family-hull log-volume, making the theta/family-hull equality reflexive at this boundary. Both localized-cover forms now also have additive-payload variants: Lean constructs the paper trace, the derived Step (xi) source, and the remaining non-Step audit from the lower additive-Haar arithmetic-degree payload. The same lowering now also applies to the underlying preferred Step (xi) route ladder, from the raw Step (xi) source through the concrete-localized, principal-product, localized-cover, and transported scalar-parameter layers. The representable, locally representable, coordinate scalar-image, and real-line coordinate wrappers inherit the same additive-payload reconstruction, so these routes no longer need a separate non-Step audit when the additive-Haar payload is available. At the Additive-Haar local C_Θ boundary, the constructed Remark 3.9.5 hull/determinant source can now be built directly from the localized vector-bundle determinant source; hence the determinant-source handoff to the Step (xi) local-term constructor is reflexive. The remaining equality at this local boundary is the genuine calibration of the canonical C_Θ scale with the finite Step (xi) sum. The same constructed-trace handoff has now been propagated to the finite-extension valuation-unit-ball compact-open norm-square boundary, using the localized determinant, Hodge-evaluation, theta/family-hull, and Ob7 construction chain already required by that strict route. A calibrated variant of this boundary now replaces the raw Hodge theta-monoid/localized-adjusted sum equality by the Hodge/family-hull equality and derives the adjusted-sum equality from the localized Ob3/Ob5 identity. A further restriction-calibrated variant replaces the placewise Hodge canonical-degree/local-prime equalities by an anchored local-global restriction source together with one global Hodge canonical-degree calibration. The strongest finite-extension variant now also moves the remaining localized possible-region comparison down to the p-adic finite source itself: the equality after the unit-ball/additive-Haar/compact-open/norm-square projections is derived by the source-core projection theorem. The principal-product finite-extension source lowers this once more: its localized possible-region field is fixed definitionally to the Theorem 3.11 principal-product possible-image family, so the strongest public wrapper no longer takes a p-adic possible-region equality as an independent input. A calibrated p-adic finite hull-cover source now also derives the pointwise Ob3 localized-region log-volume identities from local vector-bundle calibrations, instead of accepting those identities as fields of the finite-extension localized decomposition source. Its family-hull-indexed refinement derives the cover identity, pairwise disjointness, and finite-additive volume sum from an index map on the principal-product family hull. A cover-local measure-model refinement lowers the p-adic volume input further: the localized cover sum is derived from the log-volume measure model attached to the calibrated p-adic regions, and the preferred public finite-extension route can consume this measure-backed cover source directly. A direct-product refinement removes the supplied p-adic family-hull index: Lean constructs it from the local-factor cell cover of the principal-product family hull, using local-factor separation to prove the fiber and disjointness properties. A tensor-measure refinement removes the supplied p-adic cover-local measure model: direct-product cell log-volume identities and their finite tensor-cell summation formula construct the measure model used by the preferred public route. The p-adic tensor data has also been lowered to the source-core localized decomposition: Lean derives the tensor-cell log-volumes and tensor-sum formula from the adjusted determinant log-volume identities carried by that finite-extension hull/vector-bundle source. The direct-product local factors in this p-adic branch are now charted by local-ring/vector-bundle factor calibrations, so Lean projects their local-ring equality and direct-summand log-volume identities instead of taking raw factor regions at the public boundary. These chart calibrations have in turn been lowered to valuation-ball factor calibrations: the p-adic local-factor regions are now supplied by valuation-ball/additive-Haar data and then projected to the charted local-ring interface. A further principal-product compact-open-factor variant replaces the direct Hodge/family-hull equality by a localized adjusted-determinant sum calibration; Lean recovers the family-hull equality from the localized Remark 3.9.5 decomposition theorem. The additive-Haar principal-product branch now also has a cover-derived localized hull source: instead of taking the localized-region containment in the family hull and the family-hull log-volume sum as standalone fields, it assumes the explicit cover identity $\phi(\bigcup_i \Theta_i) = \bigcup_\beta H_\beta$ and the finite additivity formula for this cover; Lean derives the old containment and volume-sum interface from these lower hull-cover statements. This boundary has now been lowered again: each H_β carries a local vector-bundle calibration, is presented as a fiber of an index map on the family hull, and Lean derives the cover identity, pairwise disjointness, finite-additive cover sum, and pointwise $\log \mathrm{vol}(H_\beta)$ determinant identity after identifying that local vector bundle with the additive-Haar determinant localization. The strict positive determinant contribution has now been separated from the valuation-unit-ball normalization. A principal-product additive-Haar route constructs the paper trace internally and derives the positive summand from the source-level inequality $1 < \mu(U)$. A stronger local-field branch now constructs such a strict factor explicitly: for a finite extension of $\mathbb{Q}_p$, the inverse base-prime dilation of the normalized unit ball is identified as $p^{-1}O_K$, and the source-core Haar-modulus theorem proves $\mu(p^{-1}O_K) = p^{[K:\mathbb{Q}_p]} > 1$. Conversely, Lean proves that the older unit-ball-positive boundary is contradictory: the same unit-ball source forces $\mu(O_v) = 1$, hence normalized Haar log-volume 0. The principal-product hull source has also been connected to the lower upper-semi quotient formalism of Remark 3.9.5(v): the concrete hull source now projects to a bounded-family hull quotient source, proves that this quotient hull is the canonical/principal family hull, and proves that any two nonempty quotient-indexed possible regions have the same collapsed quotient image. For equality-orbit choices, Lean packages this with region equality, log-volume equality, quotient-image equality, and containment in the family hull. The principal-product source boundary has been lowered again in source-core: a scalar-parameter direct-product hull source now constructs the full Remark 3.9.5 principal product-hull system and its minimal intersection theorem. Product-parameter representability is no longer a global assumption: Lean derives it from coordinate scalar-image data. Once local coordinate points can be extended to product points, the coordinate landing/preimage laws prove that each coordinate product region is the image of the local integer factor under the coordinate scalar map. Lean then uses the coordinate region itself as the local scalar parameter and assembles these choices into the product scalar parameter. The intersection scalar selector is obtained from the represented direct-product intersection parameter. This replaces the principal intersection law by the direct-product hull theorem plus explicit coordinate scalar-image data. This scalar construction has now been transported onto the public Step (xi) carrier: a generic equivalence transport theorem carries principal $\lambda \cdot O$ hull systems across carrier equivalences, and the transported scalar-parameter source constructs the Point(`target`)-level principal product hull source consumed by the localized theta-equals-family-hull route. Thus the generic route no longer needs a standalone principal product-hull source on the public carrier; it needs coordinate scalar-image data and an explicit carrier equivalence. In the one-coordinate real-line specialization, even this carrier equivalence is not supplied: Point(`target`) is canonically equivalent to its real coordinate and to the product carrier Unit $\to \mathbb{R}$. The strict public coordinate and real-line endpoints now also construct the Step (xi) paper trace internally from the same coordinate scalar-image hull, localized determinant, theta-family-hull, and Ob7 data, so these endpoints no longer expose either a bare public principal-hull source or a standalone paper-trace argument. Their additive-payload variants now go one step further: Lean rebuilds the derived Step (xi) source and the remaining non-Step audit from the additive-Haar arithmetic-degree payload, matching the stricter principal-product constructed-trace boundary. The valuation-ball selected exact-Ξ boundary has also been tied to this coordinate source: Lean converts a coordinate scalar-image valuation-ball cover to the selected scalar-parameter cover by taking the local scalar parameter to be the coordinate region itself, proves that the induced holomorphic hull system agrees with the direct-product hull system, and then reuses the selected radius-cover exact-Ξ transport theorem. The coordinate source has also been lowered to a sharper local image boundary: an exact formula $H_v(\lambda_v) = \lambda_v O_v$ constructs the older coordinate landing and preimage laws, and the valuation-ball public audit now has an exact-image variant that projects through the coordinate valuation source before transporting exact-Ξ to the public carrier. Below this boundary, Lean now constructs a source-facing scalar-parameter hull system from the Hodge-theater/log-theta lattice key itself: the parameter is the tuple consisting of the Hodge theater, history label, theta-pilot lattice coordinate, and coric datum, and the local region $H_v(\lambda_v)$ is the coordinate projection of the formula-backed theta region along the public carrier equivalence. This boundary has now been lowered again: a realized local image source states that each formula-backed theta region is the realized image of its local integer factor, and that projection of the realized point is the corresponding local scalar multiple. Lean derives the projected scalar-image equality from these two local image laws and then feeds the existing scalar-parameter direct-product hull source. Independently, the exact-image direct-product formula can now be specialized to each projected theta region, constructing the same package-level coordinate scalar-image source without a standalone projected equality. For the selected valuation-ball family-hull route, the implementation now uses the more faithful scalar boundary: a nonzero local multiplication source $\lambda_v \cdot O_v$ constructs the selected scalar-parameter valuation source, and the unit-ball valuation wrapper inherits this selected source. The resulting endpoint identifies the selected scalar image with the valuation-ball direct-product cell union and with the adjusted Ob3/Ob5 determinant handoff. The same selected-scalar source now constructs the possible-image exact-$\Xi(P_B)$ source from progressively local cover data: indexed cell containment, single local-factor containment, and finally compact-open valuation-ball-radius containment, where the last step uses the additive-Haar normalization audit $B_{\beta,v} = B_{\beta,v}(r_{\beta,v})$. The nonzero-scalar and valuation-unit-ball wrappers inherit this radius-to-exact-Ξ route. Thus the exact-Ξ calibration is no longer an independent field at this boundary. The public audit packages the selected hull/cell-union equality, calibrated log-volume, normalized determinant handoff, and transported exact-Ξ source. This is still selected-parameter evidence; the remaining public-route obligation is to construct those valuation-ball containments from transported theta/valuation local exactness and to derive the realized theta-region local image laws from still lower analytic Frobenioid/log-Kummer data. For the stricter localized route, the final normalized localized determinant bound by Θ_{signed} is no longer supplied as a standalone inequality: Lean derives it from the identification of Θ_{signed} with the constructed localized family-hull log-volume. The record-canonical Remark 3.9.5 hull/determinant bridge now also feeds this record: once the paper-derived selected q-region log-volume is identified with the record bridge q-pilot log-volume, the q-region-to-normalized- determinant bound and the determinant-to-Θ_{signed} bound are both projected from the bridge. The bridge-backed paper-source constructor now derives this log-volume identification from lower alignment data: equality of the paper hull operator with the bridge hull operator, and equality of the bridge q-pilot region with the source-spine selected Theorem 3.11 possible image. The constructed-source version projects the bridge and hull operator from the constructed Remark 3.9.5 source itself, leaving only the q-pilot region/source-spine selected possible-image identification at this boundary. The Ob7 source link has now been lowered to the same record bridge: the record-canonical Remark 3.9.5 bridge supplies the q-region and determinant source, while the coric $\Theta^{\times\mu}$ invariant supplies the prime-strip/log-Kummer equality. In the same-index constructor-built possible-image route, equality of record images with source images derives both the selected q-region identification and the quotient-family union alignment. A further source-selected constructor forms the constructed Remark 3.9.5 source with the source-spine selected possible image as its q-pilot region, so this identification is definitional on that route; the remaining explicit condition is containment of the selected possible image in the record possible-image union. This selected-source boundary now also has a stricter Ob3/Ob5 determinant form: instead of receiving a raw hull/determinant compatibility record, it can receive the record-canonical family-hull compatibility source, or the still lower adjusted determinant log-volume source, and Lean derives the old compatibility projection internally. The pointwise source-region route and the current deepest retained (Ind2) action-packet transport route both have adjusted log-volume variants, so the public local source spine can pass through the measured adjusted-source boundary. That measured boundary now has a Hodge-calibrated constructor: the package measure equality is projected from the measured source, and the Θ_{signed}-to-family-hull equality is derived from 'IUTStage1HodgeFamilyHullLogVolumeCalibration', instead of being supplied as a raw downstream argument. This is still a boundary above the localized Ob3/Ob5 arithmetic-vector-bundle data used by the older compatibility routes, but the theta equality at that boundary is now Hodge-derived rather than asserted. The quotient-hull side of these variants has also been lowered: the route may now take the Theorem 3.11 possible-image quotient compatibility for the source images, and Lean forms the Remark 3.9.5 equality-quotient hull/log-volume compatibility by applying the selected holomorphic hull operator. Thus the deepest local selected route no longer needs to expose a standalone equality-quotient hull compatibility record when the lower possible-image quotient source is present. This has now been pushed down to the unified Theorem 3.11 source spine: the route can derive the possible-image quotient compatibility from the theta-pilot possible-image source together with the vertical log-Kummer packet-aligned finite-procession source. The deepest Ob3/Ob4 action-packet determinant route now performs this source-spine projection too, constructing the equality-quotient hull/log-volume compatibility before entering the later adjusted Ob3/Ob5 refinement. The concrete one-sided Theorem 3.11 constructor now performs the same lowering directly from the theta-pilot class possible-image source and the record/source image equality, so it no longer exposes a raw `PossibleImageQuotientCompatibility` argument for the $(\mathrm{Ind}_1), (\mathrm{Ind}_2)$ image quotient law; the (Ind_3) upper-semi comparison remains part of the typed-core log-volume input. The concrete primitive packet now exposes this one-sided source directly from its own possible-image source, vertical log-Kummer source, gluing torsor, and selected q-choice. A comparison audit now relates this vertical packet source to the stricter fiber (Ind_2) action-packet transport source used by the theta-region route, proving agreement of the record possible-image family and the selected q-region as a target set. A further symmetry-label wrapper projects the fiber action-packet source from lower label-transport data tied to the packet's vertical log-Kummer indeterminacy source. The wrapper can now also be constructed from the common Theorem 3.11 possible-image witness, the equality-quotient possible-image family, and the explicit symmetry-label, upper-semi, and procession transports. The packet-facing public completion route has a corresponding possible-image-witness entry point, so it no longer exposes a preassembled symmetry-label wrapper or an arbitrary fiber (Ind_2) action-packet transport source. On this route the explicit quotient-compatibility argument disappears; the remaining source-side comparison is the record/source possible-image trace supplied by the action-packet transport layer. The determinant upper-bound input has also been lowered on the same source-spine routes: the Ob4 tensor-power bound is derived from the exact-theta calibration identifying Θ_{signed} with the record Ob3/Ob5 family-hull log-volume. Thus the strict action-packet route now uses adjusted determinant log-volume data plus exact family-hull theta calibration, rather than a standalone tensor-power estimate. The Ob7 prime-strip/log-Kummer record can now be read from a single coric $\Theta^{\times\mu}$-style source link aligned with the same quotient-hull bridge; its retained statement includes the local Gaussian/environment equalities and unit-character equalities from the Ob7 source endpoint. The strongest selected route now constructs this source link from the constructed Remark 3.9.5 bridge and a coric prime-strip invariant, after transporting determinant compatibility across the record-union/equality-quotient-family alignment. In the same-index source-spine case, this alignment is now derived from the equality of the record possible-image family with the Theorem 3.11 source image family; the selected route also derives the selected-q containment in the record union from the same equality. A lower selected route now takes the paper-shaped pointwise trace that the record possible image attached to each concrete Hodge-theater/log-theta choice equals the corresponding source-spine possible image; Lean assembles this trace into the full family equality before applying the quotient-union theorem. The selected Ob3/Ob4 route, the adjusted Ob3/Ob5 source-region route, and the exact-theta source-spine route now also have product-formula Ob7 variants: the determinant/global prime-strip equality is proved from finite local adjusted-summand calibrations and the finite prime-strip product formula, while the coric unit-character data constructs the $\Theta^{\times\mu}_{\mathrm{LGP}}$ invariant before entering the existing log-Kummer Ob7 source link. These routes therefore no longer need the determinant/global Ob7 equality as an independent input. This pointwise trace is now itself derived from the existing theta-pilot class-region source once its class formula is identified with the Hodge-theater source spine's theta-pilot class-image family. The route now also accepts record-level theta-pilot class formulas and Hodge-theater/history/log-theta-lattice formulas, projecting them through the class-region source before assembling the Step (xi) comparison. Below the lattice formula layer, it can consume the same-lattice-key image law, which constructs the lattice formula from representative choices before comparison with the source-spine class images. One level lower, it can consume typed theta-pilot fiber-transport invariance and derive the same-key image law internally. Below that, the selected route can now consume the (Ind1), (Ind2) equality-quotient pullback and, on the current strongest local source route, the retained post-procession (Ind2) action-packet transport source; Lean derives the quotient image pullback, the fiber-invariant lattice image law, and the record/source trace before entering the Step (xi) hull/determinant comparison. The adjusted Ob3/Ob5 determinant-log-volume variants have now been added at the class-region, class-formula, lattice-formula, lattice-image-law, and typed fiber-transport levels, so these intermediate routes inherit the record-canonical determinant compatibility from the adjusted log-volume source rather than exposing a raw hull-approximant compatibility record. The action-packet theta-formula comparison with the source-spine class images can now also be derived from the equality of the record possible-image family with the unified Theorem 3.11 source image family; it is no longer a separate input on the exact-theta action-packet route. A further constructed-record variant installs the unified source-spine image family directly as the record's theta-pilot possible-image family, so the record/source image equality becomes definitional; the remaining image-side calibration is that these source-spine images are the package target-region family. This calibration has now been lowered from a whole-family equality to the pointwise theta-pilot class-region trace $\mathcal{R}_{\mathrm{class}}(\mathrm{class}(c)) = \mathcal{R}_{\mathrm{target}}(c)$; Lean derives the source-image/target-region family equality internally before installing the source-spine record. One layer lower, the class-region trace can now be produced from a target-region-backed theta-pilot source: a representative section for theta-pilot classes, a proof that target regions depend only on the theta-pilot class, and explicit target-region point witnesses. Lean constructs the witness-bearing class-indexed possible-image source from these data and threads it through the exact-theta Step (xi) action-packet route. This has now been lowered once more: the target-region class-invariance law can be derived from invariance under the concrete theta-pilot lattice key, namely Hodge theater, history label, log-theta lattice coordinate, and coric datum. This lattice-key boundary has itself now been lowered to a target-region lattice formula: each package target region is identified with a formula indexed by the Hodge theater, history label, theta-pilot lattice coordinate, and coric datum, and Lean derives the key invariance, class trace, full source-image/target-region family equality, and Step (xi) source-spine handoff from that formula. The formula is now also produced from the concrete lattice-image law already present in the Theorem 3.11 source spine: record possible images are evaluated by the lattice formula, and the multiradial record's target-region realization identifies those record regions with the package target regions. The exact-theta Step (xi) source-spine route can therefore consume the lattice-image law directly, without an independently supplied target-region lattice formula. The same package target-region formula is now projected further down from typed theta-pilot fiber transport, from the (Ind1), (Ind2) equality-quotient pullback, and from the retained post-procession (Ind2) action-packet transport source. Thus this target-region formula boundary follows the current deepest local Theorem 3.11 action-packet source rather than stopping at a standalone lattice-image law. The action-packet boundary itself has now been lowered one step: Lean projects the retained packet/action transport source from place-audited nonarchimedean Ism or archimedean order-two action-entry label-transport data, together with explicit identifications of the audited packet tensor states with the post-procession source and target local tensor states. The label-transport source is now constructed one layer lower from the retained action-entry source, source and target tensor symmetry label equations, and transport of the direct-summand symmetry kind. The same tensor-label data now also projects to the place-audited possible-image/fiber audits for the nonarchimedean and archimedean branches, so the lower action data carries both the packet transport and the equality-quotient/fiber witness information. This boundary has now been lowered again: the tensor-label equalities may be read from a single direct-summand packet SymmetryLabelTransportSource, and the corresponding concrete fiber source projects through the tensor-label route internally. Record-level possible-image point witnesses now also factor through the class-indexed Theorem 3.11 possible-image source and the record/source image equality, so the newest symmetry-label fiber source no longer needs independently supplied witness points. The same record/source image equality now gives all-choice nonemptiness for the additive-Haar arithmetic-divisor and record Ob3/Ob5 constructors, and builds the central Theorem 3.11 possible-image source-data record with its nonemptiness audit, rather than requiring a separate downstream nonemptiness field. The exact-theta source-spine/action-packet Step (xi) route has also been lowered at the hull/determinant obligation boundary: the package bridge equality, q-pilot positivity, and normalization are now projected from a single 'IUTStage1SourceHullDetObligations' object, with an endpoint recording the induced record-canonical selected source-spine bridge. The same exact-theta source-spine route now has a localized hull/vector-bundle entry point: after an explicit identification of the localized possible-image family with the source-spine 'recordThetaPossibleImage' family, Lean constructs the record Ob3/Ob5 adjusted determinant source internally and records the induced finite adjusted-sum, normalized determinant, and bridge-synchronization facts at the endpoint. This localized handoff has also been lifted through the tensor-label, label-transport, and symmetry-label transport sources, each of which projects the Fiber-Ind2 action packet internally, and through the concrete target-region lattice-formula source, which installs the package target-region possible images before entering the localized determinant constructor. It now also consumes the concrete lattice-image law itself, deriving the target-region lattice formula before entering the same localized hull/vector-bundle path. At the principal valuation-ball boundary, the symmetry-label packet now has a source object that takes a source-core Remark 3.9.5 principal valuation-ball cover together with the fiber-transport-backed family-union comparison and projects both the theta-region principal source and its principal hull. This removes a lower possible-region matching datum at the source layer. The same fiber-backed source now also proves that the representative selected over a theta-pilot class has the same typed-core log-volume as every concrete choice in that class, using the constructed post-procession (Ind2) transport audit rather than a new comparison field. This equality is now projected through the principal valuation-ball source and the compact-open realized-exact source, so the same source object that supplies the local compact-open realization also carries the representative/log-volume audit. The packet/symmetry-label public route now consumes this source object directly: the theta-region principal source and selected q-choice are projected from the same packet source before entering the constructed Step (xi) completion audit. The possible-image-witness variant now constructs the packet-indexed symmetry-label wrapper before entering this principal valuation-ball route, so the preassembled wrapper is no longer public at this surface either. The same lowering now reaches the HDD-source audit route: the principal-product Ob3/Ob5 measure-Hodge-gap HDD source remains explicit there, but the symmetry-label wrapper is constructed from possible-image witness data before the HDD source is audited against the theta-region principal hull. The same witness-derived entry point now reaches the constructed-HDD route as well: Lean builds this HDD source internally from the theta-region source, packet selected q-choice, constructed qualitative bundle, measure calibration, side conditions, and summand-Hodge data, so neither a preassembled symmetry-label wrapper nor an independent principal HDD source remains at that route surface. The witness-derived lowering now also reaches the localized finite-place data route: the localized determinant source, main-log contribution, Haar defect, and summation identities remain explicit, but the symmetry-label wrapper is again constructed from possible-image witness data before the localized C_Θ source is built. The remaining principal-product source still visible there belongs to the restriction-calibrated local arithmetic boundary. The newest localized calibration variant combines both lowerings: after constructing the symmetry-label wrapper from possible-image witness data, it lowers the two calibration inputs to one theta-region calibration source. Its endpoint records pointwise pre-ledger volume as the principal holomorphic-hull log-volume and the Hodge theta-monoid degree as the finite adjusted determinant summand sum, and Lean projects the older measure and summand-charted calibrations from these formulas. The local arithmetic side is now lowered in parallel, including on the witness-derived surface: the restriction-calibrated Frobenioid/residue-module source, the structure-sheaf/direct-summand synchronization, and the nonzero norm-square Ob7 summand first define an audited local handoff source, and the possible-image witness route constructs the symmetry-label wrapper before consuming this handoff. The same witness-derived branch now projects the principal hull used by local arithmetic from the theta-region source, instead of accepting an independent principal-product hull argument, and its fiber-backed variant constructs the theta-region source from the principal valuation-ball product-hull cover and transported family-union comparison after deriving the symmetry-label wrapper from possible-image witness data; the theta-region-defined variant then projects those two inputs from a single principal theta-region valuation-ball source on the same witness-derived surface, and the pointwise metric variant now carries that witness-derived lowering through the constructed-log-shell/metric-zero valuation-ball source; the pointwise calibration variant projects the theta-region measure/summand calibration from the same witness-derived surface; the pointwise local arithmetic variant also projects the restriction-calibrated Ob7 handoff from that surface's pointwise source data, and the localized Step (xi) variant projects the determinant/summation/Haar-defect source from the same witness-derived pointwise boundary; the IUT IV arithmetic-divisor variant projects that localized source from the local evaluation ledger on the same surface, and the Theorem 1.10 valuation-ball variant projects that ledger from the valuation-ball additive-Haar formula-gap source on the same surface; the Hodge-formula variant also constructs the pointwise measure/summand calibration from explicit formulas on that witness-derived surface, and the local-analytic variant checks the valuation-ball local-analytic source on the same surface before projecting to the Theorem 1.10 valuation-ball route. The strict pointwise local-analytic route now constructs the packet-facing Ob7 source from this product handoff while projecting the formula-gap matched IUT IV source from the local analytic valuation-ball source. Its compact-open companion constructs the principal pointwise valuation-ball source from compact-open log-Kummer map data and uses the same product-handoff local arithmetic fields; a pointwise formula-gap companion now reconstructs the local-analytic Step (xi) wrapper from the same localized Theorem 1.10/IUT IV source used by the signed C_Θ route at this local arithmetic boundary, and the formula-realized evidence route performs the same compact-open and Ob7-handoff construction before entering the pointwise Theorem 1.10/IUT IV packet comparison. Its product-handoff version packages the local-arithmetic fields into the typed Ob7 handoff at the public boundary. The local-arithmetic-degree version also constructs the pointwise Theorem 1.10/IUT IV localized source itself from the localized local-term Step (xi) data and formula-gap arithmetic-degree identities. Its synchronized compact-open variant constructs that localized local-term source from determinant synchronization and finite-place arithmetic-degree data before entering the same signed route. The lower constructor route can rebuild that pointwise source from the restriction-calibrated residue-module/unit-ball/additive-Haar source plus the two projection facts before the localized route is entered. The compact-open constructed-synchronization variant also builds the determinant/scale synchronization source from the localized determinant source and its two constructor-built comparison identities. The explicit-formula compact-open variant then constructs the measure/summand formula record itself from the three displayed formula identities before the same route is entered. A determinant-calibrated refinement lowers the Hodge summand boundary further: it takes the charted Hodge calibration to the normalized theta-region determinant, and Lean derives the determinant log-volume equality and the finite adjusted-summand formula from the record-native determinant source. The realized-exact variant constructs the compact-open log-Kummer map source from realized compact-open theta-region image exactness before applying that route. The restriction-calibrated variant constructs the product Ob7 handoff from the lower local-arithmetic data and derives the Theorem 1.10 formula-gap valuation-ball source from local-analytic arithmetic-divisor data before entering the same path. The current compact-open map route now consumes that lower pointwise restriction-calibrated local-arithmetic source directly and constructs the Ob7 handoff internally. Its p-adic finite-extension refinement also projects the ordinary localized Step (xi) determinant from the unit-ball/valuation-Haar compact-open norm-square localized determinant layer at this public boundary. The synchronized p-adic finite refinement bundles that projected determinant with the constructor-built determinant and canonical C_Θ-scale identifications as a single source object before entering the local-upper endpoint. These synchronization facts remain lower source data, but they no longer appear as two separate public route arguments. The formula-gap variant now consumes the Theorem 1.10 valuation-ball additive-Haar formula-gap source and projects the local-analytic arithmetic-divisor ledger internally for the residual calculation. The current local-analytic residual refinement lowers this boundary again: the public endpoint supplies the Theorem 1.10 valuation-ball local-analytic arithmetic-divisor ledger and residual Haar bound, and Lean constructs the formula-gap residual source internally. The localized residual refinement also removes the p-adic determinant/scale synchronization record from the strongest endpoint: it now supplies the finite-extension localized Step (xi) determinant source and the two constructor-built determinant/canonical-scale comparison facts, from which Lean constructs the synchronization object. A principal-product constructor pushes this determinant boundary lower again by projecting the p-adic localized Ob3/Ob4 determinant source from the Theorem 3.11 principal-product localized hull-vector-bundle decomposition before entering the same local-analytic residual route. The corresponding public route now threads this principal-product residual source through the source-hypothesis endpoint: the residual input is tied to the packet's source-data-with-target-regions object and to the principal hull projected from the internally constructed theta-region valuation-ball source. Its realized-exact variant projects the compact-open log-Kummer map source from compact-open realized exactness before entering the same principal-product residual handoff; the localized principal-product source now also separates the finite-place index universe from the principal-hull parameter universe. A local-arithmetic-handoff variant now constructs the pointwise restriction-calibrated Ob7 source from the product-level local arithmetic handoff before entering that residual endpoint. A stronger possible-image source-gap variant now starts instead from compact-open local Kummer map data: the local realization map, selected-region section, valuation-factor compatibility, and compact-open soundness are enough for Lean to derive the correspondence, image, possible-region exactness, and log-shell exactness chain before entering the signed C_Θ route. The compact-open graph source now also constructs this map package directly: left-totality supplies the section, graph realization supplies the map compatibility, and graph soundness supplies the possible-region law. The corresponding public source-gap endpoint now uses that constructor internally, so this branch exposes graph data rather than the compact-open map package. It is lower again at the compact-open image and possible-region levels: Lean builds the canonical graph from the compact-open log-shell image containments, then derives those containments from the Remark 3.9.5 equality identifying the selected possible region with the realized compact-open local factor. The pointwise local-coordinate source now also projects to this compact-open log-shell image boundary by reconstructing log-shell exactness from local representatives and the converse realization law, then applying compact-open Haar normalization. It now continues through the canonical compact-open graph to construct the possible-region exact, log-Kummer image, correspondence, and map packages that the public Step (xi) route consumes. The route is now also lower at the log-Kummer image-law and correspondence levels: the two image laws imply possible-region exactness, and a compact-open log-Kummer correspondence source supplies those image laws by left-totality, realization compatibility, and compact-open soundness. Thus this public source-gap branch can be entered from correspondence data rather than an already packaged equality of possible regions. The image-law branch now also has a named constructor `compactOpenRealizedExactSourceOfCompactOpenLogKummerImageSource`, which factors the same image-law, possible-region, log-shell, graph/map, correspondence, and principal-realized-exact chain into a reusable compact-open realized-exact source. The corresponding constructor `compactOpenRealizedExactSourceOfCompactOpenLogKummerCorrespondenceSource` lowers this one step further: left-totality, realization compatibility, and compact-open soundness of the log-Kummer correspondence first produce the image laws, then the same realized-exact chain applies. The scalar family-hull branches now consume image-law or correspondence data directly instead of treating compact-open realized exactness as primitive. The family-hull theta branch uses this constructor in the endpoint `preferredPublicConcretePacketExplicitDeterminantFormulaCompactOpenLogKummerImageLocalArithmeticH` and the record-family-hull q-scale branch uses the analogous endpoint `preferredPublicConcretePacketExplicitDeterminantFormulaCompactOpenLogKummerImageLocalArithmeticH` These source-with-symmetry scalar public boundaries are thereby lowered from compact-open realized exactness to compact-open log-Kummer image laws. The packaged local-degree source-gap branch now uses the correspondence constructor directly in `preferredPublicConcretePacketExplicitDeterminantFormulaCompactOpenLogKummerCorrespondenceLocalAr` so this broad branch no longer exposes compact-open realized exactness before the local-degree source is consumed. The next lowering on the same branch, `preferredPublicConcretePacketPossibleImageWitnessSymmetryLabelCompactOpenLogKummerCorrespondence` also removes the preassembled symmetry-label transport wrapper: Lean constructs it from the possible-image witness, the equality-quotient possible-image family, and the symmetry-label, upper-semi, and procession fiber transport laws before applying the compact-open correspondence route. Thus this packaged branch now exposes the Theorem 3.11 possible-image/transport data itself, not a separate wrapper. On the canonical M3 boundary those possible-image and transport laws are now constructed; this older analytic branch receives them through the explicit legacy adapter while its Step (xi) enrichments remain. A measure/summand-calibrated variant, `preferredPublicConcretePacketPossibleImageWitnessSymmetryLabelCompactOpenLogKummerCorrespondence` then lowers the same packaged branch below the pointwise determinant-formula package: the determinant formula is reconstructed from theta-region measure calibration and summand-charted Hodge/family-hull calibration before the local-degree source is consumed. A further pointwise arithmetic/formula variant, `preferredPublicConcretePacketPossibleImageWitnessSymmetryLabelCompactOpenLogKummerCorrespondence` also reconstructs the product Ob7 local-arithmetic handoff from the pointwise residue-module/unit-ball/additive-Haar restriction source and builds the theta-region calibration from explicit pointwise measure/summand formulas. The case-bounded residual variant `preferredPublicConcretePacketPossibleImageWitnessSymmetryLabelCompactOpenLogKummerCorrespondence` pushes this branch below the packaged same-index local-degree source itself: the same-index package is reconstructed from principal-product Step XI decomposition, weighted-determinant synchronization, q-normalization, formula-gap Theorem 1.10 data, and case-bounded IUT IV residuals. A local-analytic case-residual companion, `preferredPublicConcretePacketPossibleImageWitnessCompactOpenLogKummerPointwiseFormulaLocalAnalyt` lowers this public boundary again by deriving both the formula-gap matched Theorem 1.10 source and the case-bounded residual package from the valuation-ball local-analytic arithmetic-divisor source and its local distinguished, nondistinguished, archimedean, and +1 case inequalities. The constructed-hull/determinant variant `preferredPublicConcretePacketPossibleImageWitnessCompactOpenLogKummerPointwiseFormulaLocalAnalyt` then fixes the same branch's source-obligation gap from the primitive constructed-SHE bundle and `IUTStage1Remark395ConstructedHolomorphicHullDeterminantSource.toHullDetObligations`. The principal-HDD variant `preferredPublicConcretePacketPossibleImageWitnessCompactOpenLogKummerPoin` instead reads that gap from the principal product-hull CanonicalHDD direct common-container source carried by the Remark 3.9.5 construction. A formula-gap residual variant now lowers the IUT IV input again: it takes the valuation-ball additive-Haar formula-gap residual source and projects the local-analytic residual ledger internally. A projected-principal-product refinement now constructs the p-adic principal-product finite formula-gap residual package from the product-level local-arithmetic handoff, the determinant/scale equalities, and the formula-gap residual data before entering that endpoint. Its local-analytic companion promotes the Theorem 1.10 valuation-ball local-analytic arithmetic-divisor ledger to the formula-gap matched IUT IV source internally before the same principal-product projection. The formula-gap principal-product source now also constructs the q-normalized projected-principal local-analytic source directly: Lean first projects the formula-gap Theorem 1.10 data to the local-analytic ledger, then applies the principal-product canonical-scale synchronization theorem to obtain the normalized theta identity and the signed dichotomy. The principal-product localized local-analytic source now also defines the finite-place Haar-normalization defect directly as the residual $U_v - \mathrm{StepXI}_v - M_v$, where U_v is the IUT IV local arithmetic-upper contribution, StepXI_v is the projected localized determinant summand, and M_v is the local main-log contribution. Lean proves the resulting $\mathrm{StepXI}_v + Haar_v = a_v(D_v + C_v) + E_v - M_v$ equality by expanding the arithmetic-upper ledger and separately recovers the total Haar lower bound from the source residual hypothesis. Both the projected-principal local-analytic source and its formula-gap refinement now expose the constructor-built localized C_Θ source directly, including its endpoint, IUT IV handoff audit, signed bound, and dichotomy. A q-normalized p-adic localized residual source now lowers the generic scale boundary: it records $\theta_{\mathrm{signed}} = (\sum_v \mathrm{adeg}_v^{\mathrm{adj}})(-q_{\mathrm{signed}})$, and Lean derives the canonical finite-sum scale equality from q-pilot positivity. Conversely, the determinant/scale synchronization source now derives this normalized theta identity from the canonical finite-sum scale equality and q-pilot positivity; the generic localized local-analytic residual source therefore constructs the q-normalized source internally. A projected-principal specialization applies the same derivation to the principal-product p-adic localized determinant projection; the principal-product canonical-scale source now constructs the corresponding q-normalized projected-principal source internally before reading off its endpoint and dichotomy. The reduced projected-principal local-analytic endpoint now consumes this q-normalized source instead of a direct canonical-scale equality. The determinant side has also been lowered one level: an adjusted-determinant synchronized q-normalized source identifies the constructed principal HDD Ob3/Ob4 adjusted determinant source with the adjusted determinant source projected from the p-adic finite localized Step (xi) object, and Lean derives the weighted determinant equality used by the localized C_Θ handoff. The p-adic finite principal-product hull-decomposition source now also derives the finite adjusted-sum formula from the family-hull log-volume decomposition, so this branch can obtain the q-normalized theta identity from the family-hull comparison rather than as a separate adjusted-sum assertion; the q-normalized principal-product local-analytic source has a constructor for this family-hull lowering. The same adjusted-determinant source now also exposes the $C_\Theta \geq -1$ companion endpoint by projecting through the q-normalized local-analytic handoff. Its local-arithmetic-degree constructor keeps the full adjusted-determinant synchronization while deriving the residual Haar lower bound from the packaged local arithmetic-degree residual source. The same source now also has a primitive Ob3/Ob4 constructor, `ConstructorBuiltPrincipalProductPadicFiniteAdjustedDeterminantQNormalizedLocalizedLocalAnalyticR` which derives the full adjusted-determinant equality from equality of the localization family, anchor, and positive tensor power. The localization-family input is now lowered one source layer: Lean proves that the adjusted localization projected from a localized vector-bundle entry is determined by the underlying localized bundle, the structure-sheaf log-volume, the raw adjusted log-volume, and the positive determinant weight, and it promotes pointwise localized vector-bundle equality plus anchor and tensor-power equality to equality of the adjusted-determinant sources. This is lowered once more by a component-field criterion: per-place equality of the underlying localized bundle, structure-sheaf log-volume, raw adjusted log-volume, and determinant weight, together with anchor and tensor-power equality, suffices to identify the adjusted-determinant sources. The current lower criterion opens the localized bundle itself: equality of the local-ring label and direct-summand log-volume family, together with the structure-sheaf, raw-adjusted, weight, anchor, and tensor-power equalities, identifies the adjusted-determinant sources. A norm-square refinement gives the same projected adjusted-determinant synchronization from equality of direct-summand norm families. The current lower criterion pushes this through the Haar-normalized local-field layer: the additive-Haar determinant theorem reconstructs the raw adjusted log-volume from the normalized Haar log-volume sum and the structure-sheaf term, and the valuation-unit-ball and p-adic finite-extension unit-ball variants transport the same Ob3/Ob4 synchronization through the local unit-ball normalization data. The same full adjusted-determinant source now projects directly to the projected-weighted local-analytic route; in the same-index case, the summand, anchor, and positive tensor-power fields are derived from the full Ob3/Ob4 equality by `ConcretePacketProjectedPrincipalProductSameIndexWeightedDeterminantQNormalizedLocalArithmeticDeg` rather than supplied as three independent weighted shadow fields. The same-index local-arithmetic route now also has a family-hull variant of this constructor: it derives the q-normalized adjusted-sum identity from the principal-product family-hull comparison and the p-adic finite localized hull decomposition. A further record-family-hull q-scale constructor derives the unscaled θ_{signed}-equals-record-family-hull equality from the pointwise determinant/Hodge calibration. The record-family q-scale compatibility is now derived from the adjusted-sum q-normalization, determinant/Hodge calibration, and the p-adic finite localized hull decomposition, so the remaining scalar source obligation on this branch is the paper-level derivation of that adjusted-sum q-normalized theta/log-volume comparison. The side-condition surface now has both levels: the family-hull endpoint constructs the same-index local arithmetic-degree source from full adjusted-determinant synchronization and the family-hull theta comparison, while the refined record-family-hull q-scale endpoint first derives the unscaled theta equality from determinant/Hodge calibration and then proves the same signed Corollary 3.12 dichotomy and $C_\Theta \geq -1$ lower bound. A heterogeneous projected-weighted determinant source now handles the endpoint case in which the principal HDD determinant side and the localized p-adic side have different hidden γ-indices: it compares the projected weighted determinant summands, anchor, and positive tensor power over the common β-index, then derives the weighted determinant equality before entering the q-normalized handoff. The public realized-exact local-arithmetic route now consumes this heterogeneous source and projects the q-normalized residual handoff internally, so the determinant equality is no longer a public endpoint input on that route. At the same heterogeneous local-analytic boundary, the residual Haar inequality can now be constructed from the packaged local arithmetic-degree residual source rather than supplied as a raw total bound. This local-analytic endpoint now also has a side-condition-sourced variant: q-pilot positivity and source normalization are supplied as `IUTStage1SourceSideConditions`, and Lean reconstructs the named side-hypothesis record internally before applying the heterogeneous projected-weighted route. The heterogeneous projected-weighted source now also exposes the $C_\Theta \geq -1$ companion endpoint by projecting through the same q-normalized local-analytic handoff, matching the signed endpoint at the non-same-index determinant boundary. The IUT IV side of this heterogeneous boundary has now been lowered as well: the ProjectedWeightedDeterminantQNormalizedFormulaGapResidualHaarSource variant consumes the Theorem 1.10 formula-gap arithmetic-divisor source, derives the local-analytic ledger internally, and exposes both the signed dichotomy and the $C_\Theta \geq -1$ endpoint at the same projected determinant boundary. Its constructor now also derives the formula-gap residual Haar inequality from the packaged local arithmetic-degree residual source, rather than taking that finite-sum inequality as a separate source field. The same constructor now feeds a direct endpoint theorem returning both the signed dichotomy and the $C_\Theta \geq -1$ companion bound from that local-degree payload. This lowering is now visible on the realized-exact local-arithmetic public route itself: the formula-gap projected-weighted wrapper first constructs the local-analytic projected source internally and then enters the existing heterogeneous endpoint, so the local-analytic IUT IV ledger is no longer the route-level input on that path. The same public wrapper now has the $C_\Theta \geq -1$ companion theorem, and both the signed theorem and this companion have now been lifted to the direct side-condition surface. Thus the public route can expose the signed dichotomy and lower-bound endpoint at one formula-gap projected determinant boundary without a separate side-hypothesis wrapper. A constructor-level public endpoint lowers this boundary once more: instead of taking the projected formula-gap source as a route argument, it takes the principal-product local Step (xi) decomposition, the projected weighted summand/anchor/tensor-power identifications, the q-normalized theta identity, the Theorem 1.10 formula-gap valuation-ball source, and the local arithmetic-degree residual payload. Lean derives the finite residual Haar lower bound from that payload, constructs the projected formula-gap source internally, and returns both the signed dichotomy and $C_\Theta \geq -1$. The same constructor-level endpoint now also has a side-condition-sourced variant: q-pilot positivity and source normalization are supplied as ‘IUTStage1SourceSideConditions‘, and Lean rebuilds the named side-hypothesis record internally before entering the constituent route. Its local-degree side-condition form combines both lowerings: the direct side-condition surface no longer takes the residual Haar lower bound as primitive, but derives it from the local arithmetic-degree residual payload. The projected side-condition surface now also has a case-bounded form: the valuation-ball distal/non-distal/archimedean residual source is projected internally to the local arithmetic-degree residual payload before the signed dichotomy and $C_\Theta \geq -1$ endpoint are applied. The stricter same-index concrete source now records the case where the local-arithmetic handoff, the p-adic localized Step (xi) determinant source, and the IUT IV local-analytic ledger share the endpoint β-index. Lean now projects the q-normalized principal-product residual source from this concrete source and derives the signed dichotomy without passing through the older public wrapper. This same-index route is now also exposed as the preferred public endpoint for the shared-β case, so the public theorem surface consumes the constructed same-index source directly rather than a separate localized C_Θ source, determinant equality, or raw signed comparison. The same signed endpoint now also has a side-condition-sourced variant: Lean rebuilds the named side-hypothesis record from `IUTStage1SourceSideConditions` before applying the constructed same-index source route. Its route audit records the q-normalized endpoint, the localized C_Θ handoff audit, the derived IUT IV signed upper bound, and the resulting Corollary 3.12 dichotomy from the same source object. The companion goal-evidence audit gives the corresponding relative nonvacuity statement: an inhabited same-index source inhabits the route-audit proposition and yields the same signed endpoint. This does not claim construction of the foundational Hodge-theater/Frobenioid data themselves. A canonical-scale same-index companion now removes the raw normalized theta identity from this local-analytic source boundary. Its public source is the principal-product localized residual package carrying the determinant equality, canonical C_Θ-scale equality, Theorem 1.10 local-analytic ledger, and residual Haar lower bound; Lean then applies the principal-product determinant/scale synchronization theorem to construct the q-normalized projected-principal source internally. Thus this route obtains $\theta_{\mathrm{signed}} = (\sum_v \mathrm{adeg}_v^{\mathrm{adj}})(-q_{\mathrm{signed}})$ from canonical scale synchronization rather than postulating it as a same-index field, and the same internally q-normalized route supplies the $C_\Theta \geq -1$ companion endpoint. This canonical-scale route now also has signed and $C_\Theta \geq -1$ side-condition-sourced public endpoints: Lean rebuilds the named side-hypothesis record from `IUTStage1SourceSideConditions` before entering the same scale-synchronized local-analytic source route. The same two endpoints now have source-obligation-gap forms, so this same-index lower-comparison surface can project q-positivity and source normalization from `IUTStage1SourceObligationGap` instead of exposing the raw side-condition record when the gap is available. A formula-gap same-index companion now lifts this replacement through the Theorem 1.10 formula-gap boundary. Its source carries the principal-product formula-gap residual package; Lean projects the local-analytic ledger and then uses the same canonical-scale route to obtain the signed endpoint, again without a raw formula-gap theta-normalization field. The corresponding $C_\Theta \geq -1$ endpoint is now read from this same internally q-normalized scale-synchronized route. The formula-gap canonical-scale route now has the same side-condition-sourced signed and $C_\Theta \geq -1$ variants, so the public surface can consume `IUTStage1SourceSideConditions` rather than a separate side-hypothesis wrapper at this boundary. Its signed and $C_\Theta \geq -1$ variants are also lowered through `IUTStage1SourceObligationGap`. The case-bounded version of this scale-synchronized route now constructs the formula-gap residual package from the scale-synchronized local-analytic source, the valuation-ball formula-gap source, and the explicit distinguished, nondistinguished, archimedean, and distinguished +1 Theorem 1.10 case bounds. Thus the valuation-ball case split feeds the formula-gap endpoint without reintroducing a raw same-index theta-normalization field. The scale-synchronized local-analytic residual source itself now has a lower constructor from local arithmetic-degree residual data. Its determinant equality and canonical C_Θ-scale equality still enter at this boundary, but the total Haar residual lower bound is no longer a separate field: it is read from the packaged local arithmetic-degree residual source, whose pointwise $StepXI_v + Haar_v$ identity has already been summed over the finite place index. The corresponding $C_\Theta \geq -1$ endpoint is also exposed directly from this internally q-normalized scale-synchronized source. This lowering has also been lifted to the concrete same-index public route. Lean now constructs the projected principal-product p-adic finite Step (xi) source from the realized-exact local-arithmetic handoff and assembles the same-index scale-synchronized local-analytic source from the remaining determinant/scale synchronization facts, the Theorem 1.10 local-analytic ledger, and the packaged local arithmetic-degree residual source. Thus the signed dichotomy and $C_\Theta \geq -1$ endpoint are available without taking the scale-synchronized local-analytic source itself as an atomic public input. The determinant side of this constructor has now been lowered further: Lean derives the constructor-built weighted determinant equality from the three component equalities for weighted summands, anchor, and positive tensor power against the projected localized Step (xi) determinant source. The canonical C_Θ-scale equality remains the explicit scale-synchronization input for this constructor path, but the canonical audited boundary is now the scale-synchronized source itself: the whole determinant-source synchronization and canonical scale synchronization are the remaining source facts, while the three component equalities are only one checked way to construct the former. At the packaged q-normalized local arithmetic-degree boundary, Lean now also derives this canonical scale equality from the theta/log-volume identity $\theta = \sum_v StepXI_v \cdot (-q)$ together with q-pilot positivity before entering the scale-synchronized local-analytic route. The same-index formula-gap refinement now lowers the IUT IV side of this source: it consumes the Theorem 1.10 formula-gap matched arithmetic-divisor source and projects the local-analytic ledger internally before entering the same q-normalized route. At this boundary the residual Haar lower bound is now also derived from local arithmetic-degree data: Lean proves that the pointwise identity $StepXI_v + Haar_v = a_v(D_v + C_v) + E_v - M_v$, together with the total Haar-defect lower bound, gives the formula-gap residual inequality used by the same-index endpoint. The reusable local source IUTStage1IUTIVLocalArithmeticDegreeResidualSource packages this payload independently of the same-index route: it exposes the local arithmetic-upper identity, the arithmetic-upper finite sum as $\sum_v(StepXI_v + Haar_v + M_v)$, the main-log finite sum, the residual lower bound, and an endpoint audit from the same local degree and Haar-defect data. The same-index namespace now has the corresponding residual-field constructor from this packaged payload, so the formula-gap source can fill its residual inequality from the local arithmetic-degree residual source rather than from separate raw Haar-defect and pointwise equality inputs. The same payload now also constructs the constructor-built localized Step (xi) local-term C_Θ source from the determinant/scale synchronization source and the IUT IV local evaluation ledger, eliminating the separate local main-log, Haar-defect, arithmetic-upper sum, and main-log sum arguments on that constructor path. Lean now further lowers the residual Haar summation itself. A local lower-weight ledger records weights w_v with $\sum_v w_v = 1$ and $w_v \leq Haar_v$ pointwise; finite summation gives $1 \leq \sum_v Haar_v$. This lower-weight residual source is converted to the older local arithmetic-degree residual package and is consumed by the adjusted-determinant q-normalized residual route before the signed C_Θ handoff. The same finite-summation lowering now propagates through the principal-product scale-synchronized local-analytic source and the projected-weighted formula-gap source, so those C_Θ consumers no longer take the summed Haar residual inequality as their primitive residual input. The lower-weight ledger is now itself derived at the pointwise layer: Lean puts weight 1 at the distinguished place and 0 elsewhere, proves that the finite weight sum is 1, and uses pointwise residual nonnegativity plus the distinguished +1 residual to build the lower-weight residual package. The finite Haar-defect lower-bound source has the same distinguished-place construction, so the lower-weight ledger is no longer independent once a normalized-place Haar lower bound is available; this projection now reaches the p-adic-normalized case and arithmetic-gap sources directly. The adjusted-determinant route now combines this residual lowering with the primitive Ob3/Ob4 localization, anchor, and positive tensor-power synchronizations, deriving both the full Ob3/Ob4 equality and the global residual Haar inequality internally before the q-normalized local-analytic handoff. On the same-gamma projected-weighted branch, the same combined adjusted-determinant source is now projected to the weighted summand, anchor, and positive tensor-power equalities, so those shadow fields are no longer separate inputs there. A stricter residual layer IUTStage1IUTIVLocalArithmeticDegreeDefinedResidualSource now defines the Haar defect itself as

$$\mathrm{localArithmeticUpper}_v - \mathrm{StepXI}_v - \mathrm{mainLog}_v.$$

Lean proves the pointwise identity $\mathrm{StepXI}_v + \mathrm{Haar}_v = a_v(D_v + C_v) + E_v - M_v$ by unfolding the IUT IV local arithmetic-upper contribution, and then constructs the older packaged residual source and the localized Step (xi) local-term C_Θ source from this defined-residual input. Thus the Haar-defect function and pointwise equality are no longer independent data at this lowered layer; the remaining numerical input is the global lower bound for the defined residual sum. A pointwise lower source now reduces that global input to local nonnegativity of every defined residual together with one distinguished local residual contribution at least 1; Lean derives the finite summed lower bound by splitting the finite sum into the distinguished term and the remaining nonnegative terms. The same-index formula-gap residual theorem now also consumes this defined-residual source directly, projecting internally to the packaged residual source before entering the existing q-normalized route. The parallel same-index source ConcretePacketProjectedPrincipalProductSameIndexWeightedDeterminantQNormalizedDefinedResidualSource uses the same determinant and theta-equality data as the local-degree source but exposes the defined-residual payload at its boundary, and Lean projects it to the existing local arithmetic-degree source internally. This projection is now exposed all the way to the named preferred public endpoint and its route audits: the defined-residual same-index source constructs the formula-gap source, the q-normalized local-analytic source, the signed Corollary 3.12 dichotomy, and the public-boundary inventory by internal projection to the local arithmetic-degree route. The same projection now also reaches the scale-synchronized local-analytic endpoint: the local arithmetic-degree stage derives the canonical C_Θ-scale equality from the q-normalized theta/log-volume identity and q-pilot positivity, so the defined-residual boundary does not carry that scale equality as a separate public input. Thus the packaged residual source is no longer a public input at this endpoint; the remaining local-estimate trust boundary is the lower bound for the globally summed defined residual. The next parallel source, ConcretePacketProjectedPrincipalProductSameIndexWeightedDeterminantQNormalizedPointwiseDefinedResidua lowers this boundary at the same public endpoint: it exposes the pointwise nonnegativity/distinguished-residual source, constructs the summed defined-residual source internally, and then enters the defined-residual public route. The endpoint therefore no longer takes the global residual lower bound as a raw same-index input; what remains is to derive the pointwise local nonnegativity and distinguished contribution from the nonarchimedean, non-distal, and archimedean IUT IV local estimates. This lower local-estimate boundary has now been attacked directly. The source IUTStage1IUTIVLocalArithmeticDegreeGapBoundedPointwiseResidualSource derives the pointwise residual source from the local arithmetic-gap inequalities

$$\mathrm{StepXI}_v \leq \mathrm{localArithmeticUpper}_v - \mathrm{mainLog}_v$$

for all places, together with one distinguished $\mathrm{StepXI}_{v_0} + 1$ version of the same inequality. Lean proves the residual nonnegativity and the distinguished residual lower bound by ordered real arithmetic. The further source IUTStage1IUTIVTheorem110ProcessionBoundPointwiseResidualSource uses the case-defined Theorem 1.10 procession upper bound: once localized Step (xi) is bounded by that procession bound, the existing Theorem 1.10 formula source supplies the comparison with $\mathrm{localArithmeticUpper} - \mathrm{mainLog}$, and Lean constructs the pointwise defined-residual source internally. The strongest same-index public route has since been rewired past the pointwise residual wrapper through the valuation-ball case-bounded same-index source described below; the remaining local-estimate work is to derive those distinguished, nondistinguished, and archimedean case comparisons from the still lower IUT IV estimates rather than from a supplied case-bounded residual source. The valuation-ball formula-gap layer now has the same lowering: the IUT IV source IUTStage1IUTIVTheorem110ValuationBallAdditiveHaarFormulaGapMatchedArithmeticDivisorEvaluationSource projects directly to a formula arithmetic-divisor source, and IUTStage1IUTIVTheorem110ValuationBallProcessionBoundPointwiseResidualSource uses that projection to derive the same pointwise residual source from the valuation-ball additive-Haar Theorem 1.10 data. This has now been split into the actual local cases visible in IUT IV Theorem 1.10. The source IUTStage1IUTIVTheorem110ValuationBallCaseBoundedPointwiseResidualSource replaces the uniform comparison with the projected procession bound by the distinguished nonarchimedean bound, the nondistinguished nonarchimedean zero bound, the archimedean bound, and one distinguished +1 inequality. Lean then reconstructs the projected case-defined procession inequality and projects to the valuation-ball procession residual source and the pointwise defined-residual source internally. The same case-split source now also feeds the localized Step (xi) local-term C_Θ constructor: Lean first projects the valuation-ball local cases to the pointwise residual source, then to the defined residual source, and finally constructs the localized C_Θ source without taking the pointwise residual wrapper as a separate input at that handoff. The closed formula-gap variant constructorBuiltLocalizedStepXILocalTermCThetaSourceOfValuationBallFormulaGapCaseBoundedResidualSourc lowers the same handoff one step further: its local-analytic and Theorem 1.10 C_Θ trace records are projected from the valuation-ball additive-Haar formula-gap source, rather than being supplied as separate constructor arguments. The local-analytic companion constructorBuiltLocalizedStepXILocalTermCThetaSourceOfValuationBallLocalAnalyticCaseBoundedResidualSou pushes this lower again: it constructs the formula-gap matched Theorem 1.10 source from the valuation-ball additive-Haar local-analytic arithmetic-divisor source before entering the same case-bounded C_Θ handoff. The new local-analytic case-source form constructorBuiltLocalizedStepXILocalTermCThetaSourceOfValuationBallLocalAnalyticCaseSourceResidualSourc removes the formula-gap-indexed case residual from this localized handoff as well: the distinguished, nondistinguished, archimedean, and distinguished +1 inequalities are stated directly over the local-analytic valuation-ball source, and Lean constructs both the formula-gap matched source and the older case-bounded residual source internally. The case-split residual source also now projects directly to the defined residual source and the packaged local arithmetic-degree residual source, so downstream constructors can consume the summed residual and Haar-defect package without reintroducing a pointwise residual argument. The newest local-analytic refinement lowers this boundary once more. The source IUTStage1IUTIVTheorem110ValuationBallLocalAnalyticProcessionBoundPointwiseResidualSource does not carry the three expanded case inequalities. It carries the single comparison of localized Step (xi) with the projected case-defined Theorem 1.10 procession upper bound, plus the distinguished +1 comparison and the distinguished-kind witness. Lean expands the local case definition to recover the older local-analytic case-bounded source and then applies the same localized C_Θ constructor through constructorBuiltLocalizedStepXILocalTermCThetaSourceOfValuationBallLocalAnalyticProcessionBoundResidual Thus the remaining Step (xi)-to-procession comparison is still a genuine source boundary, but the separate distinguished/nondistinguished/archimedean case fields are no longer independent inputs at this local-analytic handoff. That comparison has now been factored into its paper-facing components. The source IUTStage1IUTIVTheorem110ValuationBallLocalAnalyticProcessionComparisonPointwiseResidualSource introduces a procession-normalized local log-volume, identifies the localized Step (xi) weighted adjusted log-volume with it, and separately records the procession-normalized upper-semi estimate against the projected Theorem 1.10 procession bound. Lean derives the previous Step (xi)-to-procession source and the localized C_Θ constructor constructorBuiltLocalizedStepXILocalTermCThetaSourceOfValuationBallLocalAnalyticProcessionComparisonRes from these two ingredients. The upper-semi estimate has now been lowered again by IUTStage1IUTIVTheorem110ValuationBallLocalAnalyticProcessionalEstimatePointwiseResidualSource: Lean identifies the normalized local procession value case-by-case with the distinguished Step (v) exact procession bound, the nondistinguished Step (vi) zero tensor bound, or the archimedean Step (vii) metric-container log-volume, then derives the projected Theorem 1.10 processional upper bound from the formalized Proposition 1.4/1.5 inequalities. The remaining local boundary is the Step (xi) identification with this normalized procession quantity together with the distinguished one-unit residual inequality. A companion route, IUTStage1IUTIVTheorem110ValuationBallLocalAnalyticProcessionalArithmeticGapPointwiseResidualSource, keeps the processional estimates for the ordinary local bounds but sends the distinguished one-unit term through the arithmetic-minus-main residual inequality used in the Haar-defect summation. Thus the localized C_Θ handoff no longer requires the stronger assertion that the distinguished +1 margin lies below the local procession formula. This boundary has now been lowered one step further by IUTStage1IUTIVTheorem110ValuationBallLocalAnalyticProcessionalPointwiseResidualSource:

it takes the pointwise defined Haar-residual source $1 \leq \text{arithmeticUpper} - \text{StepXI} - \text{mainLog}$ at the distinguished place, identifies StepXI with the procession-normalized local log-volume, and derives the distinguished arithmetic-minus-main $+1$ comparison internally before projecting to the arithmetic-gap route. It also projects to the lower-weight local arithmetic-degree residual package, so the distinguished-place indicator ledger is available directly on the processional local-estimate route. The next Lean layer, IUTStage1IUTIVLocalArithmeticDegreePointwiseHaarDefectResidualSource, identifies the defined residual with the local Haar-normalization defect via the arithmetic-degree identity $\text{StepXI}_v + \text{Haar}_v = a_v(D_v + C_v) + E_v - M_v$. Its processional wrapper IUTStage1IUTIVTheorem110ValuationBallLocalAnalyticProcessionalHaarDefectResidualSource therefore constructs the pointwise residual source from local Haar-defect nonnegativity and one distinguished defect lower bound, rather than taking the pointwise residual inequalities themselves as primitive; it and the p-adic unit-ball processional Haar-defect source now project through the same lower-weight residual package. A p-adic unit-ball refinement now lowers this Haar-defect boundary: IUTStage1FinitePlacePadicUnitBallHaarIndexDefectSource defines the local defect as $p_v^{[K_v:\mathbb{Q}_{p_v}]}$ from the unit-ball Haar-character normalization source, proves local nonnegativity and the distinguished one-unit lower bound, and records the source Haar endpoint $\mu(O_v) = 1$, $\mu(p_v O_v) = p_v^{-[K_v:\mathbb{Q}_{p_v}]}$. The new IUTStage1IUTIVLocalArithmeticDegreePadicUnitBallHaarDefectResidualSource and its processional local-analytic wrapper project to the older Haar-defect route before entering the localized C_Θ constructor. This boundary is now lowered once more by the Step (xi) synchronization source: it assumes the local calibration

$$StepXI_v = A_v - M_v - p_v^{[K_v:\mathbb{Q}_{p_v}]},$$

where A_v is the already-constructed local IUT IV arithmetic upper term. Lean unfolds $A_v = a_v(D_v + C_v) + E_v$ to recover the expanded arithmetic-degree identity internally. This synchronization is now further lowered to the formula matching equalities $\text{StepXI}_v = a_v(D_v + C_v)$ and $E_v = p_v^{[K_v:\mathbb{Q}_{p_v}]} + M_v$. The stricter audited bridge from the formula-gap matched valuation-ball arithmetic-degree/p-adic source now projects those source-backed formula pieces to the canonical Step (xi) constructor: the Step (xi) equality comes from localized determinant weight and raw-log-volume calibration. The prime-error split is now lowered further: it is constructed from the additive-Haar local analytic arithmetic-divisor evaluation source plus the p-adic identity $E_v = \delta_v + M_v$. The valuation-ball refinement builds this identity into the prime-error term itself, projects to the valuation-ball formula-gap matched prime-error source, and is now constructed from the arithmetic-degree-controlled local arithmetic source by combining $E_v = \text{defect}_v + M_v$ with $\text{defect}_v = \delta_v$. This is now composed one layer lower from the formula-gap matched arithmetic-degree/p-adic source plus the synchronized Proposition 1.4/1.5 arithmetic-degree comparison source, and that comparison source is now projected from the local Step (v)/(vii) degree-identification formula source. The Record-Ob3/Ob5 arithmetic-divisor component route now composes this with its own local-degree projection, so the p-adic split can be constructed from component data plus valuation-ball p-adic-defect/main provenance. The current lower Record-Ob3/Ob5 source consumes that valuation-ball p-adic-defect/main object directly, so the valuation-ball evaluation source, the Haar-index defect source, and the equality $E_v = \delta_v + M_v$ are projected rather than supplied independently. Finite sums, nonnegativity, formula-gap construction data, and arithmetic-divisor calibration are read from the valuation-ball local-estimate package rather than restated on the p-adic split record. The arithmetic-degree and formula-gap formula sources consume this constructed split before projecting to the canonical p-adic Step (xi) formula boundary. Thus the remaining trust boundary is the derivation of the arithmetic-degree-controlled local arithmetic source from the full IUT IV local estimate construction, not the Haar-defect lower bound or an independent synchronization record. The same case split has now been lifted to the same-index signed boundary by ConcretePacketProjectedPrincipalProductSameIndexWeightedDeterminantQNormalizedCaseBoundedResidualSou It keeps the determinant and theta-equality fields of the pointwise same-index route, but its residual payload is the valuation-ball Theorem 1.10 case split itself. Lean projects this source to the pointwise residual source, then to the q-normalized local-analytic source, where it constructs the signed C_Θ bound object. The named case-bounded public endpoints now expose both the signed Corollary 3.12 dichotomy and the companion $(-1 : \mathbb{R}) \leq C_\Theta$ lower bound, so the pointwise residual package is no longer the last public same-index residual input on that route. The same route is now pinned by endpoint-level evidence and public-boundary inventory audits: Lean records that the case-bounded valuation-ball source first constructs the pointwise residual source, then the defined residual and q-normalized comparison data, before reading off the dichotomy. The boundary has now been lowered once more by the source ConcretePacketProjectedPrincipalProductSameIndexWeightedDeterminantQNormalizedLocalArithmeticDegreeSo it has the same same-index determinant and theta-equality fields as the formula-gap source, but replaces the raw residual inequality field by the packaged local arithmetic-degree residual source. Lean constructs the older formula-gap source and hence the local-analytic q-normalized route from this local-degree source, and the same namespace exposes the resulting signed Corollary 3.12 dichotomy directly from that local-degree source. Its route audit now also exposes, from the same local-degree source, the constructed q-normalized endpoint, localized C_Θ audit, signed IUT IV upper bound, and dichotomy. The named preferred public endpoint now has the corresponding local-degree version, so the signed endpoint can be invoked without presenting the older local-analytic source as the public input. There is also a side-condition-sourced local-degree endpoint: Lean rebuilds the named side-hypothesis record from `IUTStage1SourceSideConditions` and returns both the signed dichotomy and the companion $C_\Theta \geq -1$ bound from the same local-degree source, using the formula-gap projection for the dichotomy and the scale-synchronized projection for the lower bound. The next wrapper removes that packaged local-degree source from the public surface: it takes the three weighted determinant component identifications, the q-normalized theta/log-volume identity, the Theorem 1.10 formula-gap valuation-ball source, and the local arithmetic-degree residual payload, then constructs the local-degree source internally before proving the same two outputs. The source-gap form specializes this expanded route at `IUTStage1SourceObligationGap.sideConditions`, and the possible-image witness refinement now constructs the packet-indexed `WithSymmetryLabelTransport` from the Theorem 3.11 possible-image witness, equality-quotient possible-image family, and the three fiber transport laws before entering that same local-degree route. This removes a preassembled symmetry-label wrapper from this branch, while the derivation of the witness and transport laws from the full log-theta procession remains a paper-level source obligation. The same possible-image construction now also feeds both scalar-lowered branches: the family-hull theta route, whose scalar input is the family-hull log-volume comparison rather than the direct q-normalized adjusted-sum identity, and the record-family-hull q-scale route, where determinant/Hodge calibration derives the unscaled record-family hull equality before the endpoint consumes only q-scale compatibility with the principal-product family hull. Neither branch requires a preassembled symmetry-label wrapper at this public boundary. Both possible-image scalar branches now have compact-open log-Kummer image variants: the family-hull theta branch enters the theta/log-volume comparison after deriving compact-open realized exactness from image laws, and the record-family-hull q-scale branch does the same before the q-scale comparison. Both possible-image scalar branches now also have correspondence-level variants: Lean extracts image laws from the compact-open log-Kummer correspondence source before entering the theta/log-volume or q-scale comparison. The weighted local-degree possible-image branch has also been lowered below the compact-open realized-exact source: Lean now accepts realized valuation-ball exactness and reconstructs compact-open realized exactness using the local Haar-normalization identity compactOpenSubset = valuationBall(compactOpenRadius) before applying the same possible-image source-gap endpoint. A further log-shell refinement lowers this input once more: Lean accepts the two paper-facing equalities $\Theta_{\mathrm{region}} = \mathrm{realized}(\mathrm{logShell})$ and logShell = valuationBall, composes them into realized valuation-ball exactness, and then follows the compact-open route above. The p-adic unit-ball Step (xi) synchronization source now projects directly to that local arithmetic-degree residual payload: the Haar defect is the p-adic unit-ball index $p_v^{[K_v:\mathbb{Q}_{p_v}]}$, and Lean obtains the generic residual endpoint by composing the p-adic Haar-index residual, pointwise Haar-defect residual, and summed local arithmetic-degree residual constructors. Thus, on this slice, the residual package is no longer the primitive mathematical assumption; the local Step (xi) synchronization is now the next boundary to lower. This synchronization has now also been lowered to a p-adic arithmetic formula matching source. Instead of taking the synchronized upper-contribution identity directly, the new constructor takes the two local equalities $\mathrm{StepXI}_v = a_v(D_v + C_v)$ and $E_v = p_v^{[K_v:\mathbb{Q}_{p_v}]} + M_v$. Since the local arithmetic upper contribution is definitionally $a_v(D_v + C_v) + E_v$, Lean reconstructs the p-adic Step (xi) synchronization and the expanded Haar-residual identity from these formula pieces. The new bridge from the formula-gap matched valuation-ball arithmetic-degree/p-adic source maps the existing source-backed formula tower into this canonical constructor, so the remaining local-estimate obligation is now the underlying determinant calibration and prime-error split, not a separately supplied Step (xi) synchronization record. The calculation is now exposed directly as well: Lean proves that the local defined residual is exactly the p-adic unit-ball index and then sums these identities to obtain the total residual Haar lower bound used by the C_Θ comparison. The arithmetic formula-matching source now projects directly to the generic local arithmetic-degree residual package, so the canonical residual consumer can use the two formula equalities without exposing the intermediate p-adic Step (xi) synchronization source. The additive-Haar bridge now composes this residual projection through the source-backed formula tower itself: the p-adic prime-error split, arithmetic-degree calibration, formula-gap matching, arithmetic-divisor-backed local-degree component, constructed Theorem 3.11 local-term source, and valuation-ball arithmetic-degree controlled local source all project directly to the same residual package. This has now been composed one layer lower: the Record-Ob3/Ob5 valuation-ball local-estimate source itself projects directly to the canonical residual package, and the component Record-Ob3/Ob5 source does so once its valuation-ball formula provenance is supplied. A lower component variant now reconstructs that valuation-ball formula-gap source internally from the arithmetic-degree-controlled local arithmetic source before projecting to the residual package. A still lower checked component variant constructs that controlled source internally from the Theorem 1.10 local-analytic ledger, the p-adic Haar-index defect source, arithmetic-degree calibration, and the projected Step (v)/(vii) comparison bounds. This route now factors through the canonical p-adic arithmetic formula-matching package, so Lean exposes the reconstructed $StepXI_v = a_v(D_v + C_v)$ and $E_v = \delta_v + M_v$ laws before passing to the residual package. The component-level p-adic defect/main projection now lowers to the same arithmetic-degree-controlled local arithmetic source: Lean reconstructs the valuation-ball formula package internally before applying the local-degree projection. For the ordinary component route this still leaves the component-to-controlled-formula synchronization at that boundary, but the new controlled Record-Ob3/Ob5 component source removes that equality as a separate input: its arithmetic-divisor local-degree component is assembled over the formula-gap package reconstructed from the controlled valuation-ball local arithmetic source, and the projection audit checks that the resulting Record-Ob3/Ob5 valuation-ball source carries this formula object definitionally.

The p-adic Haar controlled component source lowers this boundary once more: it stores the Theorem 1.10 valuation-ball arithmetic-divisor ledger, the p-adic unit-ball Haar-index defect, the arithmetic-degree calibration, and the projected Step (v)/(vii) comparison bounds, then reconstructs the controlled local arithmetic source internally before projecting to the Record-Ob3/Ob5 valuation-ball source. Thus the controlled local arithmetic package is no longer an independent input at this strict component layer. This strict source now also feeds the downstream p-adic defect/main local-estimate source and canonical local arithmetic-degree residual package directly, without reintroducing the older local-arithmetic or formula-synchronization arguments. The strict p-adic-Haar route now also proves that its p-adic-defect/main valuation-ball local-estimate endpoint is the canonical reconstruction from the arithmetic-degree-controlled local source, and the remaining p-adic-defect/main Record-Ob3/Ob5 source preserves this reconstruction through the strict p-adic-Haar projection. The new reconstruction audit bundles this with the valuation-ball endpoint, the reconstructed arithmetic-degree source, and the carried controlled Record-Ob3/Ob5 component, so this strict layer has a single checked source-reconstruction certificate. It also records the case-local Theorem 1.10 inequalities actually consumed by the corridor: the distinguished Step (v) exact-procession bound, the nondistinguished Step (vi) zero bound, the archimedean Step (vii) metric-container bound, and their recombination as the all-place case-procession arithmetic-degree inequality. At the local-degree formula comparison layer, Lean now proves that the p-adic-defect/main valuation-ball local-estimate source is exactly the object obtained by first reconstructing the arithmetic-degree-controlled local source. Thus the p-adic-defect/main local-estimate package is not an additional datum at that comparison boundary. The same layer now proves the nondistinguished Step (vi) zero-gap estimate `Iut.Stage1.Experiments.IUTStage1AdditiveHaarTheorem110StepXILocalDegreeIdentificationFormulaSour` so the local-degree audit exposes the distinguished, nondistinguished, and archimedean cases before the p-adic-defect/main projection. This now rests on the lower valuation-ball construction theorem `Iut.Stage1.IUTStage1IUTIVTheorem110ValuationBallAdditiveHaarLocalAnalyticConstructionFormulaSour` which proves the same Step (vi) zero-gap law before additive-Haar projection. It also proves the corresponding projection preservation theorem: `Iut.Stage1.Experiments.ConstructedTheorem311OneSidedIUTIVTheorem110AdditiveHaarMatchedLocalDegre` identifies the valuation-ball additive-Haar evaluation object projected from the reconstructed p-adic-defect/main source with the original valuation-ball object carried by the arithmetic-degree-controlled source. The new lowering `Iut.Stage1.Experiments.ConstructedTheorem311OneSidedIUTIVTheorem110AdditiveHaarMatchedLocalDegre` uses this preservation together with `Iut.Stage1.Experiments.IUTStage1AdditiveHaarTheorem110PadicDefectMa` to construct the canonical p-adic-defect/main controlled Record-Ob3/Ob5 component source from the local-degree formula comparison source. The corresponding formula-gap/comparison constructor also now proves `Iut.Stage1.Experiments.IUTStage1AdditiveHaarTheorem110PadicDefectMainValuationBallLocalAnalyticA` namely that reconstruction through the controlled local-arithmetic source preserves the arithmetic formula-matching source consumed by the local-degree comparison, using the supplied valuation-ball/comparison synchronization. The `Iut.Stage1.Experiments.ConstructedTheorem311OneSidedIUTIVTheorem110AdditiveHaarMatchedLocalDegre` lowering now constructs that local-degree formula comparison source from the synchronized controlled component source itself; its bridge projection `Iut.Stage1.Experiments.ConstructedTheorem311OneSidedIUTIVTheorem110AdditiveHaarMatchedLocalDegre` therefore reaches the canonical residual package without separately exposing the local-degree formula package. The matched-object variant `Iut.Stage1.Experiments.ConstructedTheorem311OneSidedIUTIVTheorem110AdditiveHaarMatchedLocalDegre` lowers this comparison layer once more: the public source carries the typed distinguished and archimedean Step (v)/(vii) local-degree identification objects and their valuation-ball/additive formula synchronization, while Lean projects the local-degree formula source and formula-matching equality internally before entering the p-adic-defect/main controlled component route. The same matched-object source now also supplies `Iut.Stage1.Experiments.ConstructedTheorem311OneSidedIUTIVTheorem110AdditiveHaarMatchedLocalDegre` so the case-local Theorem 1.10 pointwise bound and the residual package share the object-backed Step (v)/(vii) handoff. The compact-open route `Iut.Stage1.Experiments.ConstructedTheorem311OneSidedIUTIVTheorem110AdditiveHaarMatchedLocalDegre` now consumes this matched-object source directly and is audited against the older local-degree formula route after object-to-formula projection. The same matched-object source is now projected from the controlled Record-Ob3/Ob5 component source itself: Lean reads the Step (v)/(vii) object data from the arithmetic-divisor component chain and reconstructs the valuation-ball formula source from the controlled local arithmetic package. The controlled-component compact-open route then consumes that lower source directly and factors definitionally through the matched-object route. The countermodel `Iut.Stage1.IUTStage1StepXIDependencyAudit.ValuationBallProjectionWithoutPreservationToyCountermo` records why this preservation cannot be silently omitted. The remaining construction task is therefore higher: derive the synchronized controlled component/local-estimate layer itself, including its finite sums, nonnegativity, component formula synchronization, and arithmetic-divisor calibration, from the full IUT IV local-estimate data. The processional local-estimate and processional p-adic unit-ball Haar sources now project to that same residual package as well, so the residual consumer can be fed directly from the distinguished exact-procession, nondistinguished zero, archimedean metric-container, and p-adic unit-ball Haar-index data. At the p-adic unit-ball/local-analytic boundary, Lean now also proves endpoint contracts for the two projected sources: the valuation-ball local-analytic localized source and the expanded p-adic Haar-residual source are both reconstructed from the p-adic unit-ball localized source and the matching local arithmetic-evaluation ledger. The localized source itself also proves the same residual arithmetic directly: its local defined residual is the p-adic unit-ball Haar index, and summing these identities recovers the residual Haar lower bound at this boundary. A matching weakened-boundary countermodel shows why this is not a cosmetic replacement: if the synchronized index is allowed to be arbitrary, the Step (xi) identity and residual/index equality can both hold with zero residual, so the p-adic normalized-place lower law is the source-backed input that rules out the failure. The lower defect/main split has its own diagnostic: $E_v = \mathrm{defect}_v + M_v$ does not imply $E_v = \delta_v + M_v$ unless the IUT IV arithmetic defect is identified with the p-adic Haar defect. A sharper lower-boundary countermodel keeps the Theorem 1.10/p-adic Haar-index and arithmetic-degree reconstruction equations true while detaching the Record-Ob3/Ob5 component formula from the reconstructed p-adic Haar formula. A final strict-boundary diagnostic shows that this detachment reaches the new consumer surface: the component residual and reconstructed p-adic residual can differ unless the controlled source carries the formula synchronization definitionally. The same audit now records the analogous Ob3/Ob4 determinant-shadow failure: matching the weighted determinant shadow and finite-sum scale does not identify the full Record-Ob3/Ob4 source, since the anchor/localization payload may still differ. Consequently the full `recordOb3Ob4_eq_stepXI` synchronization remains a genuine Remark 3.9.5 source obligation until derived from the hull/determinant construction. Lean now also records the positive criterion: equality of the localization family, anchor, and positive tensor-power data implies equality of the full Ob3/Ob4 adjusted determinant source, via `IUTStage1Remark395Ob3Ob4AdjustedDeterminantSource.ext_of_localization_anchor_tensorPower`. It also records the projection from that source-level synchronization to the older weighted deter-

minant summand, anchor, and tensor-power equalities consumed by the current canonical route, via `IUTStage1Remark395Ob3Ob4AdjustedDeterminantSource.weightedDeterminant_summand_anchor_tensorPower`
The full adjusted-determinant q-normalized source now uses that projection to enter the projected-weighted local-analytic route without adding the three component fields as separate same-index assumptions; the same-index local-arithmetic route now has the corresponding constructor `ConcretePacketProjectedPrincipalProductSameIndexWeightedDeterminantQNormalizedLocalArithmeticDeg`
The Record-Ob3/Ob5 component source itself now has the same lowering: `RecordOb3Ob5ArithmeticDivisorBackedComponentSource.ofConcretePossibleImageWitnessSourceLocalizat`
derives its `recordOb3Ob4_eq_stepXI` field from the localization, anchor, and positive tensor-power equalities before entering the existing component constructor. The valuation-ball Record-Ob3/Ob5 source now composes that component constructor with the valuation-ball formula source, so the local Theorem 1.10 valuation-ball boundary also no longer requires the full `recordOb3Ob4_eq_stepXI` equality when those three primitive synchronizations are supplied. On the constructor-built branch, the corresponding valuation-ball constructor now also projects the `determinantSource = ob3ob4Source.toWeightedDeterminantSource` handoff from the Remark 3.9.5 constructor-built source generated by the same Ob3/Ob4 package, rather than taking that determinant equality separately. Its family-hull variant additionally replaces the direct q-normalized adjusted-sum field by the family-hull theta/log-volume comparison, and the record-family-hull q-scale refinement derives the unscaled theta equality from the pointwise determinant/Hodge calibration. The q-scale compatibility then follows from the adjusted-sum q-normalization and the p-adic finite hull-decomposition theorem, and the corresponding public side-condition endpoint threads this refinement into the signed dichotomy and $C_\Theta \geq -1$ outputs. The adjusted-determinant source itself is now constructible from the three primitive Ob3/Ob4 synchronizations by `ConstructorBuiltPrincipalProductPadicFiniteAdjustedDeterminantQNormalizedLocalizedLocalAnalyticR`
The localization-family synchronization has also been pushed to pointwise localized vector-bundle data: `IUTStage1StructureSheafAdjustedLocalizedVectorBundleSource.toAdjustedLocalizationSource_eq_`
identifies adjusted localization entries from the underlying bundle, structure-sheaf term, raw adjusted log-volume, and weight, while `IUTStage1Remark395Ob3Ob4LocalizedVectorBundleDeterminantSource.toAdjustedDeterminantSource_eq_of`
promotes pointwise localized vector-bundle equality, anchor equality, and positive tensor-power equality to equality of adjusted-determinant sources. The lower theorem `IUTStage1Remark395Ob3Ob4LocalizedVectorBundleDeterminantSource.toAdjustedDeterminantSource_eq_of`
replaces the localized-entry equality by equality of the concrete Ob3-1 component fields: localized bundles, structure-sheaf log-volumes, raw adjusted log-volumes, and weights, plus anchor and positive tensor power. The lower theorem `IUTStage1Remark395Ob3Ob4LocalizedVectorBundleDeterminantSource.toAdjustedDeterminantSource_eq_of`
opens the localized bundle into local-ring and direct-summand log-volume data. The norm-square theorem `IUTStage1Remark395Ob3Ob4NormSquareLocalizedVectorBundleDeterminantSource.toAdjustedDeterm`
performs the corresponding projection from direct-summand norm families. The additive-Haar theorem `IUTStage1Remark395Ob3Ob4AdditiveHaarCompactOpenNormSquareLocalizedVectorBundleDeterminantSou`
lowers this further to componentwise equality of compact-open Haar normalization records, deriving the raw adjusted log-volume equality from the Haar sum formula. The unit-ball and p-adic finite-extension variants `IUTStage1Remark395Ob3Ob4UnitBallValuationHaarCompactOpenNormSquareLocalizedVectorBundleDetermina`
and `IUTStage1Remark395Ob3Ob4PadicFiniteExtensionUnitBallCompactOpenNormSquareLocalizedVectorBund`
project the same criterion through valuation-unit-ball and finite-extension unit-ball local-field data. The finite-extension branch has also been lowered one step further: `IUTStage1Remark395Ob3Ob4PadicFiniteExtensionUnitBallCompactOpenNormSquareLocalizedVectorBundleDe`

derives the projected additive-Haar componentwise matching from equality of the underlying finite-extension Haar normalization factors, together with the structure-sheaf, weight, anchor, and tensor-power synchronizations. The finite-extension factor equality is itself now reduced to componentwise matching of the finite-extension Haar payload–integer unit-ball source, realization, realized-region map, compact-open radius, and Haar measure–by `IUTStage1PadicFiniteExtensionProperUltrametricHaarNormalizationSource.ext_of_components` and the determinant theorem `IUTStage1Remark395Ob3Ob4PadicFiniteExtensionUnitBallCompactOpenNormSquar` This criterion is now lower again at the constructed base-prime dilation/mass source: componentwise equality of the constructed local-field payloads projects to the finite-extension proper-ultrametric componentwise equality via `IUTStage1PadicFiniteExtensionConstructedDilationMassHaarNormalizationSource.ComponentwiseEqual.t` and the determinant theorem `IUTStage1Remark395Ob3Ob4PadicFiniteExtensionUnitBallCompactOpenNormSquar` then carries this lower source criterion to the Ob3/Ob4 adjusted-determinant synchronization. The determinant criterion is now lower at the unit-ball Haar-character source as well: componentwise equality of local sources carrying $\mu(O_v) = 1$ and $\mu(p_v O_v) = p_v^{-[K_v:\mathbb{Q}_p]}$ projects through the Haar-character, $\mathbb{R}_{\geq 0,\infty}$-modulus, constructed dilation/mass, and proper-ultrametric layers by `IUTStage1PadicFiniteExtensionUnitBallHaarCharacterNormalizationSource.ComponentwiseEqual.toConst` and the determinant theorem `IUTStage1Remark395Ob3Ob4PadicFiniteExtensionUnitBallCompactOpenNormSquar` then feeds the same Ob3/Ob4 adjusted-determinant synchronization. The same criterion has now been lowered through the quotient-coset source: Lean derives the residue-coset partition from the finite additive quotient $O_v/p_v O_v$, projects componentwise equality to the unit-ball Haar-character source via `IUTStage1PadicFiniteExtensionUnitBallQuotientCosetHaarCharacterNormalizationSource.Componentwise` and applies `IUTStage1Remark395Ob3Ob4PadicFiniteExtensionUnitBallCompactOpenNormSquareLocalizedVect` This now descends to the quotient-native residue-module source as well: the finite residue quotient $O_v/p_v O_v$ carries its residue-field module structure, Lean computes the quotient cardinality and representatives, projects to the unit-ball Haar-character layer, and applies `IUTStage1Remark395Ob3Ob4PadicFiniteExtensionUnitBallCompactOpenNormSquareLocalizedVectorBundleDe` It now descends one layer further to the residue-submodule source: the residue-field action is carried before quotienting on the additive group of O_v, the $p_v O_v$ image is represented by a submodule, Lean transports the quotient module to $O_v/p_v O_v$, and applies `IUTStage1Remark395Ob3Ob4PadicFiniteExtensionUnitBallCompactOpenNormSquareLocalizedVectorBundleDe` The same residue-submodule source now has a direct unit-ball Haar-character endpoint: after transporting the quotient module across $O_v/p_v O_v$, Lean recovers $\mu(O_v) = 1$, $\mu(p_v O_v) = p_v^{-[K_v:\mathbb{Q}_p]}$, the additive Haar-character scalar, and the all-subset base-prime dilation law without reintroducing the quotient-native residue-module object as a separate source. This lower local object now also feeds the finite-place IUT IV Haar-defect boundary: a new residue-submodule p-adic Haar-index source carries these local submodule sources at every finite place and constructs the existing unit-ball Haar-index defect source by projecting each place to its Haar-character normalization. The same lowered source now carries the relative nonvacuity audit and the public-boundary inventory audit directly: Lean constructs the old local-analytic source internally before proving that the q-normalized route, localized C_Θ audit, signed bound, and dichotomy are all obtained without extra public localized-C_Θ, raw signed-bound, hull, or calibration inputs. The same evidence is exposed at the named local-degree public endpoint by endpoint-level route and public-boundary audits, so downstream uses of the public theorem do not need to open the same-index source namespace or invoke the older local-analytic public name to recover the constructed q-normalized handoff, localized C_Θ audit, signed IUT IV bound, relative nonvacuity witness, and dichotomy. The public-boundary inventory records that localized C_Θ, raw signed comparison, principal hull, theta-region principal source, measure calibration, and summand calibration are not separate public inputs on this route. The remaining boundary is therefore the construction of the underlying same-index Hodge-theater/Frobenioid/log-Kummer source package. A further source-hypothesis refinement removes the direct side-condition package from this public boundary: the route now consumes source-labeled q-pilot positivity and normalization hypotheses and constructs the Stage 1 side-condition record internally. The case-bounded same-index boundary now has the corresponding source-obligation-gap form. Its public dichotomy and $C_\Theta \geq -1$ endpoints project the q-pilot positivity and normalization from ‘IUTStage1SourceObligationGap.sideConditionHypotheses‘ before entering the valuation-ball case split. Thus the strongest case-bounded public surface no longer names an anonymous side-hypothesis record, although the paper-level derivation of the source-obligation gap itself remains part of the lower Theorem 3.11 construction boundary. The q-normalized local arithmetic-degree route now has the same source-gap lowering at the packaged, expanded constituent, family-hull theta, and record-family-hull q-scale surfaces, and the same-index scale-synchronized local-analytic/formula-gap lower-comparison companions now have matching source-gap endpoints. The projected-principal-product q-normalized formula-gap source pair and the projected formula-gap constituent/residual endpoints now have the same lowering, including the direct residual-Haar, local-degree residual, and case-bounded residual forms. In each case Lean projects q-pilot positivity and normalization from ‘IUTStage1SourceObligationGap‘ before proving the signed dichotomy and the $C_\Theta \geq -1$ companion bound. This removes the raw side-condition record from these projected formula-gap surfaces; the remaining work lies in deriving the source-obligation gap and the determinant/IUT IV comparison inputs from paper-level data. The projected constituent case-bounded surface has also been lowered through the concrete packet: Lean constructs that source-obligation gap from ‘packet.primitiveConstructor.constructedBundle.toStructuredInputsWithSHE‘ and the Stage 1 side conditions before invoking the case-bounded signed and $C_\Theta \geq -1$ endpoints. Thus the case-bounded route no longer requires an assembled source-obligation gap as a public input at this boundary; the remaining sign/normalization input is the Stage 1 side-condition package. The hull/determinant-obligations variant now projects that package from ‘IUTStage1SourceHullDetObligations‘, so the projected case-bounded route can use the same source object for Remark 3.9.5 hull/determinant provenance and for q-pilot positivity/source normalization. Lean also provides the corresponding constructed-source specialization for the heterogeneous projected formula-gap constituent: the endpoint consumes a constructed Remark 3.9.5 holomorphic hull/determinant source directly and projects its ‘IUTStage1SourceHullDetObligations‘ before entering the same case-bounded comparison. A source-backed specialization now projects those side conditions from ‘IUTStage1SourceHullDetObligations‘: the public signed and $C_\Theta \geq -1$ case-bounded endpoints still build the source-obligation gap from the primitive constructed-SHE bundle, but they read q-pilot positivity and normalization from the hull/determinant obligation object. The remaining lower boundary for this route is therefore the construction of the source hull/determinant obligation package from the paper-level possible-image and hull data. This boundary has now been lowered once more for the same case-bounded surface: the constructed Remark 3.9.5 holomorphic hull/determinant source projects its Theorem 3.11 hull/determinant constructor and hence its ‘IUTStage1SourceHullDetObligations‘; the public signed and $C_\Theta \geq -1$ endpoints consume that constructed source directly. The same endpoint has a stricter principal-product HDD variant: the public route may now consume the Theorem 3.11/Remark 3.9.5 principal-product HDD source and read the source-obligation gap from that source's ‘sourceGap‘ field. A further internal-principal-HDD variant reconstructs this principal HDD source from the compact-open theta-region, the pointwise determinant formula calibration, and the source side hypotheses before entering the same case-bounded endpoint. The pointwise determinant calibration boundary is now lower: Lean constructs it from the pre-ledger volume/hull identity and the summand-charted Hodge family-hull calibration, then derives the determinant log-volume, normalized determinant, and family-hull consequences by the finite adjusted-summand theorem. The bundled same-index case-bounded source is now constructible from the same-index formula-gap source together with the explicit valuation-ball Theorem 1.10 case residual source: determinant synchronization, theta equality, and the formula-gap ledger are read from the formula-gap source, while the distinguished, nondistinguished, archimedean, and distinguished +1 residual comparisons are supplied by the local case source. The same constructor boundary has now been lowered below the formula-gap ledger by ofLocalAnalyticIUTIVSourceCaseBoundedResidualSource: determinant synchronization and theta equality remain explicit at this stage, but Lean constructs the formula-gap matched Theorem 1.10 source from the valuation-ball additive-Haar local-analytic arithmetic-divisor source before assembling the bundled case source. At the localized C_Θ layer the corresponding case estimates can now be supplied directly over the local-analytic valuation-ball source, and the same lowering has now been composed through the same-index case-bounded constructor ofLocalAnalyticIUTIVSourceLocalAnalyticCaseSourceResidualSource. Thus the bundled same-index source can be assembled from determinant synchronization, theta equality, the valuation-ball local-analytic IUT IV source, and case inequalities stated over that same source; Lean constructs the formula-gap case residual internally before entering the older same-index case-bounded route. The signed endpoint wrapper dichotomy_ofLocalAnalyticIUTIVSourceLocalAnalyticCaseSourceResidualSource now composes this constructor directly into the internal-principal-HDD Corollary 3.12 route, so the signed public boundary no longer needs the formula-gap case source on this branch. Its companion cTheta_ge_neg_one_ofLocalAnalyticIUTIVSourceLocalAnalyticCaseSourceResidualSource now proves the same branch's $C_\Theta \geq -1$ lower bound from the identical local-analytic source boundary. The case-bounded source now also has the projection toLocalAnalyticSource: it constructs the older same-index local-analytic source by projecting the local-analytic ledger from the formula-gap valuation-ball source and proving the total Haar residual inequality from the case-bounded pointwise residual source via the local arithmetic-degree residual theorem. Thus the explicit valuation-ball case split is known to imply the local-analytic endpoint route, rather than merely running in parallel to it. The remaining lower boundary has also been composed into signed and $C_\Theta \geq -1$ endpoints: the public route can now consume the formula-gap source and the explicit local case residual source directly, construct the bundled case source internally, reconstruct the principal HDD source internally, and then enter the case-bounded comparison. The scale-synchronized case-bounded route now removes the raw theta equality from this boundary, and the local arithmetic-degree residual constructor removes the separate total Haar residual field from the scale-synchronized local-analytic source. The weighted-determinant refinement now composes this constructor into the signed and $C_\Theta \geq -1$ endpoints: Lean derives the determinant-source equality from the summand, anchor, and positive tensor-power identifications before entering the same local-degree route. The canonical manifest now records the q-normalized local-arithmetic-degree route: the older scale-synchronized case-bounded source is no longer the canonical endpoint. The q-normalized theta/log-volume identity remains a public source obligation, but Lean uses it with q-pilot positivity to derive the canonical C_Θ-scale synchronization internally. The same scale-synchronized branch now has a local-analytic case-source form: the public IUT IV input is the valuation-ball local-analytic Theorem 1.10 source together with the local-analytic case split, and Lean constructs both the formula-gap matched source and the formula-gap case residual internally before invoking the signed and $C_\Theta \geq -1$ endpoints. The q-normalized local arithmetic-degree source now also feeds the same-index case-bounded boundary directly: Lean projects the formula-gap source from the local-degree object, attaches the explicit Theorem 1.10

valuation-ball case residual source, and then invokes the internal-principal-HDD route. Thus this case-bounded endpoint inherits the adjusted-determinant synchronization and record-family-hull q-scale lowerings already present in the local-degree source, instead of restating the weighted determinant component equalities at the public boundary. The source-obligation-gap version now composes one layer further: the endpoint takes the adjusted-determinant synchronization, record-family q-scale comparison, formula-gap source, local arithmetic-degree residual package, and explicit case residual, constructs the local-degree source internally, and returns both the signed dichotomy and the $C_\Theta \geq -1$ bound. The strongest same-index case-bounded surface has since been lowered by one more scalar step: the corresponding endpoint takes the projected localized Step (xi) adjusted-sum q-normalization directly, constructs the q-normalized local arithmetic-degree source from that identity and the adjusted-determinant synchronization, and then attaches the same explicit Theorem 1.10 case residual. Hence the record-family q-scale equality is now a derived intermediate on this branch, not a public input. The same boundary has also been tightened on the IUT IV residual side: the case-bounded valuation-ball source now supplies the local arithmetic-degree residual package internally via its residual projection theorem, so the adjusted-sum endpoint no longer exposes that residual package as a separate public input. This has now been lowered to the procession-bound form on the same branch: Lean reconstructs the expanded local case source from the projected Theorem 1.10 procession upper bound and the distinguished +1 comparison. The same endpoint also accepts the procession-comparison source, separating the Step (xi) identification with procession-normalized local log-volume from the upper-semi procession estimate before reconstructing the procession-bound source internally. It now also accepts the processional-estimate source, deriving the upper-semi comparison from the local Step (v), Step (vi), and Step (vii) estimates before reconstructing the procession-comparison source internally. The case-defined processional-estimate endpoint fixes the processional local value to the valuation-ball local-kind split itself and reconstructs the general processional-estimate source internally. The p-adic-normalized case-defined processional-estimate endpoint fixes the distinguished nonarchimedean place as the normalized p-adic unit-ball Haar-defect place before projecting to the ordinary case-defined source. The same adjusted-sum branch now also has a residual-level local arithmetic-degree handoff: Lean consumes the local arithmetic-degree residual package directly, without exposing a case-bounded or processional-estimate residual source. The p-adic-normalized arithmetic-gap endpoint uses this handoff to lower the normalized branch below the processional-estimate assumption: the p-adic source carries the Step (v)/(vi)/(vii) case values and the arithmetic-minus-main +1 gap at the normalized Haar-defect place, and Lean projects that data to the local arithmetic-degree residual package before entering the adjusted-sum comparison. This branch has now been lowered once more at the residual consumer: the p-adic unit-ball Haar-defect residual source alone projects to the canonical local arithmetic-degree residual package, so the adjusted-sum handoff no longer exposes the case-defined processional arithmetic-gap package when only the residual ledger is needed. The same p-adic arithmetic-gap branch also has a hull/determinant-obligation-backed form: Lean derives the source-obligation gap from IUTStage1SourceHullDetObligations, so this lower endpoint does not expose an additional assembled source-gap object once the Remark 3.9.5 hull/determinant obligations are present. What remains here is to derive the primitive Ob3/Ob4 localization, anchor, and positive tensor-power synchronizations, the q-normalized theta/log-volume identity, and this p-adic Haar-defect residual package from the concrete theta-region/Hodge-calibrated possible image data and the full IUT IV local estimates. The local-upper refinement defines the local Haar defect as the residual between the IUT IV local arithmetic-upper contribution, the localized Step (xi) determinant term, and the main-log summand; the expanded $StepXI_v + Haar_v = a_v(D_v + C_v) + E_v - M_v$ equality is then derived from the definition of the IUT IV local arithmetic-upper contribution. Conversely, the same-index theorem above uses this expanded local identity together with the total Haar lower bound to recover the formula-gap residual inequality. The localized-data constructor then constructs the localized C_Θ source itself from the localized Ob3/Ob4 determinant source, Haar-normalization defects, main-log contributions, and finite-place summation identities before running the same route audit. These variants remove the record Ob3/Ob5 determinant source from five higher source-spine entry points; the remaining explicit inputs at this boundary are the localized family identification, the exact theta/family-hull calibration, and the hull/determinant obligations synchronization. There is now a stricter constructor-built variant of this source-spine route: the rebuilt source-spine theta record is definitional on the Theorem 3.11 image family, and a constructor-built possible-image Step (xi) source supplies the record-canonical hull/determinant bridge, q-region, q-positivity, normalization, Ob3–Ob6 determinant bridge, and the record-bridge Ob7 input without an intervening obligations synchronization. Its product-formula Ob7 variant now takes finite weighted determinant prime-strip data attached to that same constructor-built determinant source; Lean constructs the coric $\Theta^{\times\mu}$ invariant and proves the determinant/global prime-strip equality before entering the source-spine route. The obligations-backed version of this product-formula route now constructs the possible-image constructor-built source at the source spine's selected q-choice; the selected-q equality is therefore definitional on that path, and the remaining endpoint facts are projected from the constructed determinant source, the product-formula Ob7 source, and the single hull/determinant obligation synchronization. A stricter weighted-partition variant lowers this once more: from a prime-strip lift, summand-place map, finite weights summing to one, and local Gaussian volume calibration, Lean builds the determinant product-formula source internally before applying the same source-spine/action-packet route. This weighted-partition route now has the same selected-q, obligations-backed constructor path: the possible-image constructor-built source is assembled at the source spine's selected q-choice, and the endpoint records the constructed summand-place map, conversion ratio, global finite product formula, determinant/global equality, bridge synchronization, positivity, and normalization. A constructor-sourced variant removes the ambient 'IUTStage1SourceHullDetObligations' object from this selected weighted-partition path: it builds the selected possible-image constructor directly from the record-canonical bridge equality, q-positivity, and normalization, and proves that the obligations visible at the endpoint are definitionally the obligations produced by the legacy finite-route hull/determinant constructor. The bridge equality and side conditions are still explicit lower inputs on this variant. The exact-theta selected Step (xi) paper-trace constructor now also accepts this symmetry-label transport source directly and projects the older action-packet transport internally; the source-spine record constructor has the same symmetry-label entry point. The packet-facing completion route can now construct the packet-indexed symmetry-label wrapper from the possible-image witness and explicit symmetry-label, upper-semi, and procession transports, so that wrapper is no longer forced to appear as a public theorem argument at this surface. At the public paper-trace route boundary, the remaining payload audit has now been split so the caller supplies only the non-Step-(xi) payload; the legacy Step-(xi) audit fields are derived internally from the structured Step-(xi) source. A stricter concrete-localized additive-Haar wrapper now makes this split explicit at the normalized Haar-modulus boundary. This has now been tightened again: the public wrapper may take only the seven lower arithmetic/formula fields from Theorem 1.10/IUT IV, and a further wrapper derives those seven fields from the IUT IV C_Θ source ledger itself. The Step (x) realified-packet field is projected from the vertical log-Kummer package, the IPL/SHE/APT fields are projected from the Theorem 3.11 bridge package, and the Step (xi) arithmetic-degree, localized determinant, adjusted log-volume, and endpoint payload slots are constructed from the normalized finite-summand source-route audit, which explicitly carries the projected additive-Haar, Haar-modulus, and localized-adjusted Hodge-summand expansion boundaries. In the newest normalized additive-Haar route, local additive-Haar normalization, the Theorem 1.10 arithmetic-divisor split, the finite-place Haar defect, the arithmetic-gap estimate, and the local-to-global C_Θ comparison are read from IUTIVCThetaObligations, not supplied as a separate residual record. When a constructor-built Step (xi) hull/determinant source is present and its canonical scale is identified with the IUT IV finite-place scale, the residual endpoint is strengthened further: the final C_Θ comparison is threaded through the constructor-built Step (xi)/IUT IV local-to-global audit rather than through the generic ledger proposition alone. The newer constructor-coupled arithmetic source goes one layer lower: it constructs the IUT IV finite-place ledger with the constructor-built canonical scale as the finite-place canonical scale, so the public handoff no longer asks for a separate scale equality. Its remaining numerical input is the lower arithmetic gap inequality $C_{\Theta,\mathrm{can}} + 1 \leq \mathrm{upper} - \mathrm{main}$. The finite-place local-to-global constructor now lowers this numerical input: local canonical scales, arithmetic/log contributions, and Haar normalization defects are summed to derive the arithmetic gap before the coupled IUT IV ledger is constructed. This finite-place source now has its own endpoint, and the localized C_Θ handoff records that endpoint explicitly before projecting the final signed comparison. The localized local-term constructor lowers this one step further: the local canonical scales are the weighted adjusted log-volumes of the localized Ob3/Ob4 determinant source, and each local upper term is decomposed as Step (xi) log-volume plus Haar defect plus main-log contribution. The arithmetic-divisor-backed additive-Haar source now projects the reusable IUT IV local-analytic and Theorem 1.10 C_Θ data from its matched local-degree, formula-gap, and local-global audits; the projection itself now factors through the typed additive-Haar formula-gap arithmetic-divisor source, so the Proposition 1.4/1.5 local endpoints, formula-to-gap comparisons, and local different/conductor nonnegativity bounds are read from that lower source. The strongest paper-trace audit now uses the same projections instead of rebuilding those C_Θ records inline. After the constructor-built determinant is synchronized with the localized Step (xi) determinant, this source constructs the localized local-term C_Θ source and derives the residual-payload audit, the constructor-built arithmetic gap, and the final C_Θ comparison. The same arithmetic-divisor-backed matched local-degree source now also projects into the canonical p-adic unit-ball arithmetic formula-matching boundary. Thus the p-adic Step (xi) synchronization audit can be reached from the Theorem 1.10 arithmetic-divisor local-degree branch, rather than only from the formula-gap arithmetic-degree/p-adic bridge. The constructed Theorem 3.11 local-term C_Θ source now projects through that same boundary, so the bridge can be audited after the multiradial possible-image, q-pilot, determinant, and local-global handoff data have been assembled. This stronger residual handoff now also feeds the base normalized and non-normalized additive-Haar public route constructors, so those boundaries can be instantiated from the constructor-built Step (xi) source instead of reverting to the generic IUT IV residual. The same constructor-built handoff now lifts through the intrinsic finite-extension Haar-modulus normalized and finite-summand additive-Haar routes, and through the corresponding unit-ball Haar-character normalized and finite-summand wrappers. It now also reaches the residue-module/unit-ball normalized and finite-summand wrappers; the normalized residue-module route has the same constructor-coupled no-standalone-scale- equality endpoint. This CTheta-derived arithmetic boundary has also been lifted through the intrinsic finite-extension, unit-ball Haar-character, and residue-module/unit-ball normalized public routes, so those stronger p-adic endpoints can now use the same source-derived arithmetic payload. The direct non-normalized additive-Haar route also has finite-place and localized constructor-built C_Θ entry points, so it no longer has to pass through the finite-summand wrapper merely to consume localized Step (xi) local-term data. The unit-ball normalized wrapper has its own full-boundary audit, and the strongest residue-module/unit-ball wrapper now factors through it before projecting to the intrinsic Haar-modulus normalized route. Its audit combines the Step (xi) residue-module source audit, the transported residue-module factor handoff, the projected unit-ball normalized audit, the IUT IV arithmetic-residual audit, and the definitional equality with the projected normalized public route. The parallel non-normalized unit-ball Haar-character and residue-module/unit-ball Haar-character finite-summand routes now have the same CTheta-derived arithmetic handoff: their public wrappers project to the intrinsic Haar-modulus/additive-Haar finite-summand route, and their full audits record the Step (xi) source boundary, the projected finite-summand boundary, the IUT IV residual audit, and the resulting remaining arithmetic-payload audit. On the hull/determinant side, the Ob3/Ob5 compatibility record can now be constructed from the lower equality $\mu_{\log}(\varphi(P_B)) = \widehat{\deg}_{\text{norm}}$: the concrete quotient theta-pilot constructor builds the canonical hull approximant versus weighted-determinant compatibility internally from that equality. The exact-theta variant also uses $\theta_{\text{signed}} = \mu_{\log}(\varphi(P_B))$ to derive the Ob4 tensor-power bound, so that bound is no longer a separate input on this constructor path. The constructor-built possible-image Step (xi) source now has the same lowered entry point: from a Remark 3.9.5 hull operator, adjusted Ob3/Ob4 determinant data, and these two log-volume equalities, Lean reconstructs the Ob3/Ob5 compatibility and tensor-power bound internally before entering the record-canonical hull/tensor bridge. The minimal-hull-system and product-hull variants expose this path without a prebuilt hull shadow; the bridge equality, measure synchronization, q-positivity, and normalization remain explicit lower inputs. The product-hull exact-theta endpoint now packages the source endpoint, product-hull provenance, reconstructed compatibility, derived tensor-power bound, selected-product-hull containment, and the resulting signed comparison as checked consequences of that lowered constructor. An obligations-backed variant lowers the remaining side-condition surface: bridge equality, q-pilot positivity, and source normalization are projected from a single `IUTStage1SourceHullDetObligations` object. The explicit synchronization left at this boundary is now the equality between that obligations bridge datum and the record-canonical product-hull/tensor-power datum. The concrete-localized route wrapper takes the minimal-hull source, localized vector-bundle determinant decomposition, and $\Theta^{\times\mu}$ Ob7 source directly, constructs the paper-derived Step (xi) source internally, and then invokes the preferred route. A stronger hull-side wrapper now goes below the minimal-hull argument: it constructs the concrete hull source internally from the source-spine equality-quotient possible-image family and the principal $\lambda \cdot O$ hull source, then threads that hull source through the localized determinant and Ob7 route. This principal-product route now has a strict variant in which the determinant upper bound is derived from Θ_{signed} equal to the localized family-hull log-volume. The strict principal-product localized wrapper has now been promoted to the named public Step (xi) paper-source boundary. The older route theorems still exist as compatibility layers, but the advertised public audit no longer takes an already constructed Step (xi) source. A product-formula strengthening of this public audit now also constructs the localized $\Theta^{\times\mu}$ Ob7 source from the localized determinant source, local Gaussian summand calibration, finite determinant product formula, and coric unit-character data. Thus the public route can consume product-formula data in place of a prebuilt Ob7 compatibility record. At this localized Ob7 constructor boundary, the tensor-power upper bound is derived from $\Theta_{\text{signed}} = \mu_{\log}(\varphi(P_B))$ and the localized determinant comparison, rather than supplied beside the product-formula source. There is now a determinant-calibrated public product-formula theorem whose audit boundary uses this constructor directly; the older theorem remains as a compatibility layer for routes not yet migrated. The downstream raw-adjusted, common-place, diagonal restriction, and environment-theta public wrappers now use the same constructor, so the Step (xi-a)–(xi-g)/Ob7 trace is no longer backed by a separate inline tensor-power estimate along this wrapper ladder. The public audit now has a stricter raw-adjusted variant: it consumes raw adjusted-localization Gaussian calibration data, constructs the localized product-formula source internally, and then invokes the product-formula public route. Each determinant summand is assigned a Gaussian place and raw ratio, the partition weight is computed from the Ob3 localization weight times this ratio, and Lean derives the weighted summand calibration and global product-formula equality from the Gaussian local-global Frobenioid restriction identity. This has now been lowered once more: the public route also accepts a globally normalized raw-adjusted source in which the raw ratio is computed as adjusted raw log-volume divided by the nonzero Gaussian global log-volume. The Ob3 weighted determinant sum together with determinant/global prime-strip equality proves the finite normalization. The Ob7 prime-strip boundary has also been lowered: common environment and Gaussian prime-place coordinates plus one valuation-indexed coric character now construct the prime evaluation, $\Theta^{\times\mu}$ lift, and coric invariant internally. A further public wrapper constructs the local-global collection, its environment anchor, and the local environment/Gaussian evaluation from global-to-local restrictions and local object identifiers; the local realified objects are forced by the restricted global prime log-volumes. In the diagonal finite valuation-index route, the environment and Gaussian prime-place equivalences are identity maps constructed internally. The preferred non-vacuous public diagonal wrapper now uses the additive-Haar compact-open factor boundary over the principal product hull system, together with an audited Hodge–Arakelov theta evaluation, global-to-local restriction data calibrated at the Hodge canonical full-label Gaussian degree, and the scalar equality between the Hodge theta-monoid degree and the localized adjusted determinant sum. The environment/Gaussian realified Frobenioid evaluation is constructed internally from zero-divisor ordinary/theta/log-shell copies whose unit log-volume is the Hodge canonical degree; normalization proves that the Gaussian theta divisor log-volume is this Hodge canonical degree. The Hodge evaluation identifies the canonical degree with the theta-monoid degree. The finite Hodge summands are no longer supplied as a public family: they are fixed internally to be the localized weighted adjusted determinant contributions, and the scalar equality proves that their finite sum is the theta-monoid degree. The structure-sheaf log-volume is identified with a distinguished direct summand of each localized arithmetic vector bundle. Each direct-summand log-volume is projected from the normalized log-volume of a local compact-open tensor factor constructed from additive Haar data. The additive-Haar layer now has a strict factor constructor: from the source-level inequality $1 < \mu(U)_{\mathbb{R}}$ for the selected non-structure compact open, Lean proves positivity of $\log(\mu(U)_{\mathbb{R}})/[K : \mathbb{Q}_p]$, then uses the norm-square projection to derive the nonzero direct-summand norm. This generic interface has been instantiated by a concrete finite-extension local-field source: the selected factor is the inverse base-prime dilation $p^{-1}O_K$, whose compact-openness and Haar mass $p^{[K:\mathbb{Q}_p]} > 1$ are proved in the source core and threaded through the preferred additive-Haar public wrapper. The same wrapper can now consume a family-hull-indexed calibrated localized hull-cover source, deriving the old family-hull containment, log-volume summation, and pointwise localized-region determinant identities from the index-fiber presentation, finite additivity, and local vector-bundle calibrations. A stricter variant lowers the volume-summation input from a global finite-additive hull law to a cover-local log-volume measure model for the localized regions under construction. The newest direct-product variant also constructs the family-hull index from local factor regions: the direct-product cover identity gives existence of an index, and local factor separation proves the required fiber equality. A tensor-measure refinement constructs the cover-local measure model from direct-product cell log-volumes and the tensor-cell summation formula, so the direct-product public route no longer needs an independently supplied finite-cover measure model. A source-core refinement lowers this again: the tensor cells are now the localized determinant summands from the additive-Haar source-core decomposition, and the tensor-cell finite-sum formula is derived from the source-core localized log-volume theorem together with the direct-product cover identity. The remaining input at this boundary is the geometric identification of the localized regions with the direct-product local-factor cells and their cover of the principal-product family hull. A charted local-ring refinement lowers the factor interface itself: the local factors are no longer an arbitrary region family, but the regions of per-place local-ring/vector-bundle calibration objects whose chart log-volumes are identified with the corresponding direct-summand log-volumes. The cover identity and factor separation are still explicit geometric inputs. A valuation-ball charted refinement now lowers the factor calibration one step further: each local-ring chart is projected from a valuation-ball/additive-Haar factor calibration, and Lean derives the charted local-ring cover interface from the valuation-ball regions, their separation, and the valuation-ball family-hull identity. The p-adic finite-extension branch now has the same principal valuation-ball lowering: its direct-product cell identity, local-factor separation, and family-hull equality are projected from the SourceCore principal valuation-ball product-hull cover before entering the p-adic charted route. The p-adic branch also has principal-hull and hull-formation alignment refinements: the valuation-cover/public-hull equality is derived from the principal valuation-ball hull theorem, and the source-core and valuation-cover possible-region equalities are projected from the concrete `principalProductHullFormationData` family. The p-adic calibration boundary is also lowered: the localized calibration and per-factor valuation-ball calibration are transported from the principal valuation cover across the principal-hull equality, so they are no longer supplied independently at this level. The additive-Haar principal valuation-ball refinement lowers this again: the direct-product cell identity, local-factor separation, and family-hull equality are projected from the SourceCore principal valuation-ball product-hull cover. The selected valuation-ball branch now also transports exact possible-image $\Xi(P_B)$ witnesses across carrier equivalences: Lean proves this first for the holomorphic hull operator and possible-image family, then applies it to the radius-cover exact-Ξ source. A public adapter now aligns this transported source with the Theorem 3.11 quotient family using quotient-index matching, pointwise possible-region equality, and hull-operator equality; the public exact-Ξ log-volume calibration is then derived rather than supplied. The quotient-index matching has also been lowered to concrete choice-level data: a selected valuation-ball index attached to each Hodge-theater/log-theta choice, invariance under the Theorem 3.11 equality quotient, and representatives over selected indices construct the quotient equivalence used by the adapter. Thus the local selected-q Ob5–Ob6 boundary now constructs the canonical $\Phi(P_B)$ family and the exact $\Xi(P_B)$ family from this calibration before the global C_Θ comparison; closed-family and product-hull wrappers feed the same audit. The remaining boundary is to keep lowering the additive-Haar source-core localized determinant decomposition. A principal-hull-aligned refinement removes the direct valuation-cover/public-hull equality from this boundary: Lean derives it from the lower principal valuation-ball cover's own principal-hull theorem and the equality between that principal hull system and the public Theorem 3.11 principal-product hull system. A hull-formation possible-region refinement lowers the remaining possible-region equality: the source-core and valuation-cover families are aligned with the concrete family inside ‘principalProductHullFormationData‘, and Lean derives the public ‘principalProductPossibleRegion‘ equality by projection. The p-adic source-core localized determinant alignment is now constructed from these transported valuation-cover fibers: the finite-extension source core is built from transported local calibration data and finite-extension unit-ball Haar factors, and Lean proves the projected vector-bundle and localized-region equalities consumed by the older route. The remaining lower matching datum is the factor-level equality between each projected finite-extension valuation ball and the transported valuation-cover factor. A finite-extension-backed valuation-cover wrapper now lowers the valuation-cover side itself: the lower principal valuation cover is projected from finite-extension p-adic Haar data before it enters the constructed source-core route. A hull-formation-selected intrinsic cover constructor fixes the lower p-adic cover's hull system to the public principal-product hull system and fixes its possible-region family to `principalProductHullFormationData`; the intrinsic possible-region comparison is therefore produced by construction, while the principal-hull comparison is exposed as the lower equality between the p-adic principal hull source and the public principal-product hull system. A public-principal specialization now constructs this lower p-adic principal source as the `ULift` copy of the public principal-product source and proves the product-hull intersection law by transport through the public intersection theorem. Lean now also proves that the induced holomorphic hull system is unchanged by this `ULift`: the proof reduces equality of `IUTStage1Remark395HolomorphicHullSystem` records to equality of the `isHull` predicate and `logVolume` field via the generated constructor-injectivity theorem. The intrinsic wrapper also removes the separate public finite-extension factor family: Lean transports the lower cover's finite-extension factors across the same hull-system equality used for the valuation-cover calibration fiber and derives the old factor-matching field internally. An inverse-dilation refinement lowers the analytic input further: each transported valuation-ball Haar factor is identified with the inverse base-prime unit ball of a constructed finite-extension p-adic Haar source, and Lean derives $1 \leq \mu(U)_{\mathbb{R}}$ from the strict mass theorem $1 < \mu(p^{-1}O_K)_{\mathbb{R}}$. The strict selected determinant summand is now tied to this same inverse-dilation family: the unified public route projects the selected summand from the (`positiveIndex`, `positiveSummand`) member of the valuation-cover family, rather than accepting an independent selected-factor inverse-dilation source and equality. A boundary audit theorem now exposes this dependency shape by proving both the all-factor and selected-summand inverse-base-prime unit-ball equalities from the same family. The hardened preferred wrapper lowers the theta scalar input one more step: it takes the Hodge/family-hull equality and derives the localized adjusted determinant-sum equality by the localized hull decomposition theorem before invoking the unified inverse-dilation route. This wrapper has now been lowered again: the inverse-dilation branch can start from the finite localized Hodge-summand formula and derive the Hodge/family-hull comparison from the same Remark 3.9.5 decomposition. Lean also proves the converse source-facing expansion needed by the public route: a localized-adjusted inverse-dilation source projects to the finite Hodge-summand source by unfolding the localized adjusted sum as the finite sum of weighted adjusted log-volumes. This expansion is now also available from the constructed Gaussian/Hodge inverse-dilation wrapper itself: Lean reads the constructed Hodge theta-summand family, proves its pointwise equality with the localized weighted adjusted log-volumes, and then constructs the finite Hodge-summand calibrated source directly. The corresponding public finite-summand route now enters through this constructed summand-calibrated source, rather than first projecting to the older localized-adjusted wrapper. The Haar-modulus-backed inverse-dilation wrapper now exposes the same finite-summand expansion boundary after projecting its measure-level inverse-base-prime cover to the generic inverse-dilation source; its audit records the direct constructed Hodge-summand boundary, not the older localized-adjusted expansion detour. The additive-Haar construction layer now uses this same route after it constructs the Gaussian/Hodge localized-adjusted source from compact-open factors and the strict inverse-base-prime mass theorem. The normalized structure-sheaf branch now uses the finite-summand route as well: after deriving the direct-summand handoff from compact-open normalized Haar data, the residual and IUT-IV public wrappers enter through the Hodge-summand expansion endpoint rather than the older family-hull endpoint, and the additive-Haar arithmetic-degree payload is certified by the same normalized finite-summand audit. The non-normalized additive-Haar/Gaussian-Hodge companion now has the parallel summand audit and an IUT-IV arithmetic-source wrapper for the finite-summand endpoint, so its determinant/calibration payload is no longer tied only to the older family-hull route. The residue inverse-base-prime normalized finite-summand branch now uses this same direct construction in its IUT IV and constructor-built arithmetic wrappers: Lean builds the normalized additive-Haar arithmetic obligations from the finite-summand audit instead of routing these wrappers through the older direct-summand public endpoint. The same finite-summand endpoint has now been lifted through the intrinsic finite-extension Haar-modulus, unit-ball Haar-character, and residue-module quotient wrappers, so those local arithmetic sources no longer expose only the family-hull route at their public boundary. On the normalized branch, the unit-ball Haar-character wrapper now has its own CTheta-derived endpoint and full boundary audit, and the residue-module normalized route now factors through that unit-ball audit before the intrinsic Haar-modulus normalized boundary. On the non-normalized branch, the intrinsic finite-extension Haar-modulus wrapper now constructs the same p-adic arithmetic-degree obligations directly from its projected additive-Haar source. The non-normalized unit-ball Haar-character wrapper and the residue-module/unit-ball Haar-character wrapper now also have CTheta-derived finite-summand endpoints and full boundary audits, so their remaining additive-Haar arithmetic payload is projected from the IUT IV ledger rather than supplied separately; these endpoints now build the p-adic arithmetic-degree obligations at their own projected finite-summand boundary instead of delegating that construction to a lower generic Haar-modulus wrapper. The finite-place local-global and localized Step (xi) C_Θ wrappers now use their own residual payload constructors at the same boundary, both for the non-normalized finite-summand endpoints and for the normalized structure-sheaf/direct-summand endpoints, instead of first collapsing to the coupled IUT-IV arithmetic-source projection. It now also lowers the structure-sheaf input: the hardened wrapper takes localized direct-summand calibration and derives the additive-Haar normalized compact-open equality by the additive-Haar projection theorem. A complementary non-vacuous Haar-modulus wrapper now uses the opposite, source-level direction required for the additive-Haar construction: it takes the localized structure-sheaf equality with the normalized compact-open Haar factor and derives the Step (xi) structure-sheaf/direct-summand equality from the additive-Haar localized vector-bundle decomposition theorem. The current preferred wrapper lowers the surrounding Gaussian/Hodge boundary again: the restriction local-global object, ordinary-log-volume/Hodge-degree calibration, direct-summand data, and Hodge/family-hull equality are projected from the constructed Gaussian/Hodge localized-adjusted source. It now lowers that source boundary further: Lean constructs the Gaussian/Hodge localized Ob7 source from additive-Haar compact-open factors, derives the selected strict mass from the inverse base-prime factor, projects the Haar-modulus valuation-cover side from the intrinsic finite-extension valuation-cover source, and derives the inverse-dilation Haar-modulus family from unit-ball Haar-character data. The latter is now itself constructed from residue-module quotient data: O_v/p_vO_v, $\#\kappa_v = p_v$, and $\dim_{\kappa_v}(O_v/p_vO_v) = [K_v : \mathbb{Q}_p]$ give the coset decomposition and $\mu(p_vO_v) = p_v^{-[K_v:\mathbb{Q}_p]}$. The normalized-Haar direct-summand handoff has now been propagated through this intrinsic finite-extension and residue-module route as well: the residue source supplies the structure-sheaf equality with the normalized additive-Haar compact-open factor, and Lean derives the Step (xi) structure-sheaf/direct-summand equality before invoking the non-vacuous comparison route. The transported-factor handoff audit follows the residue-module source into the localized additive-Haar vector-bundle factor: the transported valuation-ball factor and the localized determinant factor are both identified with the inverse base-prime unit ball $p^{-1}O_K$ constructed from the residue quotient data. Lean now also projects this equality to the concrete scale invariants used downstream: Haar mass $p^{[K:\mathbb{Q}_p]}$ and strictly positive normalized log-volume for both factors. This handoff is threaded through the additive-Haar residue audit, the normalized residue audit, and the strongest CTheta full-boundary audit, so the public audit object exposes the determinant-factor match directly. The equality boundary has also been lowered in two stages. A component-matched residue source asks for equality of the defining additive-Haar compact-open data fields, and Lean reconstructs the previous full source equality by additive-Haar extensionality. Below that, a scale-component residue source matches the transported factor directly against the finite-extension inverse-dilation mass source: local ring, residue prime, finite degree, realization, Haar measure, $p^{-1}O_K$, and base-prime dilation. A further valuation-ball source now lowers this once more: the transported local factor is matched before additive-Haar projection, including its valuation norm, selected valuation ball $\{x \mid v(x) \le r\} = p^{-1}O_K$, Haar measure, local realization, and pointwise base-prime dilation. Lean derives the previous scale-component source from this valuation-ball match, and then projects through the unit-ball Haar-character and intrinsic Haar-modulus wrappers, so the lower valuation ladder consumes the same concrete inverse-dilation components rather than a fresh equality at each layer. The valuation-ball source itself has also been lowered into SourceCore: from the constructed finite-extension dilation-mass source and a positive-radius identification $p^{-1}O_K = \{x \mid v(x) \le r\}$, Lean constructs the valuation-ball additive-Haar source and proves the inverse base-prime component match. Thus this part of the interface no longer needs to be entered as a single opaque equality of normalization records. Under the normed-algebra compatibility for a finite extension $K/\mathbb{Q}_p$, the radius identity itself is now derived: Lean proves from $\|\mathrm{algebraMap}(p)\| = \|p\|_p = p^{-1}$ that the inverse base-prime unit ball is exactly the valuation ball of radius p, and constructs the corresponding inverse base-prime valuation-ball source. This radius computation also exposed a hard obstruction in the surrounding Step (xi) interface: if the transported intrinsic finite-extension factor is still the valuation-unit-ball cover, then its selected radius is 1, while the constructed inverse base-prime factor has radius p. Lean proves that the corresponding normed finite-extension match source is locally contradictory, via `NormedFiniteExtensionInverseBasePrimeValuationBallMatchedSource.no_unitBallCover_factor`. The replacement local cover has now been added: SourceCore constructs a finite-extension inverse-base-prime valuation-cover packet whose local selected factors have radius p from the outset, projects it to the generic valuation-ball tensor packet, and proves the inverse-base-prime component match for every local factor. Step (xi) also has a hull-formation-selected public principal-product sibling route for this radius-p cover, so the unit-ball branch and the inverse-dilated branch are separated explicitly. The residue-module handoff has now been moved onto this inverse cover at the local-factor level: the new handoff identifies the cover's inverse-dilation mass source with the residue-module constructed mass source and derives the equality between the radius-p valuation factor's additive-Haar normalization and the residue-module inverse-base-prime unit-ball normalization. The preferred-public normalized additive-Haar route has now been migrated to this inverse-cover projection as well: its Haar-modulus valuation-cover argument is definitionally the radius-p residue-module inverse-cover source, rather than the legacy unit-ball intrinsic cover. The non-normalized additive-Haar/Gaussian-Hodge route has now been added for the same inverse-cover source: the residue-module wrapper supplies the localized structure-sheaf/direct-summand comparison, projects the Haar-modulus valuation-cover argument from the radius-p inverse cover, and then reuses the generic additive-Haar compact-open ladder to construct the Gaussian/Hodge localized-adjusted Ob7 source. Thus the normalized public wrapper is no longer the only residue-module inverse-cover entry point. The additive-Haar arithmetic-degree obligation package for this non-normalized route is now constructed as well: its Step (xi) determinant/calibration slots are read from the additive-Haar/Gaussian-Hodge finite-summand source audit, and the remaining lower arithmetic residual is projected from the IUT IV C_Θ ledger. The same non-normalized finite-summand route now also has constructor-coupled, finite-place, and localized Step (xi) C_Θ entry points through the intrinsic, unit-ball Haar-character, and residue-module/unit-ball finite-extension wrappers. This finite-summand endpoint is now exposed for the residue inverse-base-prime wrapper itself: its audit records the radius-p residue inverse-cover projection together with the inherited Haar-modulus/additive-Haar finite-summand expansion boundary. The non-normalized residue inverse-cover branch now also has constructor-built, coupled, finite-place local-global, and localized Step (xi) C_Θ handoffs, so this arithmetic layer no longer has to pass through the looser localized-adjusted/family-hull endpoint. The normalized residue inverse-base-prime wrapper reaches this finite-summand endpoint as well. Lean reconstructs the structure-sheaf/direct-summand handoff from the normalized compact-open Haar structure-sheaf equality and then projects to the generic normalized Hodge-summand expansion route. Constructor-built, coupled, finite-place, and localized Step (xi) C_Θ sources now thread this normalized residue endpoint without separately supplying local canonical scales; the intrinsic and unit-ball Haar-character normalized parent wrappers expose the same constructor entry points. This boundary has been lowered once more: the residue inverse-cover route can now take a constructed Gaussian/Hodge localized-adjusted source over the same residue-derived localized hull vector bundle, and Lean projects from it the Hodge local-global comparison, selected direct-summand data, structure-sheaf equality, and localized-adjusted theta equality required by the older additive-Haar route. The same residue corridor now has a summand-form source: the localized-adjusted theta equality is derived by summing the finite Hodge summands and identifying them pointwise with localized weighted-adjusted determinant terms. A companion restriction-local-global wrapper now derives the older public source's global object, localization, local object identifiers, and pointwise Hodge/local-prime equality from the environment-anchored realified Frobenioid restriction source. The constructed Gaussian/Hodge residue source now also factors through this restriction wrapper. The summand-form residue source factors through the same layer as well: its audit combines the finite-summand theta expansion with the projected local-global restriction boundary, so the anchored local-global source is produced by the constructed summand source rather than bypassed by the older public fields. The inverse-dilation principal-product branch now lowers this summation boundary again: a square-weight source derives the localized-adjusted theta equality from normalized finite Hodge weights together with pointwise identities between each weighted Hodge contribution and the corresponding localized weighted-adjusted determinant term. A stricter normalized-summand source now derives these weights from the constructed Hodge summands by dividing by the nonzero theta-monoid degree; Lean proves the resulting weights have total 1 and projects to the square-weight route. A raw-positive variant derives the required theta nonvanishing from local determinant positivity, and a direct-summand refinement now derives that raw positivity from the Ob3 comparison: the structure sheaf is one nonnegative direct summand, while a distinct summand is strictly positive. The square-weight branch now also projects through this direct-summand raw-positive route, so the public square presentation and the positivity normalization are no longer parallel route-level assumptions. The Haar-modulus inverse-dilation and additive-Haar public branches have now been promoted to the same strict route: their compact-open/valuation-cover constructors feed the direct-summand raw-positive normalization instead of stopping at localized adjustment. The intrinsic p-adic finite-extension, unit-ball Haar-character, and residue-module/unit-ball wrappers now expose the same strict projection, so these lower arithmetic branches no longer land only at the older localized-adjusted finite-summand endpoint. What remains below this route is to construct the underlying local square/direct-summand determinant factors and the pointwise determinant identities from the lower Hodge-theater and log-theta-lattice data. The full-boundary audit for the current route also ties the IUT IV arithmetic residual to the same public wrapper. The remaining principal-hull boundary is now lower than the public route wrapper: the coordinate and real-line public endpoints construct the principal source and paper trace internally, and the coordinate scalar-image landing/preimage laws are derived from the exact-image formula. The remaining obligation is to construct that exact local image formula from the lower Hodge-theater/log-theta-lattice data. The selected valuation-ball exact-Ξ audit now consumes the same coordinate scalar-image source, rather than a parallel selected scalar wrapper. For the selected valuation-ball hull, the nonzero-scalar multiplication route now constructs the selected scalar-parameter source and its Ob3/Ob5 handoff; the remaining work is to construct the lower transported-factor/residue-module matching data directly from the same Hodge-theater/log-theta local-field package. The valuation-unit-ball finite-extension branch remains only a normalized legacy boundary: the selected radius is forced to be 1, so $\mu(O_v) = 1$ constructs weak nonnegativity but also proves that the normalized log-volume of the selected unit-ball factor is 0. Thus the strict-positive unit-ball route is explicitly quarantined as vacuous, now also at the exact intrinsic finite-extension public endpoint. The p-adic finite norm-square/direct-summand localized-adjusted route has now been checked against the same normalization: Lean proves that the selected unit-ball direct-summand norm is forced to be 0, so the nonzero-norm version is also a quarantined unit-ball branch rather than a non-vacuous determinant route. Its wrappers through the family-hull indexed measure model, direct-product measure model, tensor-measure cover, source-core tensor cover, charted local-ring cover, valuation-ball cover, principal-hull aligned cover, transported valuation-cover calibration, constructed source-core valuation-cover, finite-extension-backed cover, and intrinsic finite-extension cover wrappers remain checked compatibility infrastructure, but not a completion witness. The explicit norm-square compatibility route `preferredPublicConcreteStepXI311312NormSquareRoute` consumes this concrete finite-extension valuation-cover data, the localized-adjusted norm-square/direct-summand source, and the explicitly lower-level additive-Haar remaining-payload audit; Lean constructs the assembled non-Step-(xi) payload internally, so this boundary no longer accepts a preassembled Step (xi) audit. The short preferred route `preferredPublicConcreteStepXI311312Route` has now been moved to the constructor-built localized C_Θ surface: it projects the IUT IV arithmetic residual and additive-Haar payload from the constructor-built possible-image holomorphic-hull/determinant source and the localized Step (xi) finite-place local-term source, rather than accepting those objects independently. The new constructor-built localized C_Θ goal-evidence audit records this short surface directly: it includes the constructor-built Remark 3.9.5 Step (xi) boundary, localized finite-place C_Θ endpoint, local-term-to-IUT-IV C_Θ handoff audit, local-to-global C_Θ audit, and the additive-Haar remaining payload derived from the localized arithmetic source. The residue inverse-base-prime projected Gaussian/Hodge additive-Haar branch, including its finite-Hodge-summand refinement, has also been lowered to this finite-place/localized constructor-built C_Θ surface, so it no longer exposes an independent IUT-IV arithmetic ledger after the summand expansion has been constructed. The same lowering now covers the residue restriction/local-global variants: the base restriction source, the projected Gaussian/Hodge restriction source, and the summand-projected restriction source all enter through the localized constructor-built C_Θ handoff. It also covers the plain direct residue inverse-base-prime additive-Haar branch and its normalized structure-sheaf companion, so those direct routes now use the localized constructor-built C_Θ handoff as well. A stricter restriction-calibrated public variant `preferredPublicConcreteStepXI311312RestrictionCalibratedRoute` now uses the environment-anchored local-global Frobenioid restriction package to derive the Hodge local-prime comparison before entering this same constructor-built localized C_Θ handoff. Its route audit now also carries the inverse-dilation strict-selected audit from the same restriction-calibrated source, so strict Haar mass, positive normalized compact-open log-volume, and the selected nonzero direct-summand norm are exposed at the public restriction-calibrated boundary rather than as a detached side result. A corresponding restriction-calibrated goal-evidence package now records this route audit together with the projected constructor-built localized C_Θ goal evidence. The handoff audit identifies the constructor canonical scale with the finite-place scale, proves the summed Step (xi) Haar-defect arithmetic gap, and derives the global C_Θ bound used by the ordered-real dichotomy. A stricter product-hull localized C_Θ wrapper now constructs the Step (xi) source from the Remark 3.9.5 product-hull system, Ob3/Ob4 adjusted determinant data, and exact theta/family-hull log-volume equalities before entering this short preferred route; its audit carries product-hull provenance together with the localized C_Θ payload derivation. The same audit now exposes the constructor-sourced Step (xi) endpoint, where the obligations object is definitionally the one produced by the legacy finite-route hull/determinant constructor. The obligations-backed variant projects the bridge equality, q-pilot positivity, and normalization from a single `IUTStage1SourceHullDetObligations` object, leaving only the identification of that bridge datum with the record-canonical product-hull/tensor-power construction explicit. The stricter canonical-HDD wrapper now moves that synchronization below the preferred route into `ProductHullCanonicalHDDCommonContainerSource`: its endpoint records that the package HDD hull/determinant bridge equals the constructor-generated obligations bridge and hence the record-canonical product-hull/tensor-power bridge, while the preferred route itself consumes this lower source object and the localized C_Θ source. Below that surface, the new `withHDDHullDetBridge` constructor builds a package whose HDD hull/determinant bridge is definitionally the record-canonical product-hull bridge, and the corresponding definitional-package audit verifies preservation of the chart and signed q/Θ endpoints. The Theorem 3.11 source record is now transported across this canonical-HDD package by `canonicalHDDRecord`; the transfer audit records that the transported record carries the constructed coric HDD bridge, preserves the possible-image family, retains the target-union equality, and keeps the separated Hodge-theater histories. The canonical-HDD preferred route now enters the standard short route through a constructor-built Step (xi) source over this transported record, so the constructed HDD bridge is part of the consumed Theorem 3.11 source data. A still stricter direct canonical-HDD wrapper now removes the preassembled `IUTStage1SourceHullDetObligations` input at this boundary: `ProductHullCanonicalHDDDirectCommonContainerSource` carries the product-hull, adjusted determinant, exact theta/family-hull calibration, q-positivity, and normalization data; Lean installs the record-canonical HDD bridge definitionally, transports the Theorem 3.11 record, and generates the hull/determinant obligations from the resulting constructor. The newest generated-full-label specialization, `GeneratedFullLabelProductHullDirectCanonicalHDDSource`, now ties this direct-HDD source to the generated Theorem 3.11 full-label quotient corridor: the multiradial record's possible images are the generated theta-class pullback, and the Step (xi) q-choice is the generated selected q-choice. The chosen-output generated full-label source, `ChosenOutputGeneratedFullLabelProductHullDirectCanonicalHDDSource`, lowers this once more by projecting that generated quotient corridor from the chosen-output primitive-packet route and then rebuilding the older generated-HDD source internally. The constructed-record version, `ConstructedRecordChosenOutputGeneratedFullLabelProductHullDirectCanonicalHDDSource`, first rebuilds the generated record's possible-image family from the chosen-output packet theorem and then lowers to the chosen-output source. The structured-bundle version, `StructuredGeneratedChosenOutputFullLabelProductHullDirectCanonicalHDDSource`, constructs the generated multiradial record from a generated SHE-structured bundle before entering the constructed-record source. The reindexed-structured version, `ReindexedStructuredGeneratedChosenOutputFullLabelProductHullDirectCanonicalHDDSource`, derives that generated bundle from the concrete structured bundle before entering the structured-bundle source. The constructed-qualitative version, `ConstructedQualitativeGeneratedChosenOutputFullLabelProductHullDirectCanonicalHDDSource`, derives that concrete structured bundle from constructed qualitative Theorem 3.11/SHE data and carries the qualitative and APT transport audits through the same reindexing. The corresponding preferred route, `preferredPublicConcreteStepXI311312ConstructedQualitativeGeneratedChosenOutputFullLabelProductHu` is definitionally the generated full-label route applied after these lowerings. These hardenings are now also available on one public surface: `preferredPublicConcreteStepXI311312RestrictionCalibratedGeneratedFullLabelProductHullDirectCanon` uses the generated full-label/direct canonical-HDD Step (xi) constructor while also requiring the restriction-calibrated local-global Frobenioid source for the p-adic Hodge local-prime comparison. Its audit pins both inputs: the generated full-label quotient corridor and direct-HDD constructor on the Step (xi) side, and the restriction-calibrated short-route audit on the p-adic side. This has now been tightened at the principal-product boundary: `ConstructedQualitativePrincipalProductHullOb3Ob5MeasureHodgeGapGeneratedChosenOutputFullLabelPro` requires the principal-product Ob3/Ob5 measure/Hodge-gap HDD source to be indexed by the constructed-qualitative generated record and synchronizes its q-choice with the selected generated q-choice. The corresponding restriction-calibrated route `preferredPublicConcreteStepXI311312RestrictionCalibratedConstructedQualitativePrincipalProductHu` enters the principal-product route through this coupled source. The direct-HDD audit now also exposes the restriction-calibrated goal-evidence package, so the inverse-dilation strict-selected branch remains visible after passing to the product-hull/HDD constructor layer. The separate current product-hull goal-evidence package records this non-vacuous principal-product route explicitly, avoiding confusion with the older quarantined norm-square compatibility audit. Its constructed-qualitative strengthening promotes the same evidence over the coupled constructed-qualitative/principal-product source, so this evidence surface no longer ranges over a generic constructor record at this point. A companion current evidence audit now couples that principal-product route to the restriction-calibrated norm-square/direct-summand projection feeding the Ob7 construction layer: Lean constructs the projected p-adic finite localized hull source, records the structure-sheaf/direct-summand synchronization and selected nonzero norm-square input, and exposes the restriction/Hodge-calibrated Ob7 source projection. This is a route-hardening step, not a completion marker, since the unit-ball norm-square branch remains diagnostic and quarantined from the inverse-dilation strict-selected branch. The same current evidence surface now has a structured-SHE version: Lean first constructs the principal-product direct-HDD source from the Remark 3.9.5 principal hull, Ob3/Ob4 adjusted determinant data, Hodge family-hull calibration, and structured Theorem 3.11/SHE side-condition package, so the source-obligation gap is no longer supplied as an assembled field at this current route boundary. A further adjusted-log-volume variant, `preferredPublicConcreteStepXI311312RestrictionCalibratedProductHullOb3Ob5AdjustedLogVolumeDirect` constructs that direct-HDD source from the record Ob3/Ob5 finite adjusted summand identity, deriving the normalized determinant equality internally instead of exposing it as a separate route input. The gap-backed variant `preferredPublicConcreteStepXI311312RestrictionCalibratedProductHullOb3Ob5AdjustedLogVolumeGapDir` lowers two more fields at this same boundary: q-pilot positivity and source normalization are projected from `IUTStage1SourceObligationGap`, while the adjusted Ob3/Ob5 source continues to derive the normalized determinant equality internally. The hodge-calibrated gap route `preferredPublicConcreteStepXI311312RestrictionCalibratedProductHullOb3Ob5AdjustedLogVolumeHodgeG` lowers the direct-HDD theta/family-hull equality as well: it is derived from the Hodge–Arakelov theta-monoid calibration of the bounded family-hull source canonically projected from the adjusted Ob3/Ob5 determinant source, together with the finite adjusted determinant endpoint. The measure-calibrated variant `preferredPublicConcreteStepXI311312RestrictionCalibratedProductHullOb3Ob5AdjustedLogVolumeMeasur` lowers the measure synchronization at the same boundary: Lean derives the Ob3/Ob5 hull-log-volume measure equality from the pre-ledger calibration to the product-hull holomorphic hull operator and the Ob3/Ob5/product-hull operator identification. The product-hull-constructed variant `preferredPublicConcreteStepXI311312RestrictionCalibratedProductHullConstructedOb3Ob5AdjustedLogV` lowers that operator-identification boundary as well: the Ob3/Ob5 adjusted source is built with the product-hull holomorphic hull operator, so the Ob3/Ob5/product-hull operator match is definitional rather than supplied as a separate compatibility field. The principal-product variant preferredPublicConcreteStepXI311312RestrictionCalibratedPrincipalProductHullConstructedOb3Ob5Adj lowers the product-hull source itself one level: the source carries the Remark 3.9.5 principal presentation by scalar multiples of the local integer region, and Lean projects the ordinary product-hull system internally. Below that public route, the constructor principalProductHullConstructedOb3Ob5SourceOfThetaRegionDefinedPrincipalValuationBallSource now derives the Ob3/Ob4 source and adjusted-summand equality from the theta-region-defined principal valuation-ball cover. Its structured-SHE variant principalProductHullConstructedOb3Ob5SourceOfThetaRegionDefinedPrincipalValuationBallStructuredS also derives the source-obligation gap from the structured Theorem 3.11/SHE bundle and side conditions. Its summand-charted Hodge variant principalProductHullConstructedOb3Ob5SourceOfThetaRegionDefinedPrincipalValuationBallStructuredS lowers the Hodge-family calibration one step further: the supplied Hodge input is the charted equality between the Hodge theta-monoid degree and the finite adjusted determinant summand sum, from which Lean derives the determinant, normalized determinant, and family-hull equalities internally. The remaining measure input is now represented by IUTStage1PreLedgerHolomorphicHullMeasureCalibration, a pointwise calibration saying that every target-region volume in the pre-ledger is the Remark 3.9.5 holomorphic-hull log-volume for the principal hull operator; Lean derives the equality of region-measure records consumed by the older route. The current product-hull/Ob7 diagnostic surface now has the corresponding theta-region constructor: Lean builds the principal-product HDD source from the theta-region principal valuation-ball data before assembling the current norm-square/Ob7 audit fields. That current audit also exposes the constructor-built Step (xi) boundary and the localized finite-place C_Θ endpoint/handoff supplied by the localized Step (xi) source; the structured-SHE and theta-region wrappers now project these same fields at their own concrete boundaries. The primitive wrapper primitiveThetaRegionCurrentProductHullPrincipalHDDSource removes the constructed-SHE bundle and transport guard from this HDD boundary by projecting them from the primitive Theorem 3.11 source constructor. The preferred public route now has the parallel theta-region route audit, and preferredPublicConcreteStepXI311312PrimitiveTheorem311ConstructedThetaRegionGoalCompletionAudit ties that route audit to the current norm-square/Ob7 evidence from the same theta-region source data while deriving the multiradial record, constructed-SHE bundle, and transport guard from the primitive Theorem 3.11 source constructor. The concrete packet route now builds this primitive source constructor through IUTStage1PrimitiveTheorem311BridgeCertificateSourceConstructor, which derives the primitive qualitative source from the transported IPL/SHE/APT bridge-certificate source. On the qualitative side, IUTStage1SourceObligationGap can now be constructed from a strengthened structured-SHE Theorem 3.11 bundle together with the q-positivity and normalization side conditions: the promoted SHE-alignment field is read from the structured SHE/common-container context rather than supplied as an independent equality, and the resulting obligations agree with the existing structured-input obligation constructor. The remaining boundary is lower: the original common-container/SHE and log-Kummer compatibility data still have to be derived from primitive paper-side Hodge-theater and log-theta-lattice constructions. The older norm-square public-surface and goal-scope evidence audits remain checked compatibility evidence, but now explicitly quarantine the p-adic unit-ball nonzero-norm boundary as well as the compact-open strict-positive boundary; they are not completion markers. The packet-facing goal-evidence refinement now projects the Theorem 3.11 source data from the concrete primitive packet and the Remark 3.9.5 principal hull from the packet/symmetry-label theta-region valuation-ball source before entering that norm-square evidence audit, so those two objects are no longer independent public inputs at this final compatibility boundary. A further restriction-calibrated packet refinement now projects the intrinsic finite-extension valuation cover and localized norm-square/direct-summand calibration from the pointwise restriction-calibrated local arithmetic source before entering the packet-facing final evidence audit; this branch remains diagnostic, since it also carries the unit-ball norm-square contradiction. The localized-C_Θ packet refinement now removes the standalone IUT IV C_Θ ledger, additive-Haar arithmetic-degree obligations, and remaining-payload witness from this same packet boundary: Lean projects them from the constructor-built localized Step (xi) local-term source before calling the restriction-calibrated packet evidence. The pointwise-localized packet refinement lowers this once more by constructing that localized C_Θ source from the pointwise localized Step (xi) determinant source, local main-log terms, Haar normalization defects, and finite-place summation identities before deriving the signed comparison and dichotomy at the packet-facing evidence layer. The arithmetic-divisor-localized packet refinement lowers that boundary again: the pointwise localized Step (xi) source is projected from the IUT IV arithmetic-divisor local evaluation ledger, whose local arithmetic-upper terms are identified with the Step (xi) weighted log-volumes, Haar defects, and main-log contributions before the same packet evidence is derived. The Theorem 1.10 valuation-ball packet refinement now lowers the IUT IV input one step further, projecting that arithmetic-divisor localized source from the valuation-ball additive-Haar formula-gap source before the packet evidence enters the arithmetic-divisor layer. The local-analytic valuation-ball packet refinement lowers the source once more: the formula-gap matched Theorem 1.10 source is projected from the local-analytic valuation-ball arithmetic-divisor source at the packet-facing boundary. The synchronized local-analytic packet refinement then constructs this local-analytic source from the constructor-built Step (xi) determinant/scale synchronization, the local-analytic arithmetic-divisor ledger, and the finite Haar/main-log comparison data. Its arithmetic-realized synchronized variant lowers this last comparison: it takes the explicit Theorem 1.10 local identity $\mathrm{StepXI}_v + \mathrm{Haar}_v = a_v(D_v + C_v) + E_v - M_v$ and derives the older Step (xi)/Haar/main-log arithmetic-upper equality by unfolding the local arithmetic contribution. The formula-gap synchronized packet refinement lowers the local-analytic arithmetic-divisor input further by projecting it from the named valuation-ball formula-gap source carrying the Proposition 1.4 log-shell and Proposition 1.5 metric comparison records. Its arithmetic-realized companion performs the same recovery of the Step (xi)/Haar/main-log equality from $a_v(D_v + C_v) + E_v - M_v$ after projecting the local-analytic ledger from the formula-gap source. Across the pointwise localized, IUT-IV arithmetic-divisor, Theorem 1.10 valuation-ball, local-analytic, synchronized, and formula-gap packet audits, the constructed finite-place local-global C_Θ endpoint is now exposed explicitly rather than being visible only at the lower localized handoff. The pointwise Theorem 1.10 valuation-ball packet route now consumes the corresponding pointwise localized source, so the determinant/scale synchronization, formula-gap source, Haar defect, Step (xi)/Haar/main-log identity, and finite Haar lower bound are projected internally at this layer. This route is now a named audit record in its own right: it records the pointwise Theorem 1.10 endpoint, the internally constructed determinant/scale synchronization endpoint, the projected formula-gap packet evidence, and the resulting signed C_Θ comparison. The localized local-term packet route constructs that pointwise source from the localized Step (xi) local-term C_Θ source and the valuation-ball formula-gap source. A compatible refinement packages the remaining local arithmetic/main-log matching as a named source layer: the valuation-ball Theorem 1.10 local main-log function is the localized Step (xi) main-log summand, and the local arithmetic upper contribution is the adjusted determinant log-volume plus Haar defect plus that summand. This replaces two anonymous equality families at the packet route by an audited compatibility source. The next arithmetic-realized refinement derives this compatibility from the expanded local Theorem 1.10 arithmetic-degree expression $a_v(D_v + C_v) + E_v - M_v$: the localized Step (xi) adjusted determinant plus Haar defect is matched with this local arithmetic-minus-main-log quantity, and Lean recovers the local arithmetic-upper equality by unfolding the Theorem 1.10 local contribution before entering the formula-gap synchronized packet audit. Lean now records this as a named arithmetic-realized packet audit, projecting the compatible packet evidence together with the localized C_Θ source, the local-term/IUT IV handoff audit, the signed C_Θ bound, and the Corollary 3.12 dichotomy directly. The formula-realized refinement goes one layer lower: it consumes the pointwise Hodge measure/summand formula source, constructs the theta-region and pointwise calibration records internally, and then enters the same arithmetic-realized packet audit. The local-arithmetic-degree refinement removes the packaged arithmetic-realization record at this boundary: it constructs that source from the main-log matching and the pointwise identity $\mathrm{StepXI}_v + \mathrm{Haar}_v = a_v(D_v + C_v) + E_v - M_v$ before recovering the localized C_Θ handoff and signed endpoint. The synchronized local-arithmetic-degree refinement goes one level lower again: from determinant/scale synchronization, the valuation-ball formula-gap ledger, the Haar defect, and the same local degree identity, Lean constructs the IUT IV arithmetic-divisor localized source and the localized local-term C_Θ source internally. Its constructed-synchronization companion now builds that synchronization record from the localized Step (xi) determinant source together with the constructor-built determinant and canonical-scale comparison equalities before entering this route. The norm-square localized determinant refinement projects this localized source from direct-summand norm-square vector-bundle data before using the same constructed-synchronization route. The p-adic finite-extension refinement now projects that norm-square determinant from unit-ball valuation-Haar compact-open local data before entering the same endpoint, and its synchronized local-upper variant packages the p-adic finite determinant together with the constructor determinant and canonical-scale sum identifications as one source object. The formula-gap local-upper variant now projects the local-analytic arithmetic-divisor ledger from the Theorem 1.10 valuation-ball formula-gap source before applying the same residual calculation. This boundary is now also represented by a typed residual-Haar source: the p-adic finite determinant/scale synchronization, the formula-gap source, and the total residual Haar lower bound are packaged together, and Lean projects this package to the localized Step (xi) C_Θ source, the local-to-global C_Θ handoff audit, the signed C_Θ inequality, and the dichotomy. The corresponding concrete compact-open restriction-calibrated public route now consumes this residual-Haar source directly, rather than taking the p-adic synchronization, formula-gap ledger, and residual inequality as separate final arguments. A further local-arithmetic handoff wrapper also consumes the product-level Ob7 handoff source, so the restriction-calibrated source, structure-sheaf/direct-summand equality, and nonzero summand norm are projected from one typed local-arithmetic package. The latest wrapper also consumes the pointwise measure/summand formula source, replacing the three public Hodge/measure formula equalities by a single source object. The current strongest compact-open wrapper consumes the named local Kummer realization map source, including its possible-region section and compact-open soundness laws. Lean forms the correspondence as the graph of this map, extracts the image laws, and then derives Remark 3.9.5 possible-region exactness, theta-region log-shell image laws, the local chart, and realized-exact compact-open sources internally. A pointwise Theorem 1.10 refinement also projects the determinant/scale synchronization and the arithmetic-degree Step (xi)+Haar identity from a single localized valuation-ball source. The construction of the arithmetic realization from foundational Hodge-theater/Frobenioid data remains a later boundary. Older public-looking route names remain implementation and compatibility layers below this citation point. The residue-module/unit-ball normalized restriction-calibrated source now also projects to the p-adic finite norm-square/direct-summand calibration source: Lean constructs the projected p-adic finite localized hull decomposition and then builds the norm-square source from the same restriction package. The remaining explicit inputs at this handoff are exactly the projected structure-sheaf/direct-summand synchronization on the p-adic unit-ball branch and the selected nonzero norm-square summand. The follow-up audit proves that this selected p-adic unit-ball norm is actually zero under the current normalization, so the handoff is diagnostic rather than non-vacuous; the additive-Haar inverse-dilation branch is therefore the side on which a non-unit selected factor still has to be constructed. That construction has now been exposed at the residue restriction-calibrated boundary: Lean projects the same residue-module source to the unified inverse-dilation localized-adjusted source and audits the selected inverse-base-prime additive-Haar factor. The audit proves strict Haar mass, positive normalized compact-open log-volume, and nonzero direct-summand norm for this selected factor, without reusing the quarantined p-adic unit-ball norm-square branch. The restriction-calibrated public route audit now includes this inverse-dilation strict-selected audit directly. Finite summation derives the Ob3-1 inequalities comparing the structure-sheaf log-volume with the localized bundle log-volume; the subtraction formula derives adjusted raw local positivity; positive determinant weights derive localized adjusted summand positivity; finite summation gives localized family-hull positivity. The theta scalar at the strict boundary is the environment theta divisor log-volume by definition. The public route now constructs the older Hodge-summand, Gaussian/Hodge, Hodge-evaluation, and Hodge-scalar sources internally from the Hodge canonical-degree divisor construction, audited evaluation, and localized adjusted-sum calibration, then derives the environment-theta/localized adjusted-sum equality; Lean then derives equality with the family-hull log-volume from the localized Ob3/Ob5 identity. This derived equality implies the family-hull/global calibration via theta/ordinary realified Frobenioid compatibility, and the localized determinant/family-hull equality then gives the determinant/global equality needed for the Gaussian-global nonzero proof. The earlier structure-sheaf/direct-summand boundary has also been lowered at the finite-extension endpoint: instead of publicly supplying 'structure-sheaf equals direct summand', the newest compact-open factor route uses the valuation-unit-ball normalized log-volume identity for the distinguished structure-sheaf factor and derives the direct-summand equality. The non-vacuous Haar-modulus/additive-Haar branch now has the same normalized-Haar-to-direct-summand handoff without passing through the vacuous unit-ball endpoint. On the residue-module/unit-ball normalized branch, the Hodge local-prime calibration has also been lowered: Lean now constructs the older normalized source from an environment-anchored local-global Frobenioid restriction source and a single ordinary-log-volume/Hodge-degree equality. Its preferred boundary audit also exposes the Step (xi-a)–(xi-g)/Ob1–Ob9 paper trace constructed from the same additive-Haar compact-open factor source, including the Ob7–Ob9 log-Kummer/vertical-shift/realified-log-volume entries. For strict positivity, the additive-Haar route now has a genuine Haar-mass-to-norm constructor, and the inverse-dilation finite-extension branch constructs a non-unit selected factor with strict Haar mass. The unit-ball finite-extension route remains as an explicit vacuity diagnostic. The earlier structure-sheaf-positive, raw-positive, summand-positive, explicit-ratio, weighted-partition, and one-summand localized constructors remain as compatibility layers. This is progress below a free-floating obligation bundle. What remains is to construct, from the source papers, the actual lower operations corresponding to:

the analytic construction of the class-indexed possible-image source,
the analytic construction of the place-audited packet/action symmetry-label transport inputs,
the analytic construction of the global-to-local restrictions and local-prime global equality,
the analytic construction of non-diagonal prime-place coordinates and the coric character,
the Hodge canonical-degree local-prime calibration, direct-summand/square determinant presentation, and point
the construction of the remaining theta-region local-field Haar factors,
and their calibration from the Hodge-theater packet data.

The strict route now derives the unit-ball tensor factors and one non-unit inverse-dilation strict factor from finite-extension local-field Haar data, and can build the record Ob3/Ob5 determinant/log-volume source from a principal-product theta-region cover identity and its finite additivity formula. The compact-open valuation-ball handoff now projects the final determinant identification from that record source at the exact possible-region, map, image, and correspondence boundaries. The exact-theta source-spine handoff can also construct its record Ob3/Ob5 determinant source from localized hull/vector-bundle data once the localized family is identified with the source-spine possible-image family; this now extends through tensor-label, label-transport, and symmetry-label transport, as well as the concrete target-region lattice formula and the concrete lattice-image law. The remaining work is to construct the full analytic valuation-cover family, its global bridge calibrations, and the Hodge-theater packet source that selects the appropriate non-unit local factors.

5.3 IUT IV estimates

The C_Θ comparison depends on IUT IV local and global estimates. The current formalization organizes the route through additive-Haar arithmetic-degree and p-adic formula objects, but the deepest estimates are still source obligations. What is now checked for the preferred route is the ordered-real and finite-sum handoff from the arithmetic gap to the canonical C_Θ-scale comparison, together with a constructor-built source-level bridge for the local-to-global handoff. In the older bridge, the constructed Step (xi) canonical scale is identified with the finite-place IUT IV scale before plus-one cancellation is applied. In the newer coupled bridge, the finite-place IUT IV scale is defined from the constructor-built Step (xi) canonical scale itself, so the exposed source assumption is the arithmetic gap inequality rather than a separate scale-identification equality. The finite-place constructor lowers this further: the arithmetic gap is derived from finite local Step (xi) summation data, namely local canonical scales, arithmetic/log contributions, local Haar defects, the local inequalities, and a global defect lower bound. The localized local-term constructor then lowers the local inequalities themselves by taking each local canonical scale to be the weighted adjusted log-volume of the localized Ob3/Ob4 determinant source and each local upper term to be Step (xi) log-volume plus Haar defect plus main-log contribution. This constructor is now fed directly by the arithmetic-divisor- backed additive-Haar source, which projects the local analytic and Theorem 1.10 C_Θ records through the formula-gap arithmetic-divisor source rather than assembling them inside the final finite-divisor route or its top-level paper-trace audit. The pointwise concrete-packet route now has the same lowering: it projects the raw IUT IV arithmetic-divisor local evaluation ledger from a valuation-ball additive-Haar formula-gap source before building the localized Step (xi) handoff. The Theorem 1.10 valuation-ball local-analytic localized source now also exposes the resulting localized C_Θ handoff audit, signed C_Θ bound, and dichotomy as source-level projections; the current product-hull evidence audit consumes these projections directly, rather than accepting an independent localized C_Θ source at that surface. The Record-Ob3/Ob5 valuation-ball source now projects to the closed valuation-ball formula-gap source and carries a DOD audit showing that the local Haar summation, arithmetic gap, signed C_Θ bound, and dichotomy are inherited without adding a raw signed comparison premise. The named-HDD compact-open boundary now projects this same valuation-ball DOD audit, making the closed C_Θ chain visible at the route-input boundary rather than only in the lower source namespace. The canonical-HDD compact-open route audit also records this DOD evidence next to the reconstructed synchronized route input, the internally constructed local-analytic Step (xi) C_Θ source and handoff audit, the derived canonical-scale bound, signed C_Θ inequality, source-bridge inequality, Corollary 3.12 dichotomy, and definitional equality to the compact-open handoff. Its Hodge-formula companion also projects the pointwise measure/summand calibration from the formula source before invoking this current product-hull handoff. The local-arithmetic companion constructs the restriction-calibrated record from the residue-module/unit-ball Haar-character source, the realified Frobenioid restriction package, and the structure-sheaf/direct-summand normalization identities before invoking the same handoff. Its constructed-Theorem 1.10 companion now assembles the local-analytic IUT IV localized source from the localized determinant, scale-synchronization, arithmetic-divisor, Haar-defect, and finite-defect data before invoking the same current-product-hull route. Its synchronization-source variant removes the separate localized determinant source and determinant/scale equalities from this public surface, replacing them by one constructor-built localized Step (xi) synchronization source. What remains to construct from first principles is this lower IUT IV source itself:

> the distinguished log-shell inclusions and numerical bounds,
> the nondistinguished zero log-volume contribution,
> the archimedean metric containment,
> the arithmetic divisor source for Theorem 1.10,
> the distinguished and archimedean formula-to-gap comparison sources,
> the local finite-place Step (xi) inequalities and Haar defect bound.

Until these estimates and the upstream source spine are constructed from the published source data, the corridor remains conditional on source-paper mathematical content.

5.4 Local arithmetic and p-adic analytic infrastructure

The implementation has moved below a purely formal real-number boundary, but it still lacks significant local arithmetic infrastructure. The remaining work includes finite extensions of $\mathbb{Q}_p$, residue fields, Haar-normalized unit balls, explicit local degrees, different and conductor terms, and the formula matching that connects those objects to the arithmetic-degree route. This is standard mathematical infrastructure in spirit, but it is not small work in Lean.

6 Blueprint and Apodeixis

6.1 Lean blueprint status

The repository contains a Lean blueprint under `blueprint/`. It has a compact prose structure: aim and scope, the source Stage 1 route, the formal interface, the route to Corollary 3.12, open source obligations, and later work. It also has a generated declaration index, `blueprint/lean_decls`, which currently lists thousands of Lean declarations.

This means the blueprint is not empty, but it is still intentionally slim. It is adequate as a navigational document for the present stage. It is not yet a complete dependency graph of the project in the strongest sense: many nodes are marked not ready, and the prose chapters do not yet expand every major Lean route into a polished theorem-by-theorem mathematical dependency tree.

The right standard for the blueprint is different from the standard for this paper. The blueprint should be a map: concise nodes, Lean links, readiness status, and dependencies. This paper should be an argument: what we are formalizing, why the type boundaries matter, what is proved, and what remains mathematically open.

6.2 Apodeixis integration plan

The project is intended to be compatible with Apodeixis as a collaboration and review layer. The practical goal is to expose the formalization in a way that supports external mathematical review:

$$\begin{gathered}\text{source-paper claim}\\ \Downarrow\\ \text{blueprint node}\\ \Downarrow\\ \text{Lean declaration or source obligation}\\ \Downarrow\\ \text{audit status and review discussion.}\end{gathered}$$

For this to work, the Lean code must avoid opaque omnibus assumptions, the blueprint must keep source claims linked to declarations, and the paper must avoid becoming a changelog. The current audit records are designed with this future use in mind. They give Apodeixis a natural unit of review: a named mathematical obligation with a precise Lean consumer.

7 Methodological Lessons

7.1 Typed boundaries are productive

The main technical lesson so far is that typed boundaries are not bureaucracy. They are mathematical tests. Once ordered real-line copies, transports, finite-label quotients, hull regions, source certificates, and upper-semi relations are assigned different types, informal shortcuts become impossible unless they are backed by explicit constructors.

This is especially useful in the 3.11–3.12 corridor because the public dispute is about whether a comparison has lost information during transport between different mathematical histories. Lean cannot decide that philosophical issue by itself, but it can prevent a formal proof from hiding the relevant movement.

7.2 Conditional routes are still useful

A conditional theorem can be scientifically useful if it names its conditions precisely. The current Stage 1 route is not a proof of Corollary 3.12 from the published IUT papers, but it has lowered the ambiguity of the problem. The remaining obligations are no longer "somewhere in the comparison". They are organized into Theorem 3.11 source construction, Step (xi) hull/determinant construction, IUT IV local-to-global estimates, and local arithmetic infrastructure.

7.3 The paper must not be a ledger

The earlier draft recorded too many implementation increments. That made it long but not more informative. The standing policy for future updates should be:

update the relevant section,
replace stale claims,
move only stable mathematical outcomes into the paper,
leave declaration-level detail to Lean and the blueprint.

This paper should remain regular research-paper length. The blueprint and generated declaration index are the correct place for detailed navigation.

8 Roadmap

The next milestones should eliminate source-obligation families rather than adding more endpoint wrappers.

Review-handoff continuation. The next broad goal is to replace the remaining analytic Step (xi) and IUT IV source interfaces by paper-level constructions. The initial-theta Hodge-theater source, concrete Hodge–Arakelov pilots, and finite Theorem 3.11 procession/log-Kummer route are now constructed. The intended next order is: lower the Remark 3.9.5 hull, determinant, and Ob3/Ob4/Ob5 synchronization fields to source data; extend the constructed IPL/SHE/APT compatibility through the analytic common container; finally formalize the IUT IV local estimate package that supplies the signed C_Θ upper bound. A reviewable completion of the 3.11–3.12 corridor should expose an endpoint whose inputs are paper-level Hodge-theater/log-theta/lattice data and whose remaining assumptions are exactly those outside the stated IUT III/IUT IV source scope.

Milestone A: Step (xi) hull/determinant construction. Continue lowering the remaining pointwise Hodge-calibration and local arithmetic fields below the constructed Remark 3.9.5/Step (xi) hull and determinant sources. The public route now constructs the principal hull, theta-region source, determinant handoff, localized Step (xi) C_Θ source, and signed endpoint from structured pointwise source objects. The latest compact-open additive-Haar variant also bundles the constructor-built determinant and canonical-scale identifications with the localized Step (xi) determinant into a single synchronization source, rather than exposing those identifications as separate public equalities. The same compact-open boundary now also has a pointwise Theorem 1.10/IUT IV variant in which the synchronization, formula-gap valuation-ball source, Haar defect, local arithmetic identity, and finite Haar lower bound are projected from one source; the p-adic formula-gap residual source performs the analogous finite-extension handoff by constructing the localized C_Θ source and signed dichotomy from the synchronized p-adic determinant and formula-gap ledger, and the concrete public wrapper now consumes that bundled source directly; the newest wrapper also replaces the three local-arithmetic restriction witnesses by the single product-level Ob7 handoff source, and then replaces the three measure/summand formula witnesses by the pointwise formula source; the current log-Kummer map wrapper further derives the correspondence, image laws, possible-region exactness, log-shell image, local chart, and realized compact-open exact sources from the named local realization map; the Record-Ob3/Ob5 synchronized companion now projects that source from the packet's canonical HDD record on the compact-open additive-Haar branch and audits the signed C_Θ comparison/dichotomy at that boundary. The next work is to derive the remaining pointwise Hodge and local-field inputs from the underlying Hodge-theater/Frobenioid data. The lower local-arithmetic boundary has also been moved, at both the named-HDD and component surfaces, from a supplied arithmetic-degree-controlled source to a constructed source obtained from valuation-ball formula-gap/p-adic data plus the Proposition 1.4/1.5 comparison synchronization; the remaining matching equality is now explicitly the equality to this constructed formula source. The product-handoff route has then been lowered one more step: the comparison package may now be projected from a local degree-identification formula source recording the Step (v) distinguished tensor formula, the Step (vii) archimedean coarse formula, and the arithmetic-degree coefficient lower bound. The concrete Record-Ob3/Ob5 component route now also combines this local-degree projection with the controlled-local-arithmetic construction of the valuation-ball formula-matching package, so neither the assembled valuation-ball formula source nor the comparison package is a separate public input at that handoff. A further p-adic Haar-index variant constructs the controlled local-arithmetic source itself from the valuation-ball local analytic arithmetic-divisor evaluation, unit-ball Haar defect, Step (xi) arithmetic-degree calibration, and the distinguished/archimedean formula bounds. Its direct controlled-product handoff now avoids the local-degree comparison reconstruction and only requires the Record-Ob3/Ob5 component equality with the p-adic Haar constructed formula source. The Record-Ob3/Ob5 component source now also projects its local degree-identification formula source internally from the arithmetic-divisor/effective-different/conductor component chain and definitionally preserves the formula core used by the comparison package. A projected-local-degree p-adic Haar route uses this projection to remove the standalone local-degree formula source and matching proof from that public surface; the remaining synchronization is the Record-Ob3/Ob5 component formula-gap equality against the p-adic Haar constructed formula source and its reconstructed comparison handoff on the local-degree branch. The controlled product-handoff branch stops before that reconstruction, so it uses only the direct component-to-p-adic-Haar formula synchronization; the audit includes a one-place countermodel showing why that remaining equality cannot be dropped. The lower p-adic Haar-index variant adds the parallel diagnostic after the controlled local arithmetic source has itself been reconstructed from the Theorem 1.10/p-adic Haar-index equations. The strict residual-consumer diagnostic then shows the downstream failure if this synchronization is not definitional: the component residual and reconstructed p-adic residual may disagree. The same projection is now available at the p-adic defect/main source itself: Record-Ob3/Ob5 arithmetic-divisor component data construct the valuation-ball p-adic defect/main local-estimate source after deriving the local-degree formula package and comparison matching internally. The stronger Record-Ob3/Ob5 valuation-ball source removes the remaining explicit component-to-valuation-ball equality from this p-adic handoff: it carries that equality as part of the valuation-ball record and projects the same defect/main source and audit directly from the packaged source. This projection now factors through a named arithmetic-degree-controlled local arithmetic source projected from the valuation-ball Record-Ob3/Ob5 data, so the p-adic defect/main constructor consumes that constructed local-arithmetic object rather than rebuilding it anonymously at the final handoff. The local-degree formula valuation-ball Record-Ob3/Ob5 source packages the formula-matched valuation-ball source, the Step (v)/(vii) local-degree formula source, and the controlled component indexed by the constructed local arithmetic object; the split bridge projects this source directly to the canonical residual package without separately exposing the local-degree formula package. The canonical residual-frontier audit is lower again: the p-adic-defect/main valuation-ball controlled Record-Ob3/Ob5 component source is now reconstructed from the matched local-degree object valuation-ball source before projecting to the canonical residual package. The local-degree formula valuation-ball source itself is projected from those typed Step (v)/(vii) objects, and neither the local-degree formula source nor the p-adic-defect/main source appears as an interface-only residual entry. The same matched-object source now also projects to the canonical case-local Theorem 1.10 arithmetic-degree bound package, so the pointwise Step (v)/(vi)/ Step (vii) upper-bound consumer and the residual ledger are fed by the same typed local-degree object layer rather than by a separately supplied local-degree formula comparison source. The same comparison layer is now lower when typed Step (v)/(vii) local-degree objects are available: Lean projects the local-degree formula source and its valuation-ball matching proof from those objects and their additive formula synchronization before constructing the p-adic-defect/main controlled source. The matched-object source is also the public local-estimate argument of the compact-open canonical-HDD inverse-base-prime route, so this lowering is now on the route spine rather than only a residual projection. The bridge now also names direct projections from that matched-object source to the canonical p-adic arithmetic formula-matching source, the local arithmetic-degree residual source, and the formula-gap/residual audit. It is also no longer a primitive object once the controlled Record-Ob3/Ob5 component is present: the controlled-component projection constructs it internally from the arithmetic-divisor-backed local-degree chain. The compact-open route has been pushed to that controlled-component boundary, so the matched-object package is now internal at that handoff. The same canonical-HDD route now also consumes the p-adic-defect/main Record-Ob3/Ob5 source directly; Lean projects the arithmetic-degree controlled local source and synchronized controlled component from that source before entering the controlled-component route. The product-handoff companion now performs both lowerings together: its public inputs are the product-level Ob7 handoff and the p-adic-defect/main Record-Ob3/Ob5 source, and Lean reconstructs the controlled component and arithmetic-degree local source inside the p-adic-defect/main canonical-HDD product handoff. Thus the strict p-adic-Haar controlled source, arithmetic-degree-controlled local source, and component/formula synchronization are reconstructed before the residual consumer is entered, without treating the p-adic-defect/main controlled source as a separate lower source boundary. The same p-adic-defect/main controlled source now also enters a compact-open product-handoff formula-realized route directly; that public route now factors through the p-adic-defect/main canonical-HDD product handoff rather than first calling the older controlled-component handoff. It now also enters the projected-local-degree formula compact-open route by projecting the local Step (v)/(vii) formula package and its comparison matching from the controlled Record-Ob3/Ob5 component chain, then using the local-degree formula controlled source to synchronize the reconstructed local arithmetic handoff. At the constructor level, the same source now also proves the projected-weighted formula-gap residual signed endpoint after one explicit equality aligns the arithmetic-degree calibration localized Step (xi) source with the principal-product p-adic finite localized Step (xi) source; the formula-gap Theorem 1.10 package and the local arithmetic-degree residual package are then projected internally rather than supplied separately. The corresponding projection audit proves, before the signed comparison is invoked, that this lower source carries the formula-gap endpoint, the valuation-ball local-analytic endpoint, and the canonical residual endpoint. Lean now names this as a direct projection from the p-adic-defect/main Record-Ob3/Ob5 boundary to the local-degree formula comparison layer, so the public route no longer has to reopen the strict p-adic-Haar route merely to exhibit the Step (v)/(vii) formula package. The synchronized controlled component layer is now the higher comparison boundary for this residual route, and the strict p-adic-Haar route remains a checked comparison route. At the local-degree formula comparison boundary, the p-adic-defect/main valuation-ball local-estimate source is now proved to be the source reconstructed from the derived arithmetic-degree-controlled local source, so it is no longer an independent package there. The same reconstruction equality is now checked for the strict p-adic-Haar route and for the p-adic-defect/main source after projection through that strict route. The new local-degree p-adic-defect/main audit records the distinguished Step (v) different/conductor bound, the nondistinguished Step (vi) zero-gap law, the archimedean Step (vii) different/conductor identification, $E_v = \delta_v + M_v$, valuation-ball projection preservation, and the lowering into the p-adic-defect/main Record-Ob3/Ob5 component in one checked object. The finite-place arithmetic-defect record is now canonicalized extensionally from the p-adic unit-ball Haar-index defect source, so the local IUT IV defect is no longer an independent numerical datum once that Haar source is fixed. The same canonicalization now lifts to the arithmetic-degree controlled local source: Lean identifies the carried source with the one reconstructed from the p-adic Haar-index constructor and the same Step (xi) calibration.

Milestone B: IUT IV local-to-global C_Θ. Continue lowering the arithmetic-divisor-backed finite-place source. The distinguished, nondistinguished, and archimedean local processional estimates now feed the local C_Θ handoff through formalized Proposition 1.4/1.5 inequality chains, and the distinguished one-unit term can now be routed through the arithmetic-minus-main residual rather than a stronger local procession formula margin. The remaining work is to construct that arithmetic residual from the underlying local-field/Haar data and push the result through the finite-place summation. At the principal-pointwise packet boundary, the localized IUT IV arithmetic-divisor handoff now has a p-adic unit-ball specialization: the Haar defect is read from the finite-place unit-ball index source, and the pointwise identity $U_v = \mathrm{StepXI}_v + \delta_v + M_v$ constructs both the older localized C_Θ source and the Step (xi) synchronization source. The preferred principal-pointwise public endpoint now consumes this p-adic source directly before projecting to the older arithmetic-divisor route. This removes one abstract defect function from that public layer. A further processional p-adic route now constructs the same p-adic handoff from the Theorem 1.10 valuation-ball local-analytic source and the local Step (xi)-to-arithmetic synchronization $\mathrm{StepXI}_v = U_v - M_v - \delta_v$, so the p-adic public route no longer needs an already-packaged arithmetic-divisor p-adic source when these processional data are available. Its synchronized variant now also reads the localized determinant source and the canonical C_Θ-scale finite-sum identity from the constructor-built/localized Step (xi) synchronization source, rather than exposing those two equalities separately. A further processional-estimate variant constructs the bundled p-adic processional synchronization from the local p-adic unit-ball Step (xi) identity and the Theorem 1.10 local-analytic processional-estimate source, with a single identification between the normalized p-adic place and the distinguished finite place. Thus this public route no longer takes the processional p-adic synchronization record as an independent source object. The normalized-place refinement removes that last identification as a public hypothesis: the processional estimate is stated directly at the p-adic unit-ball normalized place, so the distinguished nonarchimedean place is no longer chosen independently at this boundary. A further localized-local-term variant reads the same localized determinant and scale data directly from the already-derived localized local-term C_Θ source, leaving only the local-analytic processional Step (xi) identity at that boundary. The localized-local-term handoff now also has an equality-free p-adic-normalized constructor: ofLocalizedStepXILocalTermCThetaSourcePadicNormalizedProcessionalEstimateSource combines the local-term determinant/scale source, the p-adic Step (xi) identity, and processional-estimate data whose distinguished place is definitionally the p-adic normalized place. Thus the bundled processional p-adic synchronization source and the normalized-place equality are no longer separate inputs at that constructor surface. The newest pointwise Theorem 1.10 variant projects the localized determinant data and local-analytic valuation-ball ledger from the pointwise IUT IV localized source itself, so the p-adic processional endpoint no longer takes a separate localized local-term source at this layer. A new Haar-residual pointwise endpoint now rebuilds the bundled processional p-adic source from that pointwise source and the expanded p-adic unit-ball Haar-index arithmetic-degree residual identity before deriving the signed Corollary 3.12 dichotomy. The normalized pointwise endpoint remains the equality-free processional-estimate route: it uses the p-adic Step (xi) synchronization identity and p-adic-normalized processional estimates. The local-analytic pointwise refinement lowers this once more: the formula-gap valuation-ball package is no longer exposed at the p-adic public boundary; the endpoint reads the localized Step (xi) determinant object, determinant and scale comparisons, and Theorem 1.10 local-analytic ledger from the pointwise local-analytic localized source. Its Haar-residual public wrapper now removes the bundled processional p-adic synchronization from that endpoint surface by deriving it from the expanded residual source. A still lower public wrapper now separates that expanded residual source from the processional-estimate ledger, retaining only the equality that the distinguished processional place is the p-adic normalized place. The pointwise local-analytic p-adic wrapper combines this with the p-adic-normalized convention and then weakens the processional side once more: it exposes the pointwise local-analytic source, the expanded p-adic Haar-residual identity, and p-adic-normalized processional case data. The case source records the case-by-case normalized log-volume identities and the p-adic normalized-place kind, but no longer asks for the stronger distinguished +1 processional upper-bound field; the extra unit contribution is supplied by the Haar-residual lower bound. That endpoint therefore constructs the p-adic Step (xi) synchronization and distinguished alignment internally. The canonical boundary has now been lowered once more: the p-adic unit-ball/local-analytic endpoint takes a p-adic unit-ball localized Step (xi) source, a matching valuation-ball local-analytic Theorem 1.10 ledger, their local arithmetic-evaluation equality, and the normalized processional case data. Lean projects the pointwise local-analytic localized source and derives the expanded p-adic Haar-residual arithmetic-degree source internally. The normalized case layer now also projects to the lower-weight local arithmetic-degree residual package, exposing the distinguished finite place of weight 1 and the finite-sum Haar lower bound at this boundary. It is now checked as the formal weakening of the full processional estimate source: the latter supplies the stronger distinguished +1 processional upper-bound inequality, while the normalized case branch discards that field and relies on the p-adic Haar residual for the one-unit contribution. Its public pointwise-residual refinement replaces the normalized processional case input by the lower processional pointwise-residual source plus the p-adic normalized-place alignment, while still deriving the local-analytic and Haar-residual projections from the p-adic unit-ball source before entering the signed Corollary 3.12 endpoint. The arithmetic-gap refinement lowers the same surface once more: the normalized case source is constructed from the processional arithmetic-gap theorem's normalized Step (v)/(vi)/(vii) local-volume identities, with only the distinguished-place alignment left explicit at that stage. The p-adic-normalized arithmetic-gap refinement removes that equality from the public surface by indexing the arithmetic-gap theorem over the p-adic Haar-residual object itself; its distinguished place is definitionally the normalized finite place. The processional Haar-defect refinement lowers this surface again: it consumes the processional Haar-defect residual theorem directly, projects the pointwise residual theorem internally, and then follows the same p-adic unit-ball/local-analytic signed route. Thus the one-unit contribution is now read from the Haar-defect residual data rather than from an already assembled pointwise residual theorem. The p-adic unit-ball Haar-index refinement is a checked comparison route below the older generic Haar-defect surface: it consumes processional p-adic unit-ball Haar-defect residual data and projects the generic processional Haar-defect source internally. Its reconstructed variant lowers this once more by defining the p-adic Haar-residual object from the localized packet and the matching Theorem 1.10 local-evaluation equality; the normalized-place alignment is then 'rfl' at the public signed endpoint. The selected Theorem 1.10 case-defined reconstructed p-adic Haar-defect endpoint is now the compact canonical frontier for the main p-adic unit-ball/local-analytic route. Its source fixes the processional local value by the Theorem 1.10 case split and selects the p-adic Haar normalized place as the distinguished nonarchimedean case in the local-kind function, so Lean reconstructs the older estimate, normalized-case, pointwise residual, and Haar-defect handoffs internally; what remains is the paper-level derivation of the Step (xi) equality against that case-defined value. The same reconstructed local-analytic handoff now also proves the $C_\Theta \geq -1$ companion at this p-adic unit-ball boundary, so the lower bound no longer has to pass through the older public local-analytic source argument. The lower-bound projection has now also been lifted through the stronger processional p-adic source, using its embedded valuation-ball local-analytic ledger and p-adic unit-ball projection. The remaining work is to derive this smaller source package from the underlying Theorem 3.11/Hodge-theater and local-field constructions.

Milestone C: Source-faithful Theorem 3.11 reconstruction (open). The existing finite translations, quotient, preorder, coordinate bijection, and transport laws are regression models only. Completion requires the LGP Gaussian log-theta lattice, D-prime-strip processions, tensor packets, integral structures, splitting monoids, Frobenioids, genuine indeterminacy actions, vertical

Kummer isomorphisms, normalized log volumes, and horizontal equivariance from the cited IUT I–III clauses.

Milestone D: Blueprint expansion. Expand the blueprint from a slim guide plus declaration index into a richer review graph. Each nontrivial source obligation should have a node with source citations, Lean declarations, readiness status, and a short explanation of the mathematical construction still missing.

9 Conclusion

The project has reached a useful but conditional Stage 1 state. Lean verifies the final real inequality, distinguishes the disputed comparison levels, keeps $\mathrm{Ind}_1/\mathrm{Ind}_2$ equality behavior separate from Ind_3 upper-semi behavior, and threads a concrete Hodge-theater/log-theta-style source route into the current C_Θ/Corollary 3.12 endpoint. The code is not merely a prose annotation of the papers; it is an executable audit of a precise corridor.

What remains is real mathematics, not formatting. To settle the 3.11–3.12 issue inside this project, the source obligations must be discharged from Mochizuki's constructions: genuine Remark 3.9.5 hull/determinant operations, IUT IV local estimates, and the local arithmetic infrastructure connecting them. The formalization has made that task sharper. It has not made it disappear.

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